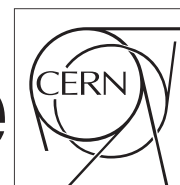


The Compact Muon Solenoid Experiment

CMS Draft Note

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Measurement of $\frac{\sigma(B_c^\pm) \times \text{Br}(B_c^\pm \rightarrow J/\psi \pi^\pm)}{\sigma(B^\pm) \times \text{Br}(B^\pm \rightarrow J/\psi K^\pm)}$ and $\frac{\text{Br}(B_c^\pm \rightarrow J/\psi \pi^\pm \pi^\pm \pi^\mp)}{\text{Br}(B_c^\pm \rightarrow J/\psi \pi^\pm)}$ at $\sqrt{s}=7$ TeV.

The CMS Collaboration

Abstract

The $B_c^\pm \rightarrow J/\psi \pi^\pm$ and $B_c^\pm \rightarrow J/\psi \pi^\pm \pi^\pm \pi^\mp$ decay modes are studied in CMS in the kinematic region where the transverse momentum of the $B_c^\pm$ meson is greater than 15 GeV/c and within the central rapidity region $|y| < 1.6$. Two ratios are measured: $\frac{\sigma(B_c^\pm) \times \text{Br}(B_c^\pm \rightarrow J/\psi \pi^\pm)}{\sigma(B^\pm) \times \text{Br}(B^\pm \rightarrow J/\psi K^\pm)} = (0.48 \pm 0.05 (stat) \pm 0.03 (syst) \pm 0.05 (\tau_{B_c})) \times 10^{-2}$ and $\frac{\text{Br}(B_c^\pm \rightarrow J/\psi \pi^\pm \pi^\pm \pi^\mp)}{\text{Br}(B_c^\pm \rightarrow J/\psi \pi^\pm)} = 2.55 \pm 0.80 (stat) \pm 0.33 (syst)_{-0.01}^{+0.04} (\tau_{B_c})$.

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PDFAuthor: G.Boz, P.Dini, S.Fiorendi, S.Malvezzi

PDFTitle: Measurement of the ratio of branching fractions B_c to J/ψ π π π over B_c to J/ψ π and of the production cross-section times branching fraction between the B_c to J/ψ π and the B^+ to J/ψ K in CMS at $\sqrt{s}=7$ TeV.

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Please also verify that the abstract does not use any user defined symbols

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1 Introductory comments

This version of the note contains the updates implemented in the analysis after the PAS was made public; it is aligned with the paper PAPER BPH-12-011. The version aligned with the public PAS BPH-12-011 is AN2012_437_v5.pdf. The main differences wrt the previous analysis are the removal of the MC-Match procedure by ΔR , discussed below, the investigation of a possible systematics due to the arbitration scheme in the $B_c^\pm \rightarrow J/\psi \pi^\pm \pi^\pm \pi^\mp$ mode (see section 10.2.5), and the re-evaluation of the systematics due to the $B_c^\pm$ lifetime uncertainty after the new LHCb measurement ([1]) (see section 9.1.5 and 10.2.6 for $R_{c/u} = \frac{\sigma(B_c^\pm) \times \text{Br}(B_c^\pm \rightarrow J/\psi \pi^\pm)}{\sigma(B^\pm) \times \text{Br}(B^\pm \rightarrow J/\psi K^\pm)}$ and $R_{Bc} = \frac{\text{Br}(B_c^\pm \rightarrow J/\psi \pi^\pm \pi^\pm \pi^\mp)}{\text{Br}(B_c^\pm \rightarrow J/\psi \pi^\pm)}$ respectively).

MC Match

In the previous version of the analysis, we used to apply a MC-Match procedure by requiring reco-GEN track matches in a ΔR cone. The $\Delta R(\text{GEN}, \text{reco})$ cut required both the pion reco track and the two muon reco tracks to be matched to the generated ones in a ΔR cone of 0.01 (for pion) or 0.006 (for muons). These values were chosen in such a way that the total number of true signal events in the signal region failing the matching was equal to the total number of signal events outside of the signal peak passing the matching. After the PAS was made public, we carried out additional studies to better investigate the impact of the MC truth match requirement on the efficiency evaluation.

BcJpsiPi channel. In the plot in Fig.1, the mass distributions of all the reconstructed events (black) and of those events passing the match requirement (green) are shown, both in a semi-log and in a linear scale. (The label “no match” means that no match cut is applied). In the full mass range, 943 events fail the match requirement, 716 of which are signal events (number obtained from the fit to the invariant mass distribution). Their invariant mass distribution is shown in Fig.2 (the right plot is the enlargement of the left one in the signal region and with the fit superimposed).

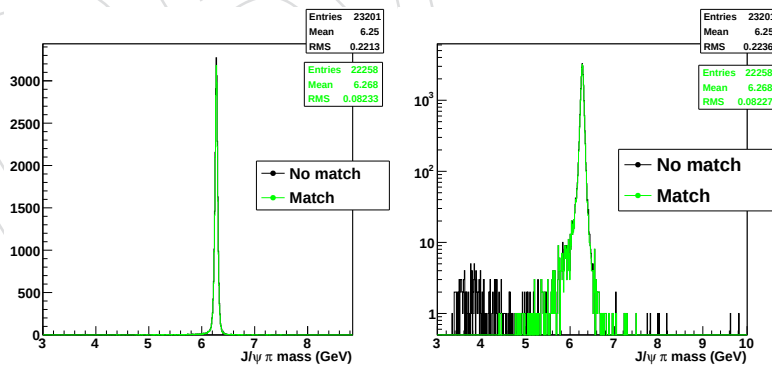


Figure 1: Invariant mass distribution of all the reconstructed events (black) and of MC matched events only (green). The plot on the left is in linear scale, the plot on the right is in semi-log scale.

The loss of events is mainly due to the requirement on muon matching (see Fig.3):

A possible reason for the “failing” events is the occurrence of final state radiation. In fact, if we exclude FSR events, we end up with 21266 reco events, only 3 of which do not pass the MC truth match cut (Fig.4).

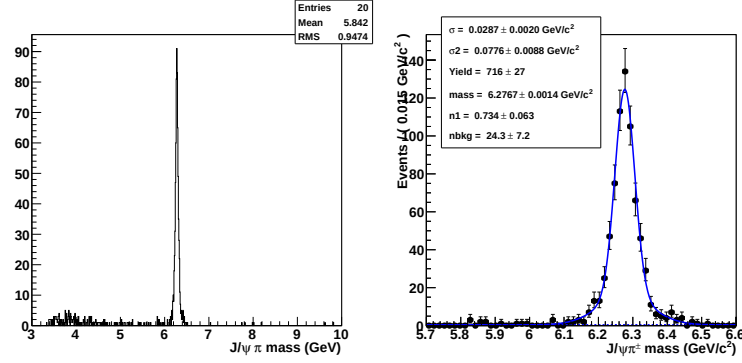


Figure 2: Invariant mass distribution of events failing the MC-match requirement.

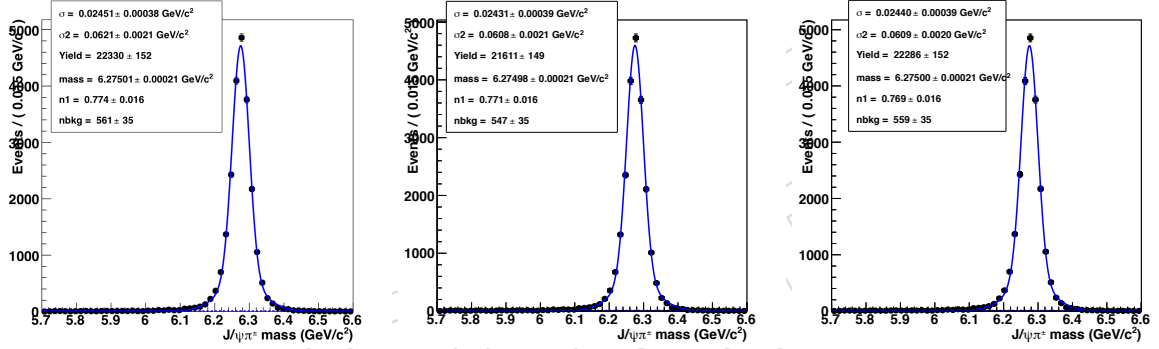


Figure 3: Invariant mass distribution for all the events (left), mu-matched events (center) and pion-matched events (right).

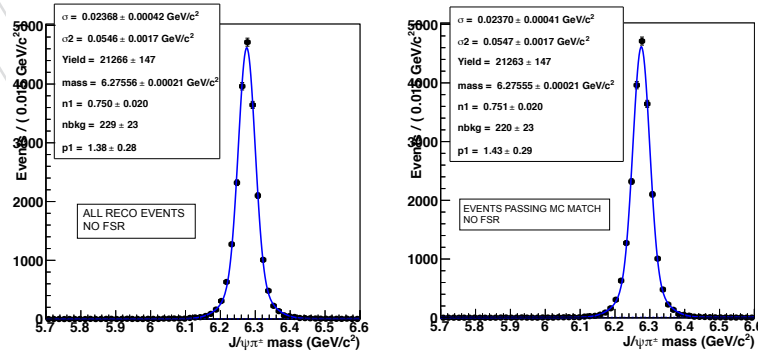


Figure 4: Invariant mass distribution of all the events with no FSR (left) and of MC-truth matched events with no FSR (right).

In order not to loose these events requiring the ΔR cut, we decided to adopt the following strategy:

- do not require any ΔR cut on the cone between the GEN and the reco track
- in order to extract the number of reco MC events, fit the invariant mass distribution of reco events with a gaussian plus a linear function; in this way, the “real” background will not be considered in the signal reco event counting

BcJpsi3pi channel. The mass distributions of all reconstructed events (black) and of those events which pass the match requirement (green) are shown in Fig.5, both in a linear and a semi-log scale.

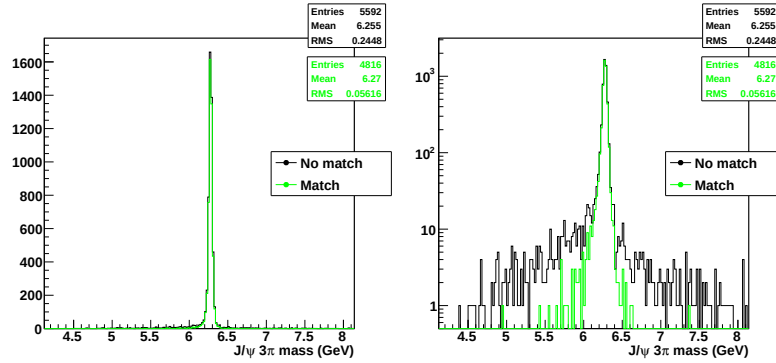


Figure 5: Invariant mass distribution of all the reconstructed events (black) and of MC-matched events only (green). The plot on the left is in linear scale, the plot on the right is in semi-log scale.

In the full mass range, 776 events fail the match requirement, 138 of which are true signal events (number obtained from the fit to the failing-events invariant-mass distribution). Their invariant mass distribution is shown in the plots in Fig.6 (the right plot is the enlargement of the left one in the signal region).

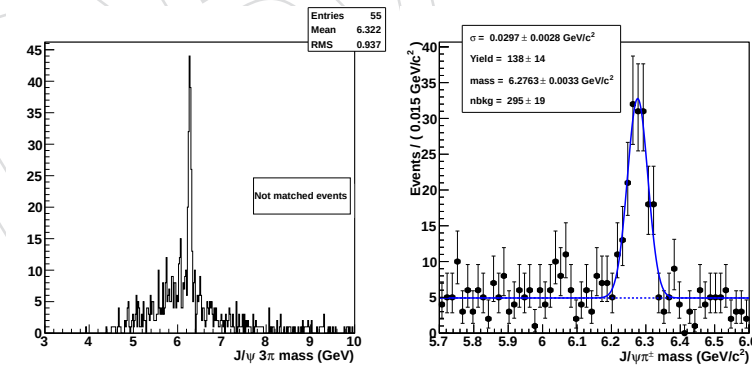


Figure 6: Events failing the MC-match requirement. The right plot is the enlargement of the left one in the signal region and with the fit superimposed.

As explained before, we decided not to apply any ΔR truth-match cut in order not to bias the efficiency calculation. The strategy in this case consist in:

- do not apply any ΔR cut
- evaluate the total number of reconstructed signal events (N_s) by fitting the reconstructed $J/\psi 3\pi$ invariant mass distribution with a Gaussian plus a flat

function

- define the signal mass range as $\pm 4 \sigma$ from the mean value of the gaussian, where σ is obtained from the fit
- extrapolate the number of background events in the signal mass range (N_b)
- when evaluating the efficiency (through the unbinned fit in 7D, see Sec.10.1), retain a reconstructed event if the $J/\psi 3\pi$ invariant mass of the candidate is within the signal mass range and a randomly generated variable between 0 and 1 is greater than the percentage of background events in the signal region ($\text{random} > N_b/N_s$).

With the above explained approach, the R_{Bc} ratio raises from 2.43 ± 0.76 to 2.55 ± 0.8 , while the $R_{c/u}$ ratio remains unchanged.

DRAFT

2 Introduction

The Heavy Flavor physics has been deeply investigated in the last decades. The B-physics program has played an important role: the Beauty Factories have shed some light onto the B^+ and B^0 mesons, while the Tevatron has led the study of higher mass B-hadrons, namely B_s , b-baryons and B_c . The advent of the LHC has opened a new era for the flavor physics and the spectroscopy of high-mass hadrons, allowing both for precise testing of the sector and for a new measurement campaign based on the copious production of Heavy Flavors at 7 and 8 TeV. One particle which deserves careful study is the B_c meson, whose small cross-section has limited, so far, its experimental investigation. A rich program of measurements is being carried out by LHCb at $2 < \eta < 5$; the complementary pseudorapidity region can be accessed by CMS. The CMS experiment, thanks to the excellent muon identification system and tracker detectors, allows for the $B_c^\pm \rightarrow J/\psi \pi^\pm$ and $B_c^\pm \rightarrow J/\psi \pi^\pm \pi^\pm \pi^\mp$ final state studies in a central $|\eta|$ region. These measurements, driven by the $J/\psi(\mu^+\mu^-)$ benchmark in the final state, will provide basic information about the $B_c^\pm$ production at 7 and 8 TeV, which are propaedeutic for any more refined experimental investigation of the $B_c^\pm$ meson. It is also interesting to study the $B_c^\pm \rightarrow J/\psi \pi^\pm$ with respect to the $B^\pm$ topological similar, more known, mode $B^\pm \rightarrow J/\psi K^\pm$. A measurement of $\frac{\sigma(B_c^\pm) \times (\text{Br}(B_c^\pm \rightarrow J/\psi \pi^\pm))}{\sigma(B^\pm) \times (\text{Br}(B^\pm \rightarrow J/\psi K^\pm))}$ provides a straightforward relative comparison between the two mesons and their production cross-sections.

3 The status of art

The $B_c^\pm$ meson is the ground state of $\bar{b}c$ system. It carries two different heavy flavors, and thus represents a unique laboratory to study the heavy-quark dynamics, complementing the $c\bar{c}$ and $b\bar{b}$ -quarkonia. The production mechanism for $\bar{b}c$ differs in an essential way from that of $b\bar{b}$, since two heavy quark-antiquark pairs must be created in a collision. While the $b\bar{b}$ pair can be created in the parton processes $\bar{q}q, \bar{g}g \rightarrow b\bar{b}$ at the order of α_s^2 , the lowest order mechanism for creating a $\bar{b}c$ system is of α_s^4 , $\bar{q}q, \bar{g}g \rightarrow (\bar{b}c)(b\bar{c})$, with the gluon-gluon contribution dominating at the LHC [2]. The first observation of approximately 20 $B_c^\pm$ events in the semileptonic channel $B_c^\pm \rightarrow J/\psi l^\pm \nu$ by the CDF Collaboration [3] demonstrated the feasibility of its experimental study. At the Tevatron only two modes have been seen: $B_c^\pm \rightarrow J/\psi l^\pm \nu$ and $B_c^\pm \rightarrow J/\psi \pi^\pm$. At the LHC new decay modes have been detected [4–8].

In the $B_c^\pm$ weak decays, both the c and b quarks compete: the b -quark decay with the c quark as spectator or the c -constituent decay with b as spectator. The annihilation process is predicted to contribute for 10 % to the final states. Experimental studies of branching ratios and form factors would help understanding the hadronic matrix elements determined by non-perturbative effects, while an accurate lifetime measurement is crucial for testing the B_c decay model. A new lifetime measurement has been recently published by the LHCb collaboration [1] in the semileptonic channel $B_c^+ \rightarrow J/\psi \mu^+ \nu X$; the only available measurement in a kinematically closed decay channel, $J/\psi \pi^-$, has been provided by the CDF collaboration [9] and is affected by an uncertainty of about 7%.

In this note results of $R_{c/u} = \frac{\sigma(B_c^\pm) \times \text{Br}(B_c^\pm \rightarrow J/\psi \pi^\pm)}{\sigma(B^\pm) \times \text{Br}(B^\pm \rightarrow J/\psi K^\pm)}$ and $R_{Bc} = \frac{\text{Br}(B_c^\pm \rightarrow J/\psi \pi^\pm \pi^\pm \pi^\mp)}{\text{Br}(B_c^\pm \rightarrow J/\psi \pi^\pm)}$ obtained with the CMS 2011 7 TeV data are discussed and compared with the LHCb measurements [4, 10].

4 Event Samples

4.1 2011 datasets and MC

The analysis is based on the 2011 data samples

- /MuOnia/Run2011A-08Nov2011-v1/AOD
- /MuOnia/Run2011B-19Nov2011-v1/AOD

and Montecarlo samples

- /BcToJPsiPi_7TeV-bcveppy-pythia6-evtgen/Fall11-HLTMuonia_PU_S6_START44_V9B-v1/AODSIM
- /BcToJPsi3Pi_7TeV-bcveppy-pythia6-evtgen/Fall11-HLTMuonia_PU_S6_START44_V9B-v1/AODSIM
- /BcToJPsiA1_7TeV-bcveppy-pythia6-evtgen/Fall11-HLTMuonia_PU_S6_START44_V9B-v1/AODSIM
- BuToJPsiK_K2MuFilter_7TeV-pythia6-evtgen/Fall11-HLTBPh20111_START44_V9B-v1/AODSIM

The considered triggers are:

- HLT_Dimuon7_Jpsi_Displaced_v1, HLT_Dimuon7_Jpsi_Displaced_v3
- HLT_DoubleMu3p5_Jpsi_Displaced_v2
- HLT_DoubleMu4_Jpsi_Displaced_v1, HLT_DoubleMu4_Jpsi_Displaced_v4
HLT_DoubleMu4_Jpsi_Displaced_v5

The analyzed integrated luminosity is 5.05fb^{-1} ; luminosity and trigger periods are summarized in Table 1 as evaluated through the pixelLumiCalc.py script.

Table 1: HLT paths and Luminosity for the signal samples:

Inst. Lumi	Run range	Int.Lumi pb^{-1}	HLT Path	L1 seed
1×10^{33}	165088-167043	728.58	HLT_Dimuon7_Jpsi_Displaced_v1	L1_DoubleMu0
1.4×10^{33}	167078-167913	255.47	HLT_Dimuon7_Jpsi_Displaced_v3	L1_DoubleMu0
2×10^{33}	170249-173198	923.83	HLT_DoubleMu3p5_Jpsi_Displaced_v2	L1_DoubleMu0
3×10^{33}	173236-178380	2243	HLT_DoubleMu4_Jpsi_Displaced_v1	L1_DoubleMu0_HighQ
5×10^{33}	178420-179889	765.35	HLT_DoubleMu4_Jpsi_Displaced_v4	L1_DoubleMu0_HighQ
5×10^{33}	179959-180252	130.57	HLT_DoubleMu4_Jpsi_Displaced_v5	L1_DoubleMu0_HighQ

The Global Tags are:

- FT_R_44_V9 for ReReco 2011A
- FT_R_44_V11 for ReReco 2011B

The json file is:

- Cert.160404-180252_7TeV_ReRecoNov08_Collisions11_JSON_MuonPhys.txt

4.2 Montecarlo samples

High statistics MC samples have been centrally produced in the CMSSW_4.4.X release to be compared with the corresponding ReReco data samples.

4.2.1 BCVEGPY

The production cross section of B_c mesons is $\sim 10^3$ less than that of the $b\bar{b}$. In `PYTHIA`, consequently, the B_c production rate would be $\sim 10^{-6}$ in a minimum bias generation. It is then necessary to enhance the event generation efficiency with a dedicated generator. The documented package `BCVEGPY`, interfaced with `PYTHIA`, is available and downloadable at the authors' web site [11][12]. In `BCVEGPY`, a matrix element generator, the generation can be stopped at parton level. The simulated quantities, such as the B_c p_T spectrum, will be compared with the data to check the agreement between the `BCVEGPY` generator and the data. An output file, containing the B_c , b and c is written in the Les Houches Accord (LHE), which can be easily interfaced with different hadronizing models.

MC samples have been produced for $B_c^+ \rightarrow J/\psi\pi^+$, $B_c^+ \rightarrow J/\psi\pi^+\pi^+\pi^-$ and $B_c^+ \rightarrow J/\psi a_1(a_1 \rightarrow \rho(770)\pi^+)$. In order to optimize the sample production, acceptance cuts have been applied at the generator level: a minimum p_T cut, $p_T(B_c) > 9$ GeV/c, and a rapidity interval $|y| < 2.5$ have been required. A $B^\pm \rightarrow J/\psi K^\pm$ sample has been also made available by the central generator team. No acceptance cuts have been required at the generation level for the B^+ sample.

The relevant generated variables have been checked: p_T and y distributions are shown in Fig.7 and 8 for the B_c and the B^+ respectively.

In order to match the acceptance region of the reconstructed events ($p_T(B_c) > 15$ GeV/c and $|y| < 1.6$, see Sec.7), transverse momentum and rapidity cuts will be implemented at a later stage of the analysis for both B^+ and B_c samples.

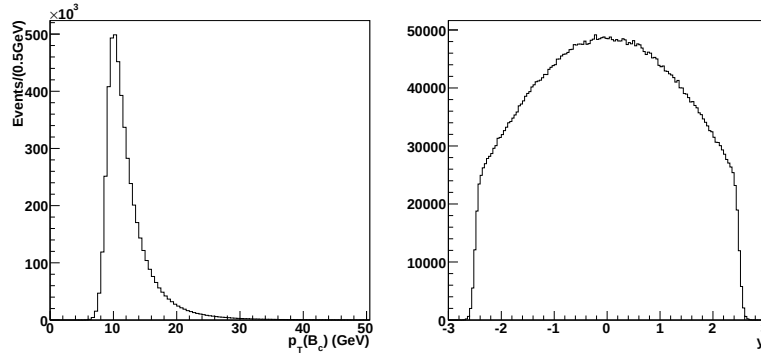


Figure 7: p_T (left) and rapidity (right) distributions of the generated B_c particle.

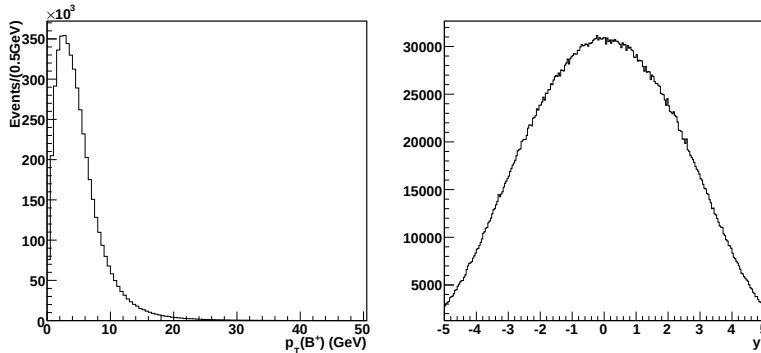


Figure 8: p_T (left) and rapidity (right) distribution of the generated $B^\pm$ particle.

4.2.2 PU reweighting

The entire machinery for pileup reweighting is documented on the web page https://twiki.cern.ch/twiki/bin/viewauth/CMS/Pileup_2011_Reweighting. Instructions for the Fall11 samples have been followed. Data have been divided in three main periods according to the beam instantaneous luminosity, as listed in Table 2.

Table 2: HLT paths and Luminosity for the three different periods as evaluated via the pixel-LumiCalc.py script

HLT path	Inst. Lumi ($\text{cm}^{-2}\text{s}^{-1}$)	Integr. Lumi (pb^{-1}) in data
HLT_Dimuon7_Jpsi_Displaced	$1 - 1.4 \times 10^{33}$	983
HLT_DoubleMu3p5_Jpsi_Displaced	2×10^{33}	923
HLT_DoubleMu4_Jpsi_Displaced	$3 - 5 \times 10^{33}$	3139

The data Primary Vertex distributions for the three instantaneous luminosity periods are compared to MC with and without PU reweighting in Fig. 9 for the B^+ and B_c samples respectively.

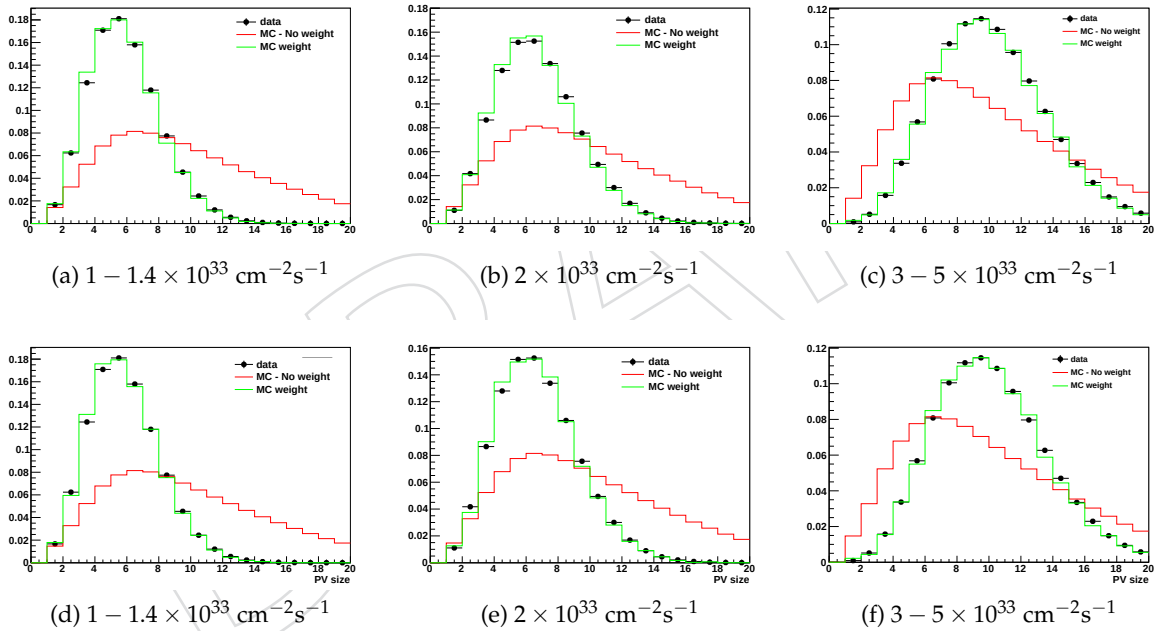


Figure 9: Primary Vertex distributions for data compared to B^+ (top) and B_c (bottom) simulation with and without PU reweighting.

5 Event Selection

5.1 Trigger and preselection

The analysis is driven by the J/ψ meson reconstruction. B_c and B^+ candidates are then formed combining it with one or three tracks.

Candidate J/ψ mesons are reconstructed from opposite-charge muons selected with the standard `softMuon` selection for the 2011 data

(see <https://twiki.cern.ch/twiki/bin/view/CMSPublic/SWGuideMuonId>).

The J/ψ candidates have to match the trigger criteria. Double muon triggers are used: the dimuon vertex is required to be displaced from the beamspot to reject the prompt J/ψ events.

Various versions of the J/ψ triggers were used during the data taking to follow the luminosity increase, i.e., tighter thresholds were applied to maintain an acceptable rate. This analysis uses

the data where `HLT_Dimuon7_Jpsi_displaced_v1(v3)`, `HLT_DoubleMu3p5_Jpsi_displaced_v2`, `HLT_DoubleMu4_Jpsi_displaced_v1(v4,v5)` were not prescaled.

All triggers apply the following cuts:

- $\cos\alpha > 0.9$, where α is the pointing angle, in the transverse plane, between the dimuon momentum and the separation between the beamspot and the dimuon vertex
- $L_{xy}/\sigma_{xy} > 3$, where L_{xy} is the transverse detachment between the beamspot and the dimuon vertex and σ_{xy} is the relative uncertainty
- $p_T^{\mu\mu} > 6.9 \text{ GeV}/c$
- Dimuon vertex probability $> 0.5 \%$
- Distance of closest approach between the two muons $< 0.5 \text{ cm}$

In addition, for the `HLT_DoubleMu3p5_Jpsi_displaced_v2`, the following tighter cuts are included:

- $p_T^\mu > 3.5 \text{ GeV}/c$
- $|\eta^\mu| < 2.2$
- Dimuon vertex probability $> 10 \%$

and for `HLT_DoubleMu4_Jpsi_displaced`

- $p_T^\mu > 4 \text{ GeV}/c$
- Dimuon vertex probability $> 15 \%$

The `HLT_DoubleMu4_Jpsi_displaced` is also seeded by a more selective muon L1 trigger `L1_DoubleMu0_HighQ` replacing the `L1_DoubleMu0`.

The muons from J/ψ candidates reconstructed in the offline analysis are required to be consistent with those which fired the trigger in a $\Delta R = \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2}$ cone of 0.5.

The behavior of the trigger turn-on curves on the main variables (p_T , L_{xy}/σ_{xy} , $\cos\alpha$) is studied both in data and MC to verify that data are well reproduced by the simulation. Plots are collected in Appendix 12.2. Trigger requirements have been tightened offline to $p_T(J/\psi) \geq 7.1 \text{ GeV}/c$ and $L_{xy}/\sigma_{xy} \geq 5$.

One or more tracks are then associated to the J/ψ meson to form a $B_c^\pm, B^+$ candidate. A check is performed to exclude tracks also belonging to the Muon Collection. Some quality cuts are applied to the tracks, namely

- number of tracker hits > 6
- number of pixel hits ≥ 2
- $\chi^2/ndof < 3$
- $|\eta| < 2.4$
- $p_T^{\pi/K} > 0.9 \text{ GeV}/c$

The three-dimensional impact parameter significance between each pion and the J/ψ vertex is set ≤ 6 to reduce combinatoric background and effect of pileup, and limit the vertexing computing time. The value $IP \leq 6\sigma_{IP}$ is inferred from MC simulation of the signal (see Fig.10).

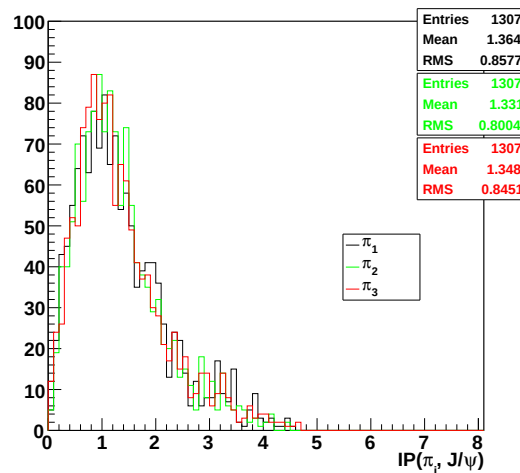


Figure 10: Significance of the 3D impact parameter between each pion and the J/ψ vertex as predicted by the simulation. The plot is made at the level of the pre-selection (after trigger cuts and the requirement of a valid five-track vertex), for those events lying in the B_c mass range.

The $B_c^\pm$ samples are generated using PYTHIA (version 6.424 [13] , with the underlying event simulated with the Z2 tune [14], unstable particles decayed via EVTGEN [15], and the detector response simulated with GEANT4 [16].

The decay vertex is reconstructed using a kinematic vertex fit [17, 18], which constrains the invariant mass of the two muons to the J/ψ nominal mass; the three (five) track vertex fit confidence level is required to be greater than 0.001. After the vertex fit, the track parameters are re-estimated at the fitted vertex, effectively using this vertex as a constraint.

The ΔR between the J/ψ and the pion track is required to be < 2.5 ; this value is obtained from the $B_c^\pm \rightarrow J/\psi \pi^\pm$ simulation (show in the left plot of Fig. 11). For the $B_c^\pm \rightarrow J/\psi \pi^\pm \pi^\pm \pi^\mp$ mode, a ΔR cut is applied on each of the three pions in the event. In particular, $\Delta R < 1$ is required for the highest p_T track (referred to as π_1), while $\Delta R < 1.6$ is required for the other two pions (π_2 and π_3). The corresponding MC distributions are shown in the right plot of Fig. 11.

In case of multiple candidates, the one with highest p_T is retained. Additional topological selections are required to improve the S/B ratio.

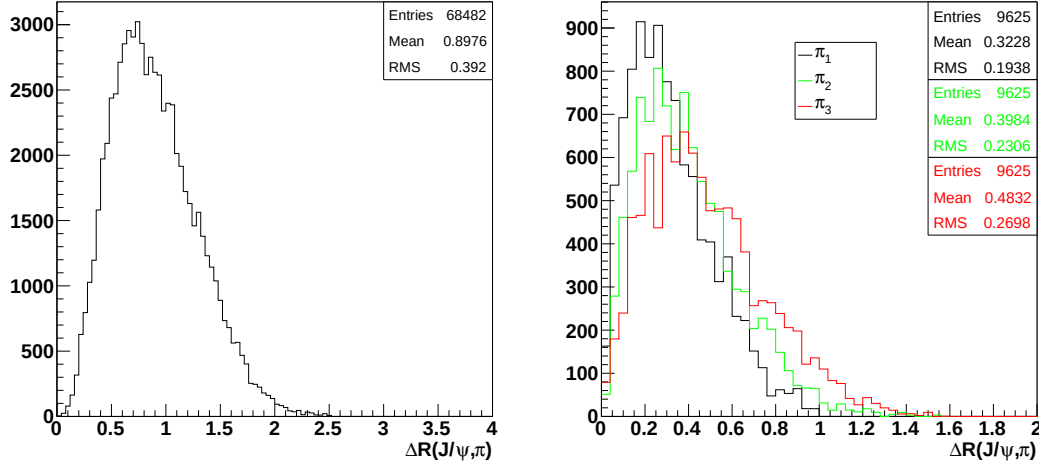


Figure 11: $\Delta R(J/\psi, \pi)$ from $B_c^+ \rightarrow J/\psi \pi^+$ (left) and $B_c^+ \rightarrow J/\psi \pi^+ \pi^+ \pi^-$ (right) MC simulation

6 The $B_c^\pm \rightarrow J/\psi \pi^\pm \pi^\pm \pi^\mp$ signal

The $B_c^\pm \rightarrow J/\psi \pi^\pm \pi^\pm \pi^\mp$ is the third and one of the more recent observed modes. The first estimate of $\frac{Br(B_c^\pm \rightarrow J/\psi \pi^\pm \pi^\pm \pi^\mp)}{Br(B_c^\pm \rightarrow J/\psi \pi^\pm)}$ was published by LHCb [4].

The search for this decay in CMS is performed just adding three-tracks, with total charge of ± 1 , to the J/ψ candidate.

The selection cuts have been optimized maximizing the $S/\sqrt{(S+B)}$ figure of merit; S is obtained from a fit to the reconstructed events of the MC sample. B is the background in the signal region inferred from the data sideband chosen in the range $[(m_{B_c} - 8\sigma_{B_c}), (m_{B_c} - 5\sigma_{B_c})]$ or $[(m_{B_c} + 5\sigma_{B_c}), (m_{B_c} + 8\sigma_{B_c})]$, where m_{B_c} is the PDG mass for the B_c meson and σ_{B_c} is obtained from the simulation.

The obtained cut set is

- $p_T(B_c) > 15 \text{ GeV}/c$
- $|y(B_c)| < 1.6$
- B_c Vertex CL $> 20 \%$
- $\cos\theta > 0.99$, where $\cos\theta = L \cdot p_{B_c} / (|L| |p_{B_c}|)$, evaluated wrt the beamspot, in the xy plane
- $p_T(\pi_1) > 2.5 \text{ GeV}/c$, $p_T(\pi_2) > 1.7 \text{ GeV}/c$, $p_T(\pi_3) > 0.9 \text{ GeV}/c$, where π_1 , π_2 and π_3 are sorted from the highest to the lowest p_T
- $\Delta R(J/\psi, \pi_5) < 0.5$ where π_5 is the vector sum of the three-pion quadrimomenta

The signal extracted from data is shown in Fig. 12. The yield is 92 ± 27 events; the signal width is $20.6 \pm 0.6 \text{ MeV}$ and is in agreement with the MC prediction ($\sim 21 \text{ MeV}$). The fit is performed through an extended unbinned maximum likelihood estimator; the signal is parametrized as a gaussian and the background as a Chebychev second order polynomial.

The evolution of the yield and of $S/\sqrt{S+B}$ as a function of the cuts has been studied to check the stability of the selection for this channel. Plots are collected in the Appendix, Sec. 12.6.1.

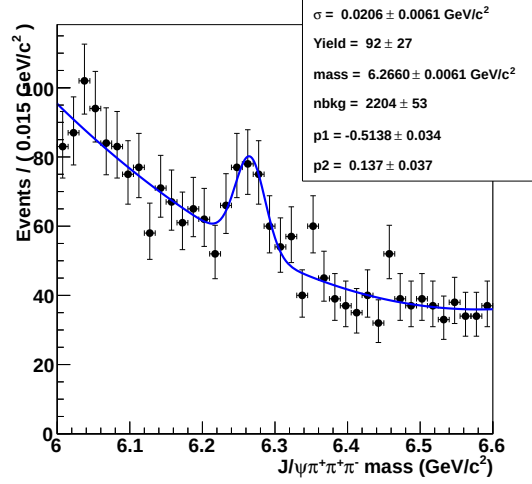


Figure 12: $J/\psi\pi^+\pi^-\pi^+\pi^-$ invariant mass distribution. The result of the fit is superimposed.

7 The $B_c^\pm \rightarrow J/\psi\pi^\pm$ and $B^\pm \rightarrow J/\psi K^\pm$ signals

The same optimization procedure has been applied to the $B_c^\pm \rightarrow J/\psi\pi^\pm$ channel; while the initial state (B_c) phase space cuts are fixed to those of the $B_c^\pm \rightarrow J/\psi\pi^+\pi^-\pi^+\pi^-$ more (i.e. $p_T(B_c) > 15 \text{ GeV}/c$ and $|y(B_c)| < 1.6$), the other cuts have been varied to optimize $S/\sqrt{(S+B)}$ for the $B_c^\pm \rightarrow J/\psi\pi^\pm$ signal. The procedure points at the following cut set:

- B_c Vertex CL $> 6 \%$
- $\cos\theta > 0.9$
- $p_T(\pi) > 2.7 \text{ GeV}/c$
- $\Delta R(J/\psi, \pi) < 1$

The corresponding $B_c^\pm \rightarrow J/\psi\pi^\pm$ signal is shown in Fig. 13. With the same cut set as for the $B_c^\pm \rightarrow J/\psi\pi^\pm$, the $B^\pm \rightarrow J/\psi K^\pm$ signal in Fig. 14 is obtained.

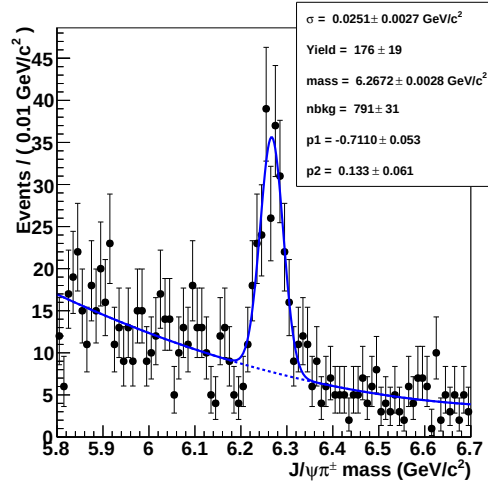


Figure 13: $J/\psi\pi^\pm$ invariant mass distribution. The result of the fit is superimposed; the lines represent the signal-plus-background fit (solid) and the background-only component (dashed).

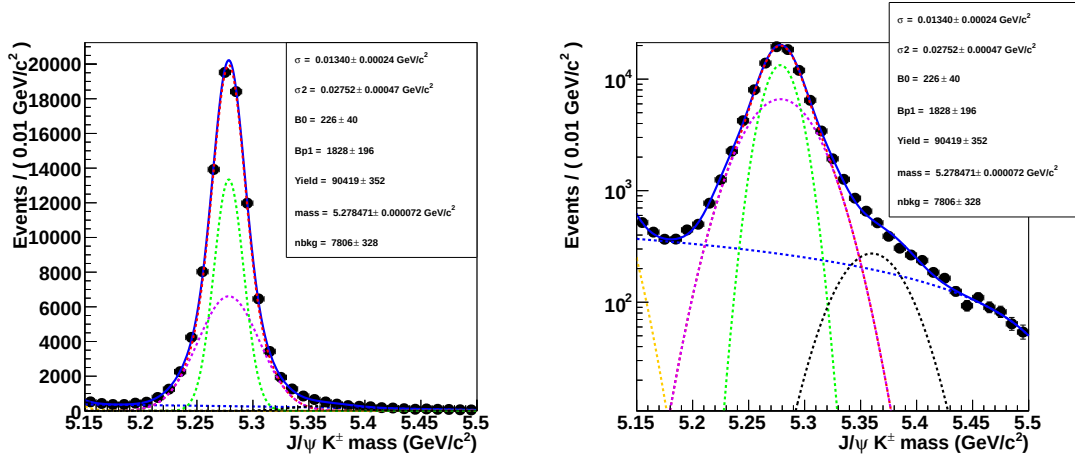


Figure 14: $J/\psi K^\pm$ invariant mass distribution with linear (left) and semi-logarithmic (right) scale. The result of the fit is superimposed with a blue solid line. The signal is parametrized with two gaussian, shown with the green and violet dashed lines. The black and yellow dashed curves represent the $B^\pm \rightarrow J/\psi \pi^\pm$ and B^0 contributions respectively.

The $J/\psi \pi^\pm$ invariant-mass distribution is fit through an unbinned likelihood fit. The resulting curve is superimposed to the $J/\psi \pi^\pm$ invariant-mass distribution in Fig.13. The signal is parametrized with a gaussian while the background is described through a second order Chebychev polynomial. The $B_c^\pm$ signal yield is 176 ± 19 and its mass is $6.267 \pm 0.003 \text{ GeV}/c^2$. The width, for this set of kinematic variables (y and p_T), is $0.025 \pm 0.003 \text{ GeV}/c^2$.

The $B^\pm$ mass invariant distribution needs to be fit with two gaussians describing the signal and a second order Chebychev polynomial for the background. The structures due to partially reconstructed decays from other B-hadrons have been studied on the centrally produced MC samples:

```
/B0ToPsiMuMu_2MuPEtaFilter.Tight_7TeV-pythia6-evtgen/Fall111-HLTBPh2011-START42.V14B-v2/GEN-SIM-RECO
/BpToPsiMuMu_2MuPEtaFilter.Tight_7TeV-pythia6-evtgen/Fall111-HLTBPh2011-START42.V14B-v2/GEN-SIM-RECO
```

The $J/\psi K^\pm$ invariant mass distribution resulting from analyzing the simulated samples mentioned above is shown in Fig. 15 for events firing the HLT_Dimuon7_Jpsi_displaced trigger. It has been fit with a double-gaussian for the $J/\psi K^\pm$ signal, and two additional gaussians, one on the left side and one on the right side of the signal, accounting for the partially reconstructed B^0 and $B^\pm \rightarrow J/\psi \pi^\pm$ contributions respectively.

The same functions have been then used to fit the $B^\pm$ mass spectrum in data in the range from $5.15 \text{ GeV}/c^2$ to $5.5 \text{ GeV}/c^2$. The background is parametrized with a second order Chebychev polynomial. The yield of the $B^\pm \rightarrow J/\psi K^\pm$ is 90419 ± 352 events and its mass is $5.27847 \pm 0.00007 \text{ GeV}/c^2$. The fit on $B^\pm \rightarrow J/\psi K^\pm$ data returns a number of events due to partially reconstructed decays from other B-hadrons of 1828 ± 196 for $B^\pm$ contribution and 226 ± 40 for the B^0 one.

For the discussion on the possible systematic error due to the choice of the fit function see Section 9.1.

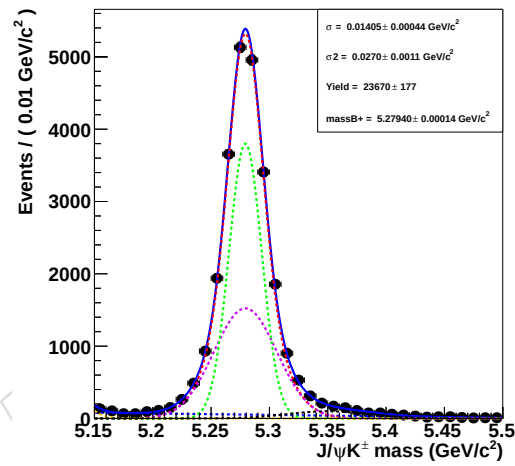


Figure 15: Fit to the $B^\pm \rightarrow J/\psi X$ and $B^0 \rightarrow J/\psi X$ MC samples. The $B^\pm$ signal is parametrized with two gaussian, shown with the green and violet dashed lines. The black and yellow dashed curves represent the $B^\pm \rightarrow J/\psi\pi^\pm$ and B^0 contributions respectively.

8 DATA and MC comparison

Before proceeding to the evaluation of the BRs, the reconstructed MC variables have been compared to those of data. They are shown in Fig. 16, where the B_c transverse momentum and rapidity and the pion p_T for the background subtracted data (error bars) are compared to those of the $B_c^+ \rightarrow J/\psi\pi^+$ MC (red).

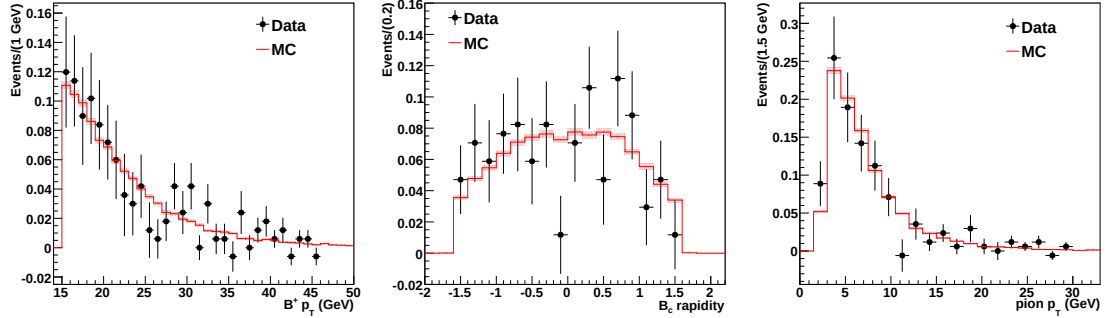


Figure 16: Background subtracted data distributions (black) of B_c p_T , y and pion p_T are compared to MC (red) for the $B_c \rightarrow J/\psi\pi$ channel. The red hashed region represents the MC statistical error.

A Kolmogorov test is performed to evaluate the compatibility of the distributions. The returned values for the probability are 0.98, 0.88 and 0.97 for the B_c p_T , y , and the pion p_T respectively.

The comparison for the $B^\pm$ signal is shown in Fig. 17. The higher statistics allowed to perform the comparison in each single luminosity/trigger period also (plots are collected in App.12.3).

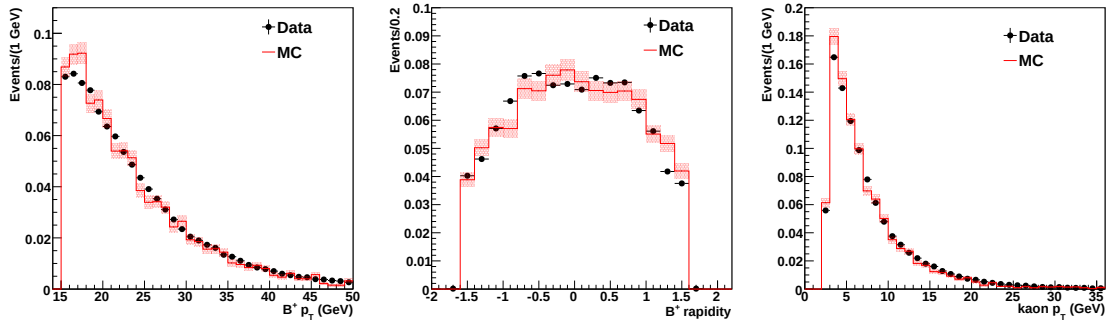


Figure 17: Background subtracted data distributions (black) of B^+ p_T , y and kaon p_T are compared to MC (red). The red hashed region represents the MC statistical error.

9 The $R_{c/u}$ ratio

The ratio

$$R_{c/u} = \frac{\sigma(B_c^\pm) \times Br(B_c^\pm \rightarrow J/\psi\pi^\pm)}{\sigma(B^\pm) \times Br(B^\pm \rightarrow J/\psi K^\pm)} \quad (1)$$

can be evaluated through the relation

$$\frac{N(B_c^\pm \rightarrow J/\psi\pi^\pm)}{N(B^\pm \rightarrow J/\psi K^\pm)} = \frac{\sigma(B_c^\pm) \times Br(B_c^\pm \rightarrow J/\psi\pi^\pm) \times \epsilon_{B_c^\pm}}{\sigma(B^\pm) \times Br(B^\pm \rightarrow J/\psi K^\pm) \times \epsilon_{B^\pm}} \quad (2)$$

where N is the number of signal events and $\epsilon_{B_c^\pm}$ and $\epsilon_{B^\pm}$ are the overall analysis efficiencies for the $B_c^\pm \rightarrow J/\psi\pi^\pm$ and $B^\pm \rightarrow J/\psi K^\pm$ signals respectively. They include geometrical acceptance, reconstruction, selection and trigger effects in the kinematic region $p_T > 15$ GeV/c and $|y| < 1.6$.

The efficiencies are evaluated as a function of the B_c ($B^\pm$) transverse momentum on the corresponding MC samples. They have been computed in 24 transverse momentum bins in the case of the B_c and 19 bins in the $B^\pm$ one; the bin size is determined by the MC available statistics (the binning definition is reported in Tab.3). The efficiency ϵ in the i th bin is evaluated as $\epsilon_i = \frac{N_i^{reco}}{N_i^{GEN}}$

where N_i^{reco} is the number of reconstructed events in the given bin and N_i^{GEN} is the number of generated B_c ($B^\pm$) events in the same bin. The efficiency behavior as a function of p_T is shown in Fig. 18 for the B_c and $B^\pm$ signals respectively.

Data are then corrected event-by-event according to the B_c ($B^\pm$) meson transverse momentum and the corresponding MC efficiency.

The efficiency corrected mass plots for $J/\psi K^\pm$ and $J/\psi\pi^\pm$ are shown in Fig. 19. The efficiency corrected yields Y_{B_c} and Y_B , obtained by fitting the distributions of Fig. 19 through an extended unbinned ML fit, are 6303 and 1320256 respectively.

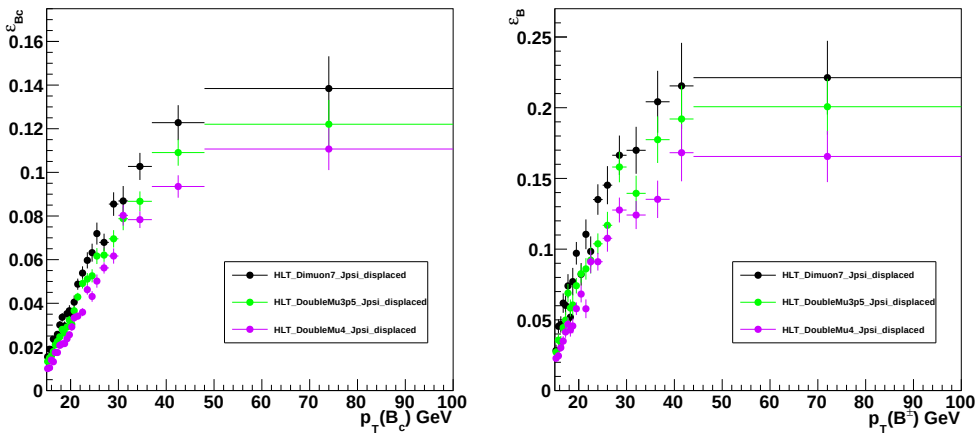


Figure 18: B_c (left) and B^+ (right) efficiencies as a function of transverse momentum.

The statistical errors on Y_{B_c} and Y_B are computed from the fit of Fig. 13 and 14, i.e. the relative error is inferred from the unweighted mass distribution fits: $Y_{B_c} = 6303 \pm 680$ and $Y_B = 1320256 \pm 5140$. The uncertainty on Y_{B_c} and Y_B due to the efficiency statistical uncertainty is accounted for in the systematic error and is evaluated through Toy MonteCarlo studies (see Sec.9.1.3).

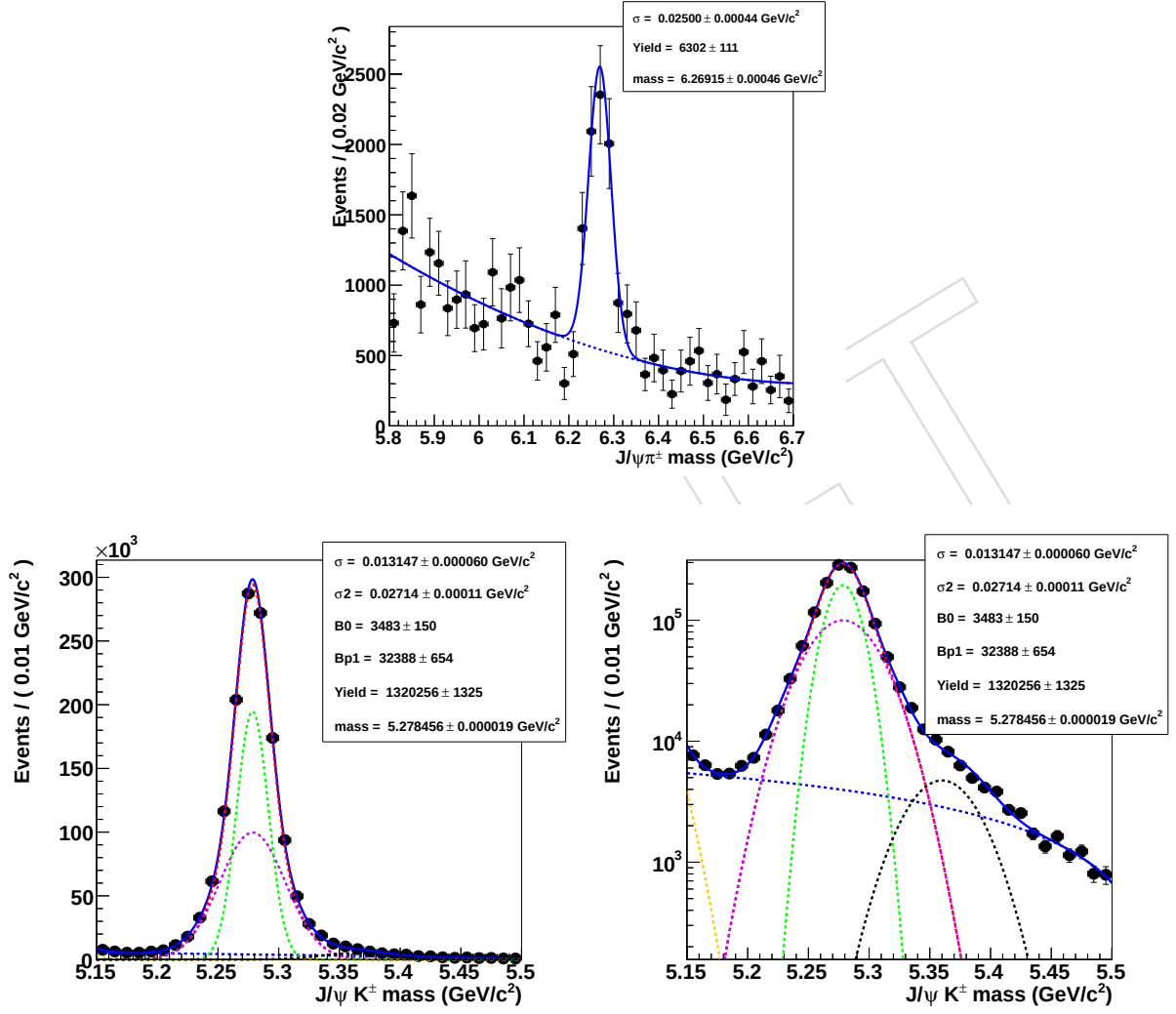


Figure 19: $J/\psi\pi^\pm$ (top) and $J/\psi K^\pm$ (bottom) efficiency corrected mass distributions. The bottom right plot shows the $J/\psi K^\pm$ efficiency corrected mass distributions using a semi-log scale.

Bin	Bin edges	
	B_c sample	B^+ sample
1	15 - 15.4	15 - 15.5
2	15.4 - 15.8	15.5 - 16
3	15.8 - 16.2	16 - 16.5
4	16.2 - 16.6	16.5 - 17
5	16.6 - 17	17 - 17.5
6	17 - 17.5	17.5 - 18
7	17.5 - 18	18 - 18.5
8	18 - 18.5	18.5 - 19
9	18.5 - 19	19 - 20
10	19 - 19.5	20 - 21
11	19.5 - 20	21 - 22
12	20 - 20.5	22 - 23
13	20.5 - 21	23 - 25
14	21 - 22	25 - 27
15	22 - 23	27 - 30
16	23 - 24	30 - 34
17	24 - 25	34 - 39
18	25 - 26	39 - 44
19	26 - 28	44 - 100
20	28 - 30	
21	30 - 32	
22	32 - 37	
23	37 - 48	
24	48 - 100	

Table 3: Binning definitions for the B_c and B^+ efficiency evaluation.

337 The ratio is evaluated by dividing Y_{B_c} and Y_B and results to be

$$R_{c/u} = \frac{\sigma(B_c^\pm) \times Br(B_c^\pm \rightarrow J/\psi \pi^\pm)}{\sigma(B^\pm) \times Br(B^\pm \rightarrow J/\psi K^\pm)} = (0.48 \pm 0.05) \cdot 10^{-2} \quad (3)$$

9.1 Systematic error evaluation on $R_{c/u}$

The systematic errors are estimated through a combination of different contributions, summarized in Table 4. Each contribution is discussed separately in the next paragraphs. Tables 5 and 6 summarize individual contributions for Y_B and Y_{B_c} .

Syst. error	Percentage	Abs. value (10^{-2})
Split sample	0	0
Fit variant	5.3	0.03
MC finite size	2.1	0.01
Efficiency binning	3.1	0.01
Total uncertainty	6.5	0.03
B_c lifetime	10.4	0.05

Table 4: Systematic uncertainty contributions for the measurement of $R_{c/u}$.

Syst. error	Percentage
Split sample	0
Fit variant	2.2
MC finite size	1.8
Efficiency binning	2.4

Table 5: Systematic uncertainties contributing to the Y_B evaluation.

Syst. error	Percentage
Split sample	0
Fit variant	4.8
MC finite size	1.1
Efficiency binning	1.9
B_c lifetime	10.4

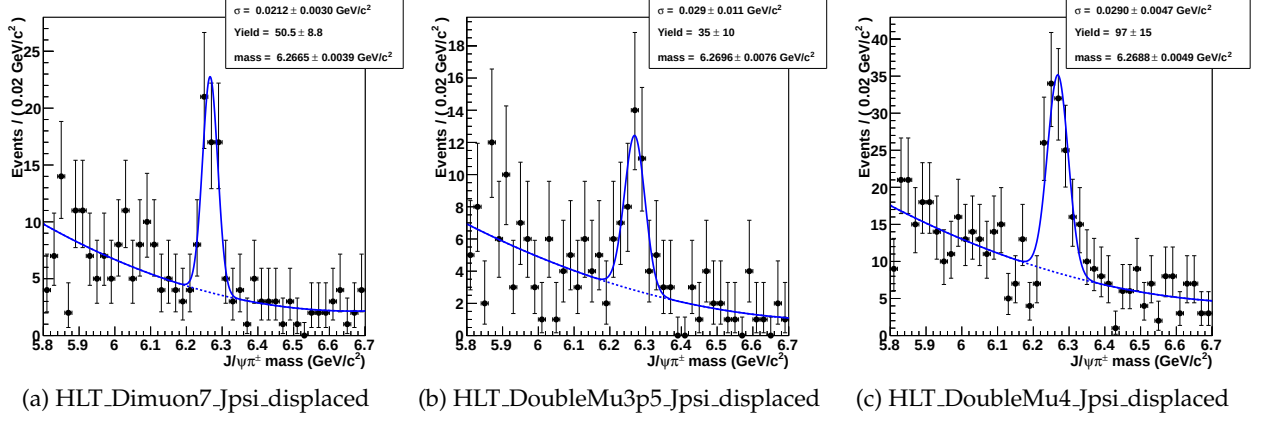
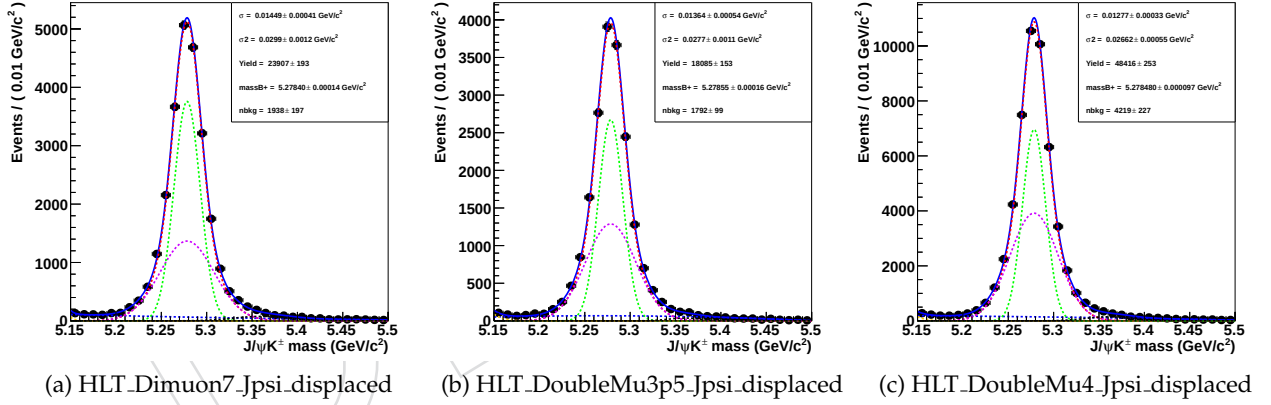
Table 6: Systematic uncertainties contributing to the Y_{B_c} evaluation.

9.1.1 Split samples

The data have been split in N sub-samples according to the different running periods, number of Pile-Up vertices and main variables of the analysis to check the consistency of the results and possible systematic effects. The study of each contribution is reported below and the systematics evaluation is obtained following the method in Appendix 12.4. No systematic uncertainty is measured, since $\chi^2/(N-1)$ is always less than 1 for each performed split sample.

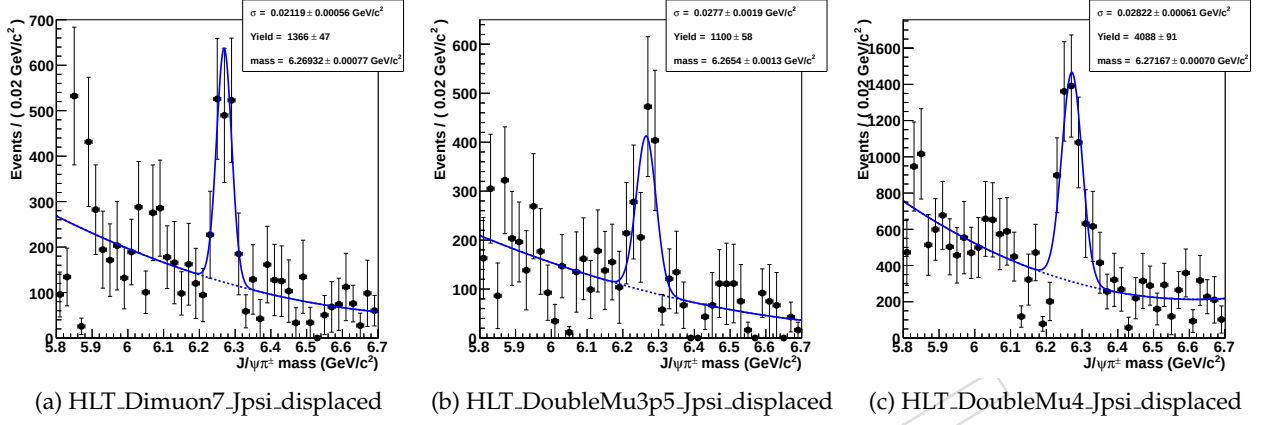
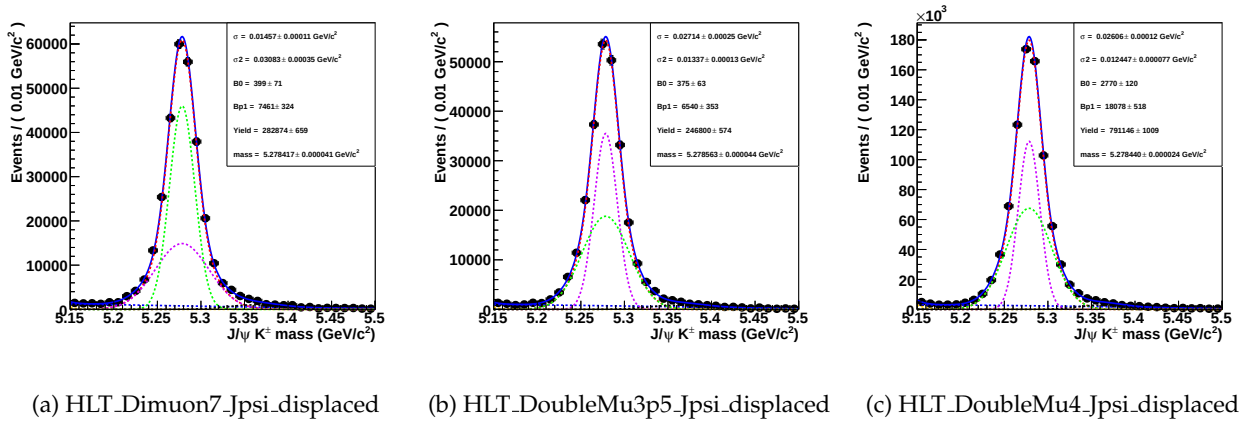
9.1.1.1 Split according to the trigger periods

The ratio is evaluated in each of the three trigger periods. The B_c and $B^\pm$ signals extracted in each of the three subsamples are collected in Fig. 20 and 21; the fits to the corresponding efficiency corrected data are shown in Fig. 22 and 23. Results of the ratio measurements are summarized in Table 7.

Figure 20: B_c signals in the three trigger periodsFigure 21: $B^\pm$ signals in the three trigger periods.

HLT path	Y_{B_c}	Y_B	$R_{c/u} (10^{-2})$
Dimuon7_Jpsi_Displaced	1366 ± 238	282874 ± 2284	0.48 ± 0.08
DoubleMu3p5_Jpsi_Displaced	1100 ± 314	246800 ± 2088	0.45 ± 0.13
DoubleMu4_Jpsi_Displaced	4088 ± 632	791146 ± 4134	0.52 ± 0.08

Table 7: $R_{c/u}$ measurement in each trigger period.

Figure 22: B_c efficiency corrected yields in the three trigger periods.Figure 23: $B^\pm$ efficiency corrected yields in the three trigger periods.

9.1.1.2 Split in high and low PU events

In order to investigate the possible dependence of the ratio measurement on the Pile-Up, the efficiency behavior as a function of the number of Primary Vertices has been studied. The MC samples have been divided in a low Pile-Up region, consisting of $1 \leq PV \leq 6$, and in a High Pile-Up region consisting of $PV \geq 7$. The efficiencies for the two regions are shown in Fig. 24 and 25 for the $B^\pm \rightarrow J/\psi K^\pm$ and $B_c^\pm \rightarrow J/\psi \pi^\pm$ signals respectively.

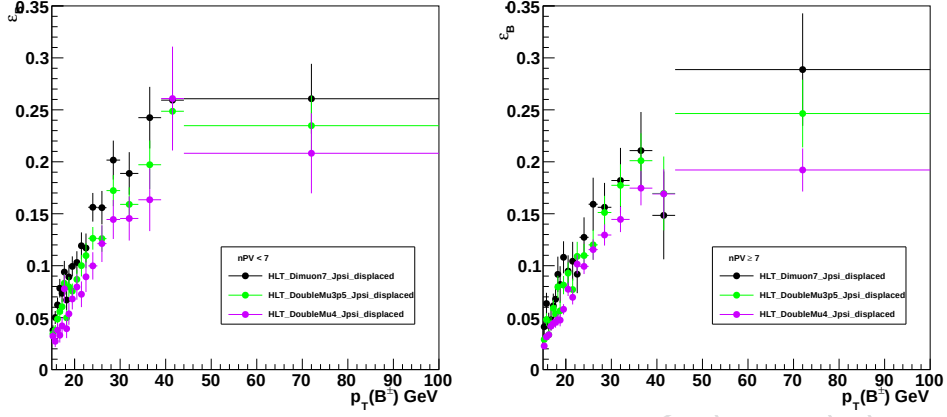


Figure 24: Efficiency for Low (left) and High (right) PU for $B^\pm \rightarrow J/\psi K^\pm$.

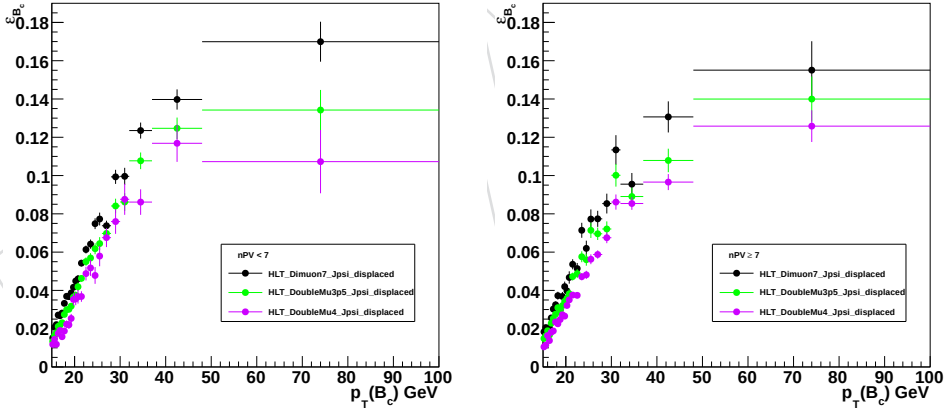


Figure 25: Efficiency for Low and High PU for $B_c^\pm \rightarrow J/\psi \pi^\pm$.

Subsequently, data have been divided according to the number of Primary Vertices in the event. The corresponding signals are shown in Fig. 26 and 27. Fig. 28 collects the efficiency corrected yields for the two PU regions for both channels. The ratio has been evaluated in the two regions as summarized in Table 8.

	Y_{B_c}	Y_B	$R_{c/u} (10^{-2})$
$nPV < 7$	2127 ± 325	424253 ± 1277	0.50 ± 0.08
$nPV \geq 7$	4043 ± 501	751218 ± 3685	0.54 ± 0.07

Table 8: $R_{c/u}$ measurement in the two PU regions.

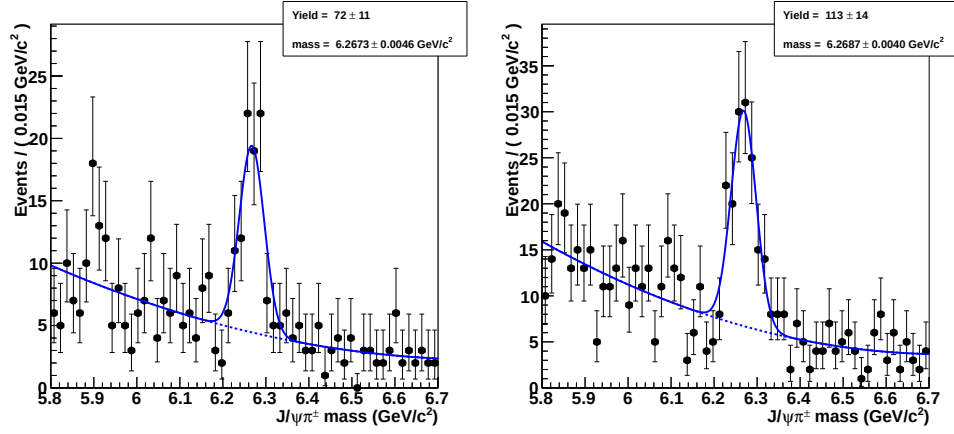


Figure 26: $B_c^\pm$ signals in the low (left) and high (right) PU region.

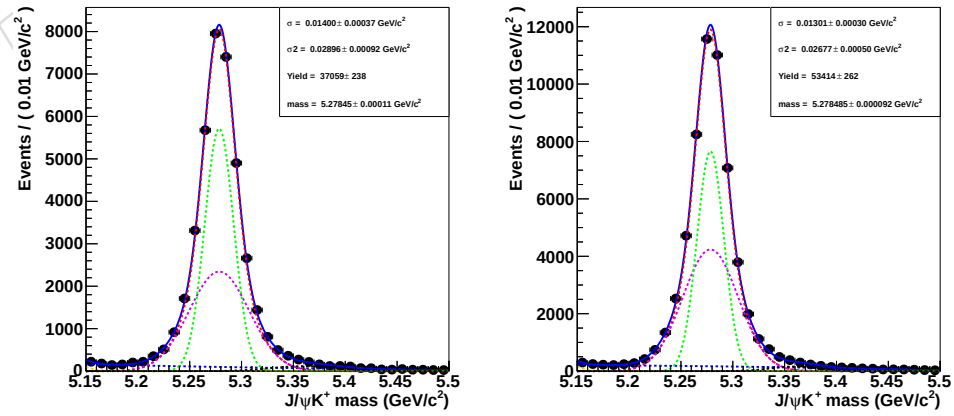


Figure 27: $B^\pm$ signals in the low (left) and high (right) PU region.

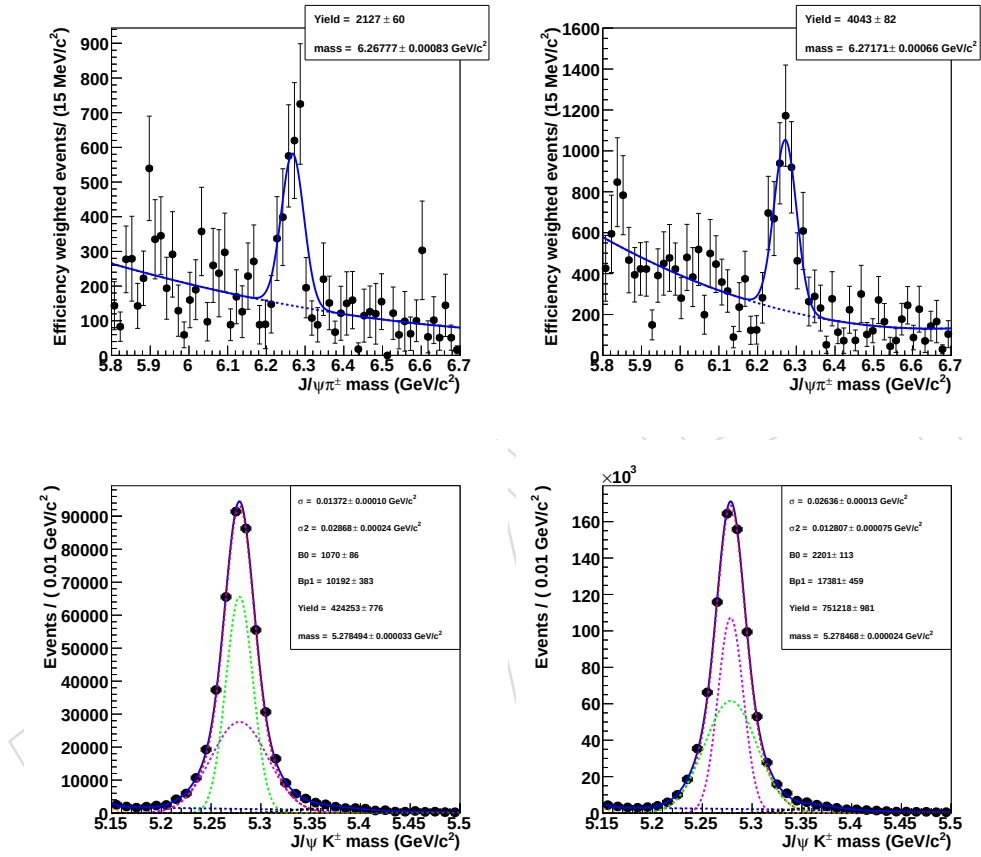


Figure 28: Efficiency corrected yields for $B_c^\pm$ (top row) and $B^\pm$ (bottom row) in the low (left) and high (right) PU region.

9.1.1.3 Split in L_{xy}/σ_{xy}

The samples have been split in two sub-samples according to the L_{xy}/σ_{xy} , namely ($5 \leq L_{xy}/\sigma_{xy} \leq 10$ and $L_{xy}/\sigma_{xy} > 10$). The cut value is chosen in order to split the sample into two sub-samples having comparable statistics. Data are shown in Fig. 29 and 30 for the B_c and $B^\pm$ signals respectively.

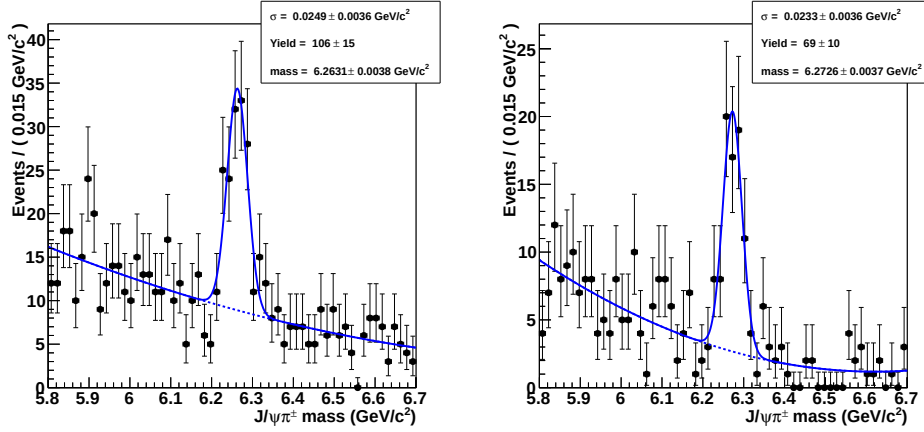


Figure 29: $B_c^\pm$ signals in the low (left) and high (right) L_{xy}/σ_{xy} region.

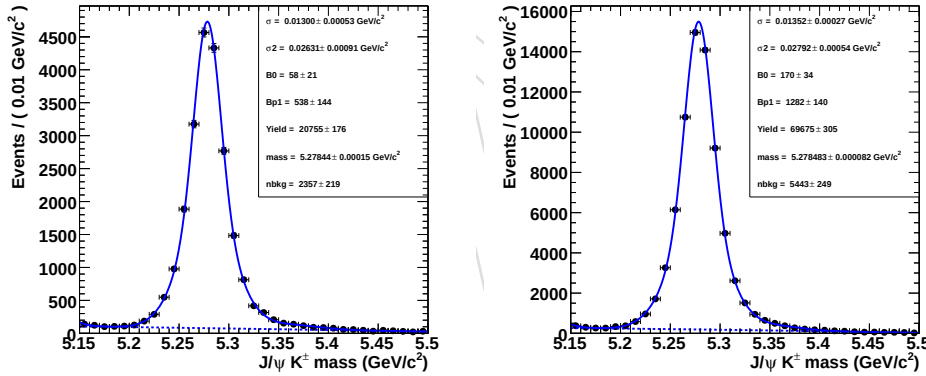


Figure 30: $B^\pm$ signals in the low (left) and high (right) L_{xy}/σ_{xy} region.

Fig. 31 collects the efficiency corrected yields for the two regions for both channels. The ratio evaluation in the two samples is summarized in Table 9.

	Y_{B_c}	Y_B	$R_{c/u} (10^{-2})$
$5 \leq L_{xy}/\sigma_{xy} \leq 10$	5487 ± 776	1223514 ± 10375	0.45 ± 0.06
$L_{xy}/\sigma_{xy} > 10$	6166 ± 894	1159447 ± 5075	0.53 ± 0.08

Table 9: $R_{c/u}$ measurement in the two L_{xy}/σ_{xy} regions.

9.1.2 Fit variant systematics

The ratio measurement in Eq.2 can be also affected by a systematic uncertainty due to the fitting procedure of the two decay channel yields. The uncertainty is accounted for by varying the fit function, the background model and the ranges of the fitted histograms.

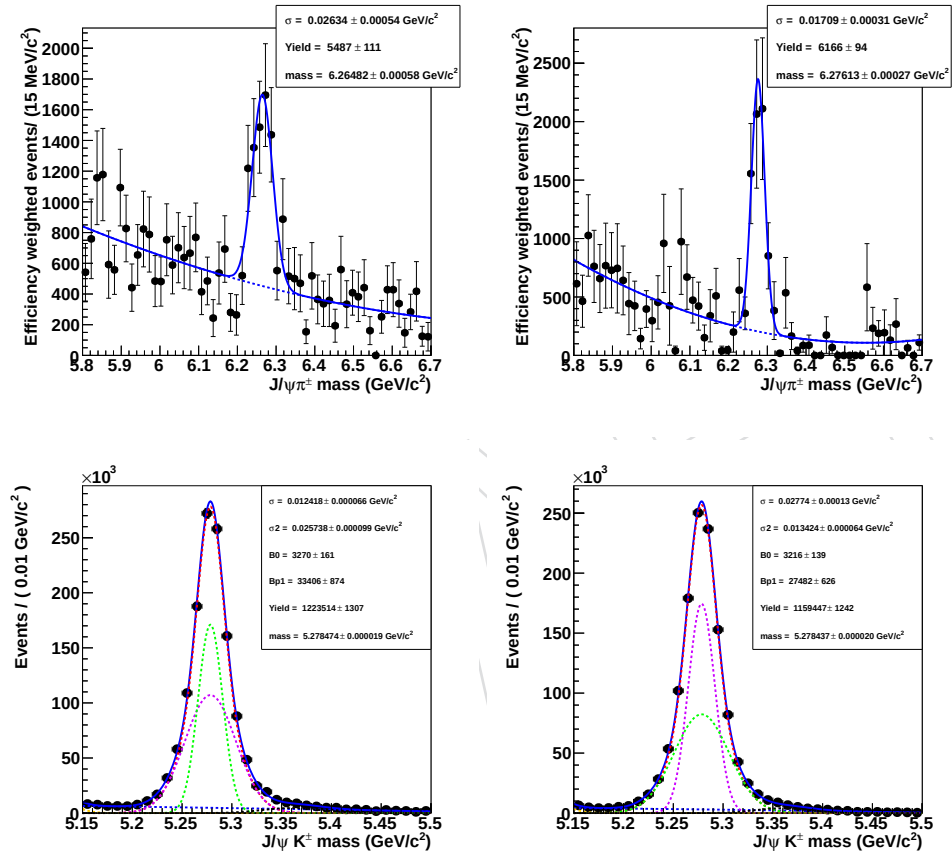


Figure 31: Efficiency corrected yields for $B_c^\pm$ (top row) and $B^\pm$ (bottom row) in the low (left) and high (right) L_{xy}/σ_{xy} region.

Following the approach in Appendix 12.5, the efficiency corrected distributions have been fit in different mass ranges and with different parametrizations. Results for Y_{B_c} are summarized in Table 10; all fit parameters are free to float, except for the widths of the two gaussians in fit (g) and their relative amount, which are fixed to MC values. The corresponding plots are collected in App. 12.5.1.1. The total uncertainty on Y_{B_c} due to the fit parametrization is 4.8%.

	Signal PDF	Bkg PDF	Range (GeV/c ²)	Y_{B_c}
a	Gaussian	1 st order Chebychev polynomial	5.8-6.7	5773 ± 607
b	Gaussian	1 st order Chebychev polynomial	6.05-6.55	6272 ± 658
c	Gaussian	2 st order Chebychev polynomial	6.05-6.55	6419 ± 742
d	Gaussian	2 st order Chebychev polynomial	5.6-6.7	6361 ± 640
e	Gaussian	3 st order Chebychev polynomial	5.8-6.7	6328 ± 673
f	Gaussian	Exponential	5.8-6.7	6209 ± 623
g	Double Gaussian	2 st order Chebychev polynomial	5.8-6.7	6790 ± 652

Table 10: Fit Variants for Y_{B_c}

As far as the $B^\pm$ is concerned, different choices for the parametrization of the signal and of partially reconstructed $B^\pm$ decays have been implemented. The fit range has also been varied to include/exclude such reflections. Details are collected in Table 11 and corresponding plots can be found in App. 12.5.1.2. The resulting systematic uncertainty on Y_B is 2.2%.

	Signal PDF	Bkg PDF	B^0 PDF	$B^+ \rightarrow J/\psi\pi^+$ PDF	Range (GeV/c ²)	Y_B
a	Crystal Ball	2 nd ord. Ch. pol	gaus	Crystal Ball	5.15-5.5	1252794 ± 7601
b	Crystal Ball	2 nd ord. Ch. pol	gaus	Landau	5.15-5.5	1307042 ± 5384
c	Crystal Ball	2 nd ord. Ch. pol	gaus	gaus	5.15-5.5	1257149 ± 5230
d	Double Gauss.	2 nd ord. Ch. pol	gaus	Crystal Ball	5.15-5.5	1328524 ± 5738
e	Double Gauss.	2 nd ord. Ch. pol	gaus	Landau	5.15-5.5	1319095 ± 4880
f	Double Gauss.	3 rd ord. Ch. pol	gaus	gaus	5.15-5.5	1295249 ± 8819
g	Double Gauss.	2 nd ord. Ch. pol	gaus	gaus	5.05-5.6	1311804 ± 4654
h	Double Gauss.	2 nd ord. Ch. pol	gaus	gaus	5.18-5.4	1324893 ± 5217

Table 11: Fit Variants for Y_B

The relative systematic uncertainties on Y_{B_c} and Y_B are then added in quadrature to obtain the systematic uncertainty on $R_{c/u}$, which results to be 5.3%.

9.1.3 Finite size of the MC sample

As described in Sec. 9, the efficiency correction is entirely derived from the simulation. The number of events in the simulation directly affects the accuracy of the efficiency determination. The efficiency uncertainty entering the evaluation of Y_{B_c} and Y_B is accounted for through Toy MC studies. 1000 toy efficiency distributions are generated by randomly varying the efficiency value in each p_T bin within its error. Data are then weighted with toy efficiencies and Y_{B_c} and Y_B are extracted. The resulting Y_x distribution is then fitted with a gaussian (Fig. 32), whose width is taken as systematic uncertainty. This results in an uncertainty on $R_{c/u}$ of 2.1%.

9.1.4 Efficiency binning

In order to estimate the possible bias due to the choice of the p_T binning for the efficiency description, three different bin choices have been tested for both channels. Binning definitions

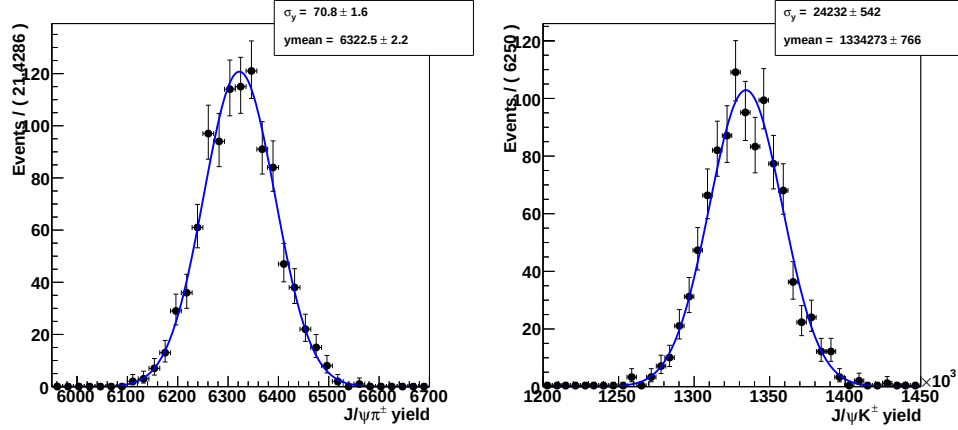


Figure 32: Distributions of the efficiency corrected yields Y_{B_c} (left) and Y_B (right) returned by the fit to 1000 pseudo-experiments.

are summarized in Table 12 and 13 for the B^+ and the B_c respectively. A maximum deviation from the central Y_B (Y_{B_c}) value of the order of 2.4% (1.9%) is found. It results in a maximum deviation from the ratio central value of 3.1%.

9.1.5 B_c Lifetime uncertainty

Our $B_c^\pm$ MC sample has been produced introducing a lifetime value of the $B_c^\pm$ meson taken from the PDG [19]. However, the LHCb collaboration has recently published a new result in the semileptonic channel $B_c^+ \rightarrow J/\psi \mu^+ \nu_\mu X$: $\tau_{B_c} = 0.509 \pm 0.008 \pm 0.012$ ps [1]. This is the most precise $B_c^\pm$ lifetime measurement published so far, and it is not covered by the PDG value plus one standard deviation. We performed our ratio measurement introducing in the MC the new LHCb lifetime value and the PDG average value minus one standard deviation. We take the difference with respect to the central value as systematic uncertainty. The uncertainty results to be $\pm 10.4\%$ and is quoted separately in the final result.

9.2 Result

In summary, the ratio $R_{c/u}$ has been measured to be

$$R_{c/u} = \frac{\sigma(B_c^\pm) \times Br(B_c^\pm \rightarrow J/\psi \pi^\pm)}{\sigma(B^\pm) \times Br(B^\pm \rightarrow J/\psi K^\pm)} = (0.48 \pm 0.05(stat) \pm 0.03(syst) \pm 0.05(\tau(B_c))) \cdot 10^{-2} \quad (4)$$

Bin	Binning1	Binning2	Binning3
1	15 - 15.6	15 - 15.8	15-15.3
2	15.6 - 16.3	15.8 - 16.4	15.3 - 15.7
3	16.3 - 16.8	16.4 - 17.3	15.7 - 16.1
4	16.8 - 17.5	17.3 - 18	16.1 - 16.5
5	17.5 - 18	18 - 18.7	16.5 - 17
6	18 - 18.8	18.7 - 19.4	17 - 17.5
7	18.8 - 19.6	19.4 - 20.1	17.5 - 18
8	19.6 - 20.5	20.1 - 20.8	18 - 18.5
9	20.5 - 21.5	20.8 - 22.6	18.5 - 19
10	21.5 - 22.5	22.6 - 23.4	19 - 19.5
11	22.5 - 24	23.4 - 24.2	19.5 - 20
12	24 - 26.5	24.2 - 25	20 - 21
13	26.5 - 29	25 - 26.5	21 - 21.8
14	29 - 32	26.5 - 28	21.8 - 22.5
15	32 - 35	28 - 31	22.5 - 23.3
16	35 - 39	31 - 34	23.3 - 24.1
17	39 - 46	34 - 40	24.1 - 25
18	46 - 100	40 - 55	25 - 27
19		55 - 100	27 - 28.5
20			28.5 - 31
21			31 - 34
22			34 - 39
23			39 - 43
24			43 - 50
25			50 - 100
Y_B	1288844 ± 5017	1292228 ± 5031	1344265 ± 5233

Table 12: Three alternative binning definitions for the B^+ efficiency evaluation and the resulting Y_B .

Bin	Binning1	Binning2	Binning3
1	15 - 15.6	15-15.3	15 - 15.8
2	15.6 - 16.2	15.3 - 15.6	15.8 - 16.6
3	16.2 - 16.8	15.6 - 15.9	16.6 - 17.4
4	16.8 - 17.4	15.9 - 16.2	17.4 - 18.2
5	17.4 - 18	16.2 - 16.5	18.2 - 19
6	18 - 18.6	16.5 - 16.8	19 - 19.8
7	18.6 - 19.2	16.8 - 17.2	19.8 - 20.6
8	19.2 - 19.8	17.2 - 17.5	20.6 - 21.4
9	19.8 - 20.5	17.5 - 17.8	21.4 - 22.5
10	20.5 - 21.5	17.8 - 18.4	22.5 - 23.5
11	21.5 - 22.5	18.4 - 19	23.5 - 24.5
12	22.5 - 23.5	19 - 19.8	24.5 - 26
13	23.5 - 24.5	19.8 - 20.6	26 - 27.5
14	24.5 - 26	20.6 - 21.4	27.5 - 29
15	26 - 27.5	21.4 - 22.2	29 - 32
16	27.5 - 29.5	22.2 - 23	32 - 36
17	29.5 - 32	23 - 24	36 - 44
18	32 - 36	24 - 25.5	44 - 100
19	36 - 45	25.5 - 26.5	
20	45 - 100	26.5 - 28.5	
21		28.5 - 31	
22		31 - 34	
23		34 - 42	
24		42 - 100	
Y_{B_c}	6186 ± 668	6267 ± 677	6225 ± 672

Table 13: Three alternative binning definitions for the B_c efficiency evaluation and the resulting Y_{B_c} .

10 The R_{B_c} ratio

10.1 Hints of resonant substructures in the $B_c^\pm \rightarrow J/\psi \pi^\pm \pi^\pm \pi^\mp$ decay mode and efficiency evaluation

The $B_c^\pm \rightarrow J/\psi \pi^\pm \pi^\pm \pi^\mp$ decay has been detected for the first time with an yield of about 100 events in LHCb and confirmed by CMS. Although some resonant substructures manifest in three and two body submasses in both the experiments, the quantitative determination of their contribution and mutual interferences in the five-body phase space would require a sophisticated amplitude analysis which is certainly out of the feasibility of the available statistics. However, the efficiency evaluation for the five-body decay of the $B_c^\pm$ could be affected by the decay dynamics and requires additional studies.

Indeed, a visual inspection of the $\pi^+ \pi^+ \pi^-$ and $\pi^+ \pi^-$ data mass projections reveals some hints of $a_1(1260)$ and $\rho^0(770)$ in the decay (Fig. 33).

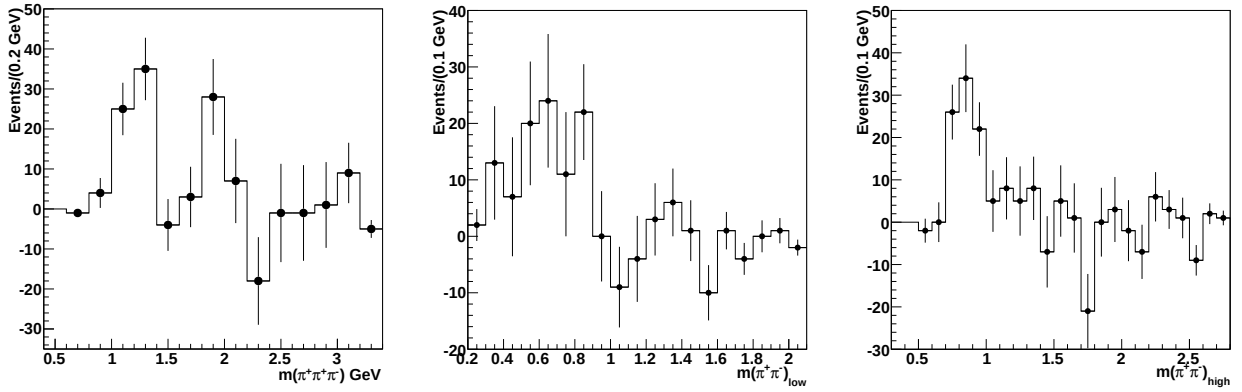


Figure 33: Sideband subtracted mass projection for $\pi^\pm \pi^\pm \pi^\mp$ (left), $(\pi^\pm \pi^\mp)_{low}$ (center) and $(\pi^\pm \pi^\mp)_{high}$. Since two same-sign pions are present in the final state, the two $\pi^\pm \pi^\mp$ pairs are identified as *low* and *high* according to their invariant mass.

Different dynamics could favor, or disfavor, different regions of the phase space; it is then important to study the efficiency as a function of a complete set of variables for the five-body final state to gauge the variation across the entire phase space. A $B_c \rightarrow J/\psi + 3\pi$ non-resonant MC has been produced to access all the phase space configurations.

A five-body decay is described by 20 variables, i.e. the 5 four-vectors of the final state. They are constrained by the 4 conservation (energy and momentum) equations and by the 5 particle masses in the final state. In addition, the initial state is isotropic (the B_c meson is a pseudoscalar) and the final state, in its rest frame, does not depend on the 3 angles describing the relative orientation. In the end there are: $20-9-3 = 8$ independent variables. The J/ψ mass constraint imposed on the two muons reduces the number of independent variables to 7. The efficiency should then be evaluated as function of 7 two-body invariant masses.

The two-body invariant masses have been inspected and the efficiency has been studied as a function of the 7 couples: $m^2(\mu^+ \pi^+)_{low}$, $m^2(\pi^+ \pi^-)_{high}$, $m^2(\mu^+ \pi^-)$, $m^2(\pi^+ \pi^+)$, $m^2(\mu^- \pi^+)_{low}$, $m^2(\mu^- \pi^+)_{high}$ and $m^2(\mu^- \pi^-)$.

The efficiency can be parametrized as a polynomial function of the type

$$\epsilon = |p_0 + p_1 \cdot x + p_2 \cdot y + p_3 \cdot z + p_4 \cdot w + p_5 \cdot r + p_6 \cdot t + p_7 \cdot v| \quad (5)$$

where $x = m^2(\mu^+ \pi^+)_{low}$, $y = m^2(\pi^+ \pi^-)_{high}$, $z = m^2(\mu^+ \pi^-)$, $w = m^2(\pi^+ \pi^+)$, $r = m^2(\mu^- \pi^+)_{low}$,
 $t = m^2(\mu^- \pi^+)_{high}$, $v = m^2(\mu^- \pi^-)$ and p_i are the free parameters to be determined via an un-
 binned maximum likelihood fit on the generated events in the 7-th dimensional space through
 a binomial probability. The absolute value is required to protect the function from assuming
 negative values.

The projections of the fit result are plotted in Fig. 34, 35 and 36; MC events are denoted with
 error bars, while the predicted event number is represented with the continuous line.

To visualize the efficiency variation in the three considered HLT paths, the efficiency function
 projections for each HLT are superimposed in Fig.37.

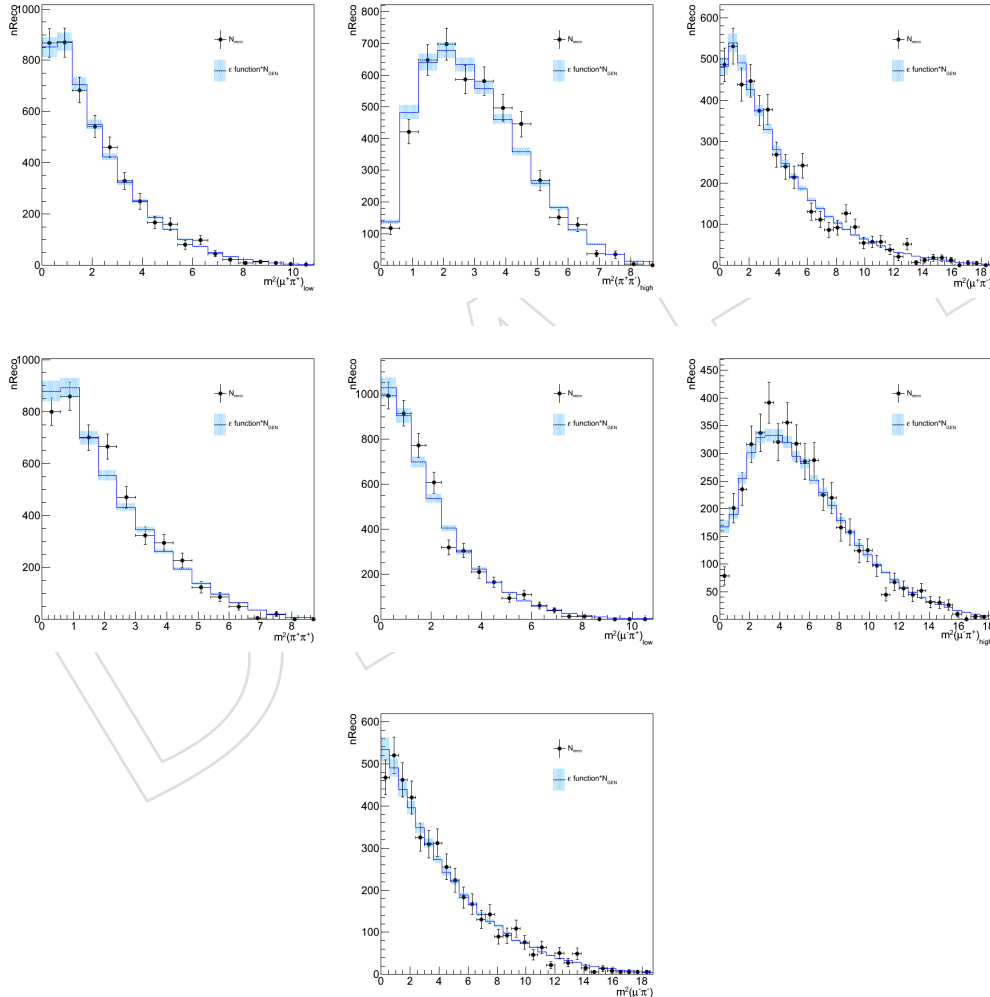


Figure 34: HLT_Dimuon7.Jpsi_displaced trigger: number of reconstructed events projected on the 7 two-body invariant masses: the black dots represent the number of reconstructed events in MC, the blue line is the expected number of reconstructed events as resulting from the 7-dimensional fit.

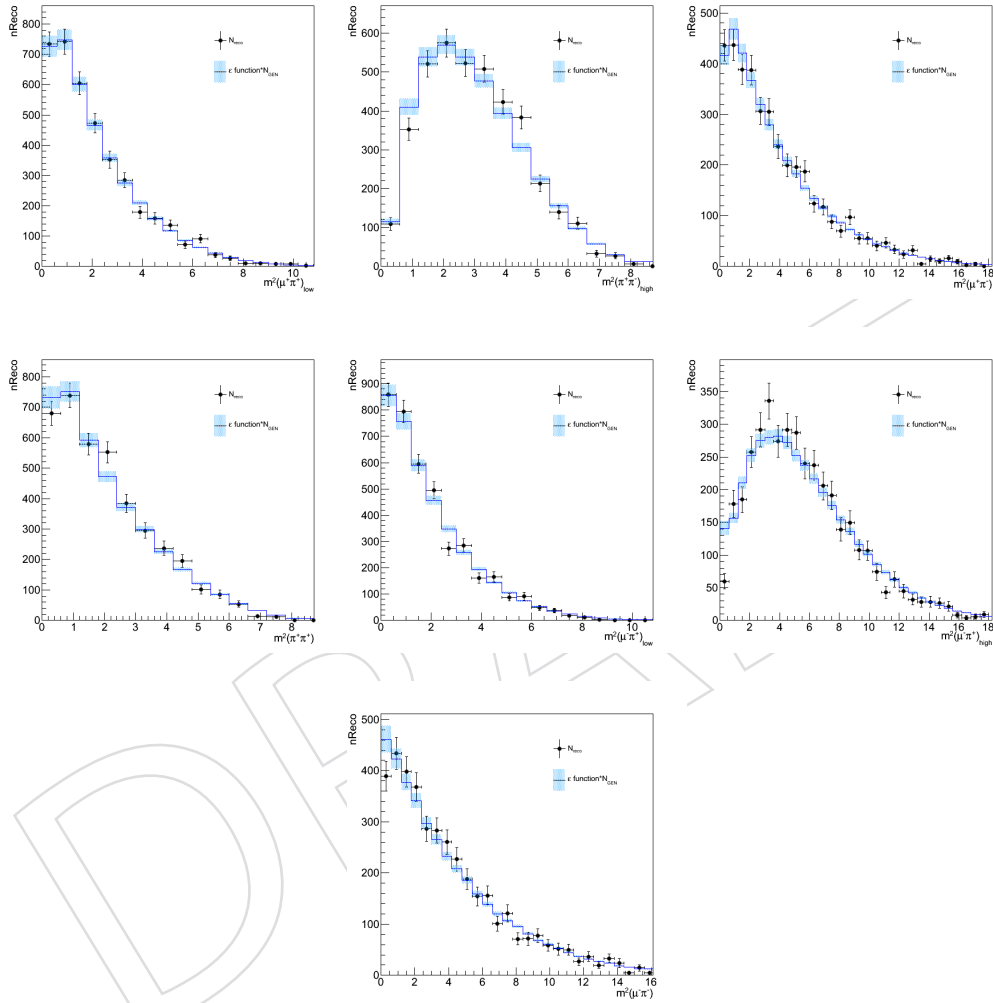


Figure 35: HLT_DoubleMu3p5_Jpsi_displaced trigger: number of reconstructed events projected on the 7 two-body invariant masses: the black dots represent the number of reconstructed events in MC, the blue line is the expected number of reconstructed events as resulting from the 7-dimensional fit.

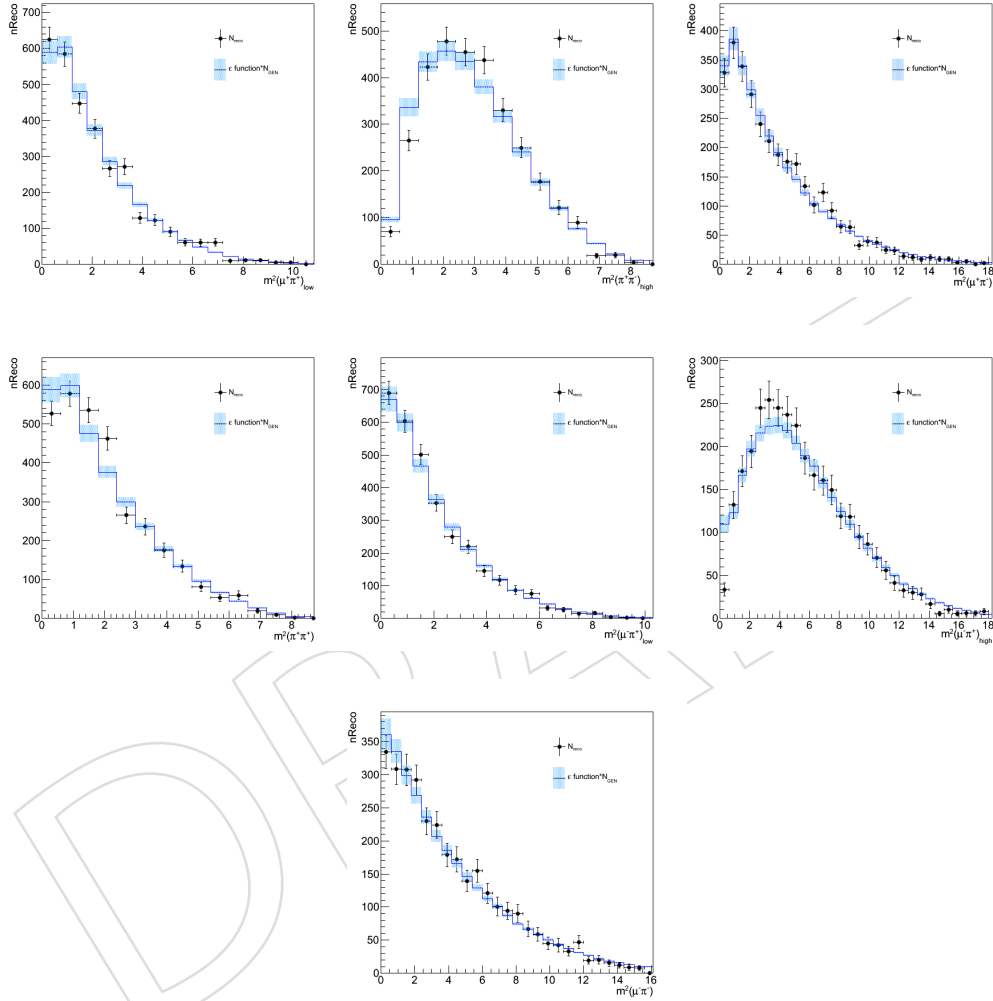


Figure 36: HLT_DoubleMu4.Jpsi_displaced trigger: number of reconstructed events projected on the 7 two-body invariant masses: the black dots represent the number of reconstructed events in MC, the blue line is the expected number of reconstructed events as resulting from the 7-dimensional fit.

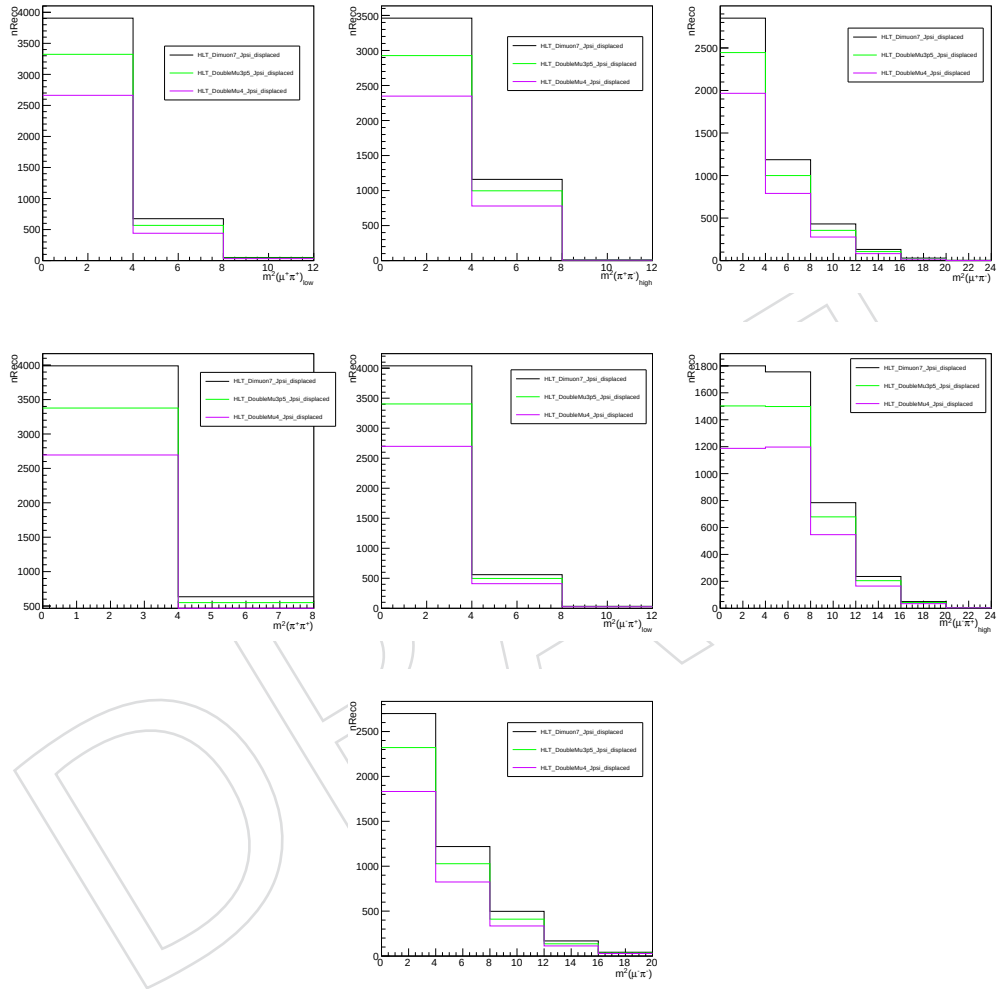


Figure 37: Projection of the expected number of reconstructed events as resulting from the 7-dimensional fit for each of the three considered HLT paths.

447 The resulting efficiency function is used to weight the data event by event. Efficiency corrected
 448 data are then fit with an extended unbinned ML fit to extract $Y_{3\pi}$ (see Fig. 38). The relative
 449 error is inferred from the fit to the unweighted dataset; $Y_{3\pi}$ is then $Y_{3\pi} = 16066 \pm 4715$. The
 450 ratio results to be

$$R_{Bc} = \frac{Br(B_c^\pm \rightarrow J/\psi \pi^\pm \pi^\pm \pi^\mp)}{Br(B_c^\pm \rightarrow J/\psi \pi^\pm)} = 2.55 \pm 0.80, \quad (6)$$

451 where the error is statistical only.

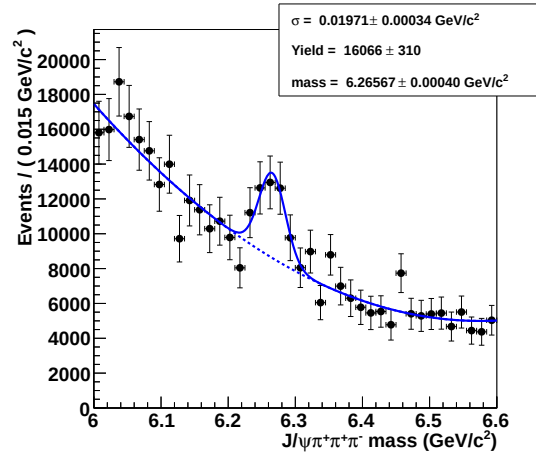


Figure 38: Fit to efficiency corrected data for $B_c^\pm \rightarrow J/\psi \pi^\pm \pi^\pm \pi^\mp$.

10.2 Systematic error evaluation on R_{B_c}

As described in Section 9.1, different sources of systematic uncertainty for the measurement of R_{B_c} are considered. Their contributions are summarized in Table 14. Table 15 lists systematic contributions to $Y_{3\pi}$.

Syst. error	Percentage	Abs. value
Split sample	0.	0.
Fit variant	9.4	0.24
MC finite size	4.1	0.10
Efficiency fit function	1.0	0.03
Efficiency binning*	1.9	0.05
Tracking efficiency	7.8	0.20
Total uncertainty	13.1	0.33
Lifetime	+1.6 -0.4	+0.04 -0.01

Table 14: Systematic uncertainty contributions for the measurement of R_{B_c} . Contributions denoted with * only affect the evaluation of Y_{B_c} and have been already described in Sec.9.1.

Syst. error	Percentage
Fit variant	8.1
MC finite size	3.9
Efficiency fit function	1.0
Tracking efficiency	7.8

Table 15: Systematic uncertainties contributing to the $Y_{3\pi}$ evaluation.

10.2.1 Split samples

The data have been split in N sub-samples according to the different running periods, number of Pile-Up vertices and main variables of the analysis to check the consistency of the results and possible systematic effects. The study of each contribution is reported below and the systematics evaluation is obtained following the method in Appendix 12.4. No systematic uncertainty is measured.

10.2.1.1 Split in high and low PU events

As previously shown in Sec. 9.1.1.2, the efficiency behavior as a function of the number of Primary Vertices has been studied in the three trigger periods. The MC sample has been divided in two PileUp regions, namely low ($1 \leq PV \leq 6$) and high ($PV \geq 7$) PileUp and the efficiency for the two regions is shown in Fig. 39 and 40.

The signals and the corresponding efficiency corrected yields for the $B_c^\pm \rightarrow J/\psi \pi^\pm \pi^\pm \pi^\mp$ mode are shown in Fig. 41. The ratio has been evaluated in the two regions as summarized in Table 16. The statistical error significantly increases due to the $B_c^\pm \rightarrow J/\psi \pi^\pm \pi^\pm \pi^\mp$ low statistics. The resulting χ^2 is greater than one; however, when applying the procedure described in Sec. 12.4, no residual systematic uncertainty is measured.

10.2.1.2 Split in B_c rapidity

The samples have been split in two sub-samples according to the B_c rapidity ($|y| \leq 0.6$ and $0.6 < |y| < 1.6$). The corresponding signals and efficiency corrected yields are shown in Fig.42 and 43. The ratio evaluation in the two samples is summarized in Table 17.

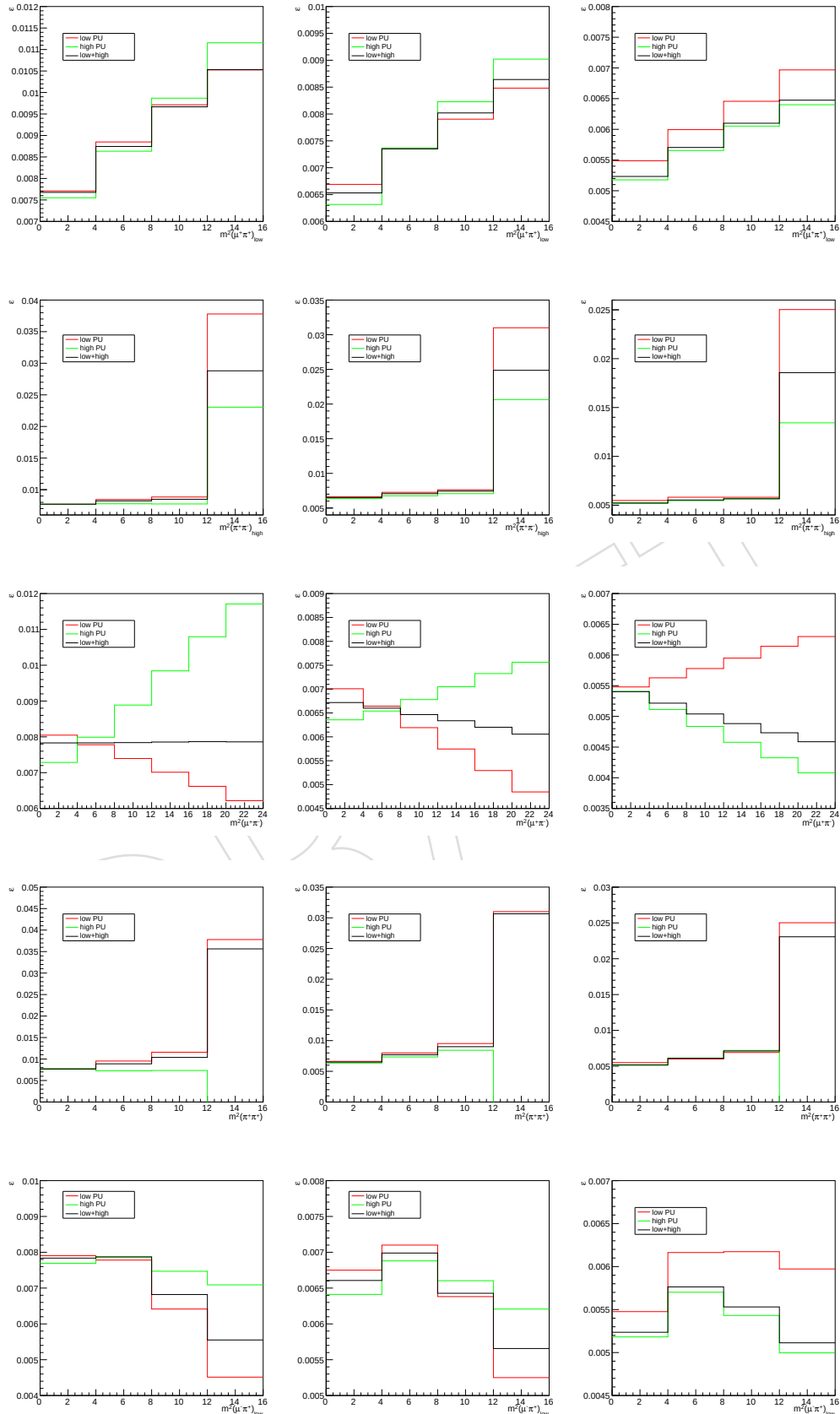


Figure 39: Projection of the efficiency function for the low (red) and high (green) PileUp regions on the first 5 considered submass pairs. The black line represents the efficiency function when no requirements on the number of PV is applied. The three plots in a row illustrate the efficiency function for the three considered HLT paths (HLT_Dimuon7_Jpsi_displaced, HLT_DoubleMu3p5_Jpsi_displaced and HLT_DoubleMu4_Jpsi_displaced respectively).

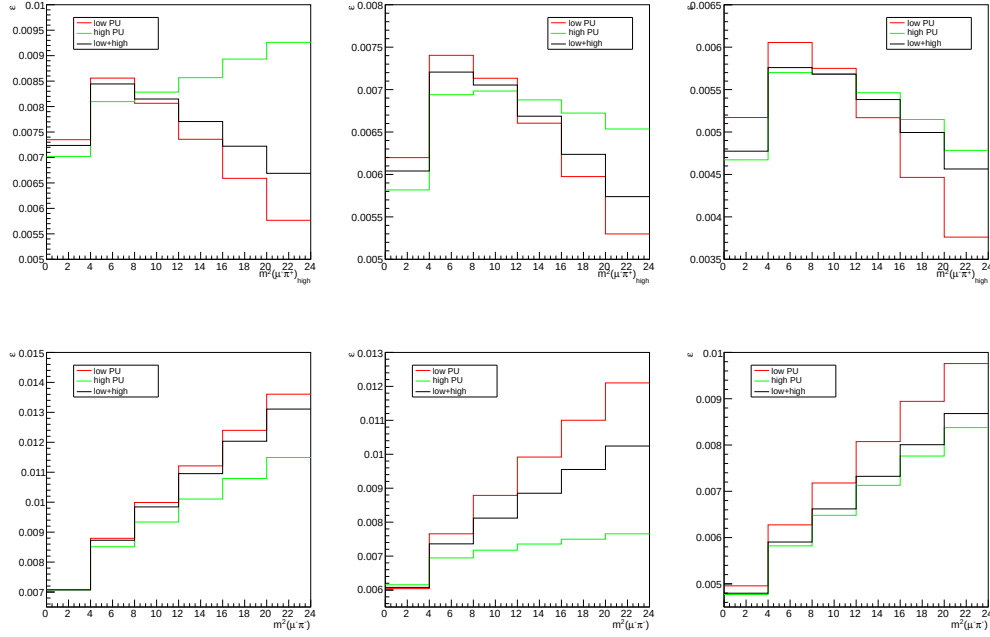


Figure 40: Projection of the efficiency function for the low (red) and high (green) PileUp regions on the last 2 submass pairs. The black line represents the efficiency function when no requirements on the number of PV is applied. The three plots in a row illustrate the efficiency function for the three considered HLT paths.

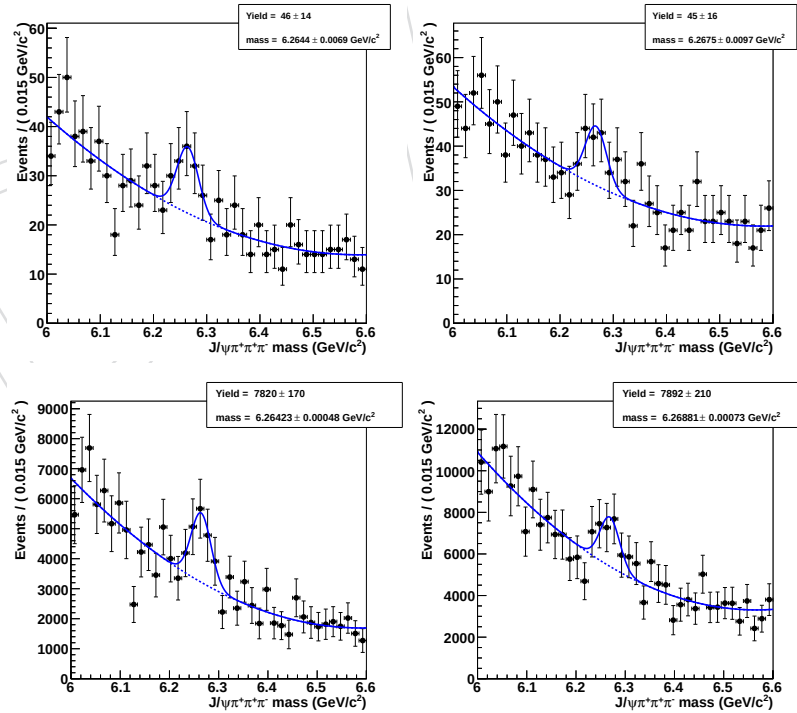


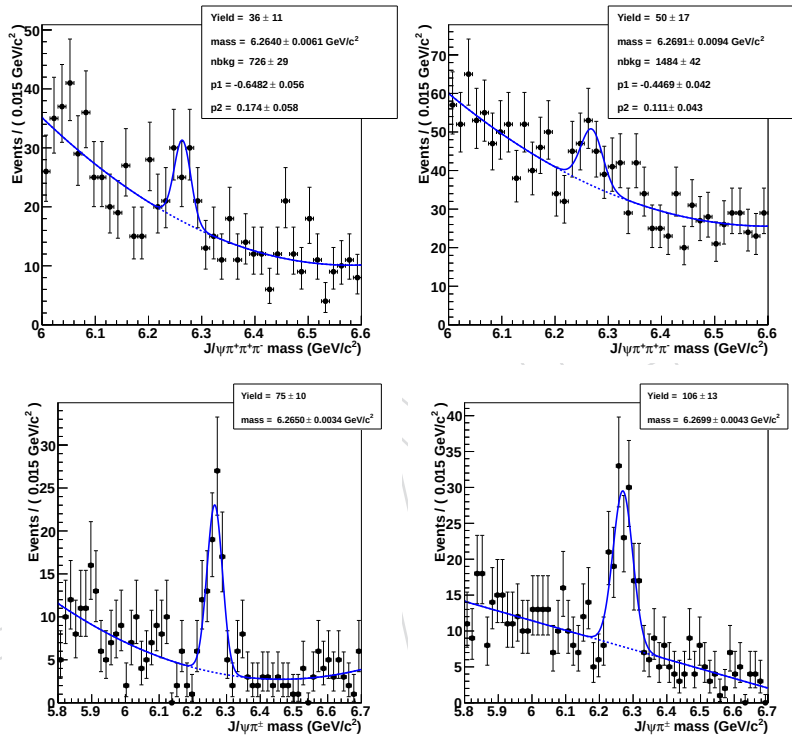
Figure 41: $B_c^\pm$ signals (top) and $Y_{3\pi}$ (bottom) in the low (left) and high (right) PileUp region.

	$Y_{3\pi}$	Y_{B_c}	R_{B_c}
nPV < 7	7820 ± 2380	2127 ± 325	3.68 ± 1.25
nPV ≥ 7	7892 ± 2806	4043 ± 501	1.95 ± 0.73

Table 16: Ratio measurement in two PU regions.

	$Y_{3\pi}$	Y_{B_c}	R_{B_c}
$ y \leq 0.6$	12286 ± 3754	5212 ± 695	2.36 ± 0.79
$0.6 < y < 1.6$	17781 ± 6046	6431 ± 789	2.76 ± 1.00

Table 17: Ratio measurement in the two rapidity regions.

Figure 42: $B_c^\pm \rightarrow J/\psi \pi^\pm \pi^\pm \pi^\mp$ (top) and $B_c^\pm \rightarrow J/\psi \pi^\pm$ (bottom) signals in the central (left) and forward (right) rapidity region.

476 The split samples turned out to be statistically consistent ($\chi^2 = 0.1$) and no additional uncer-
 477 tainty has to be accounted for due to the rapidity dependence.

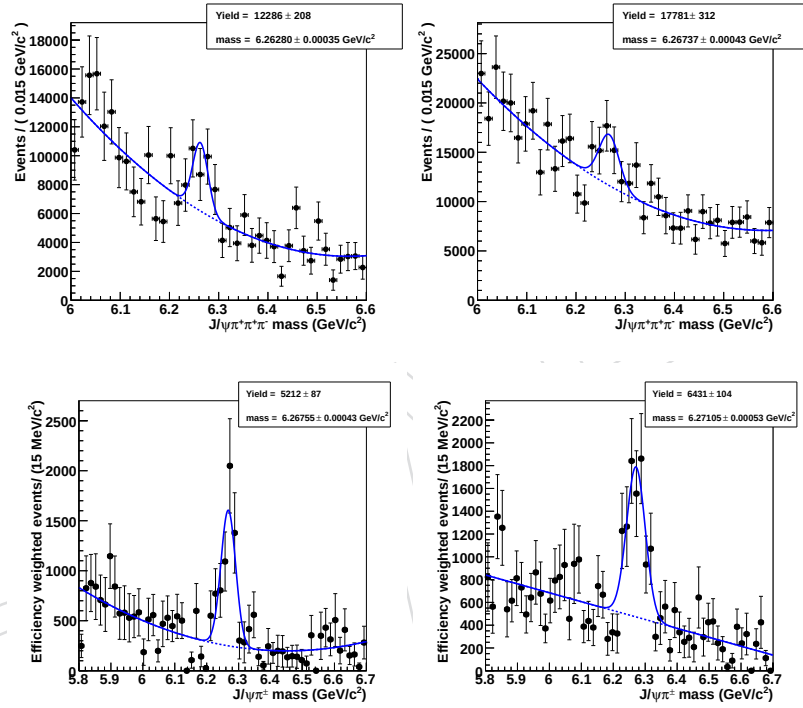


Figure 43: $B_c^\pm \rightarrow J/\psi \pi^\pm \pi^\pm \pi^\mp$ (top) and $B_c^\pm \rightarrow J/\psi \pi^\pm$ (bottom) efficiency corrected distributions in the central (left) and forward (right) rapidity region.

10.2.1.3 Split in L_{xy}/σ_{xy}

The samples have been split in two sub-samples according to L_{xy}/σ_{xy} , namely ($5 \leq L_{xy}/\sigma_{xy} \leq 12$ and $L_{xy}/\sigma_{xy} > 12$). The cut value is chosen in order to split the sample into two sub-samples having comparable statistics.

The corresponding signals and efficiency corrected yields are shown in Fig. 44 and 45 respectively. The ratio evaluations in the two samples are collected in Table 18.

	$Y_{3\pi}$	Y_{B_c}	R_{B_c}
$5 \leq L_{xy}/\sigma_{xy} \leq 12$	9696 ± 4458	5897 ± 670	1.64 ± 0.80
$L_{xy}/\sigma_{xy} > 12$	20448 ± 3916	7768 ± 1254	2.63 ± 0.66

Table 18: Ratio measurement in the two L_{xy}/σ_{xy} regions.

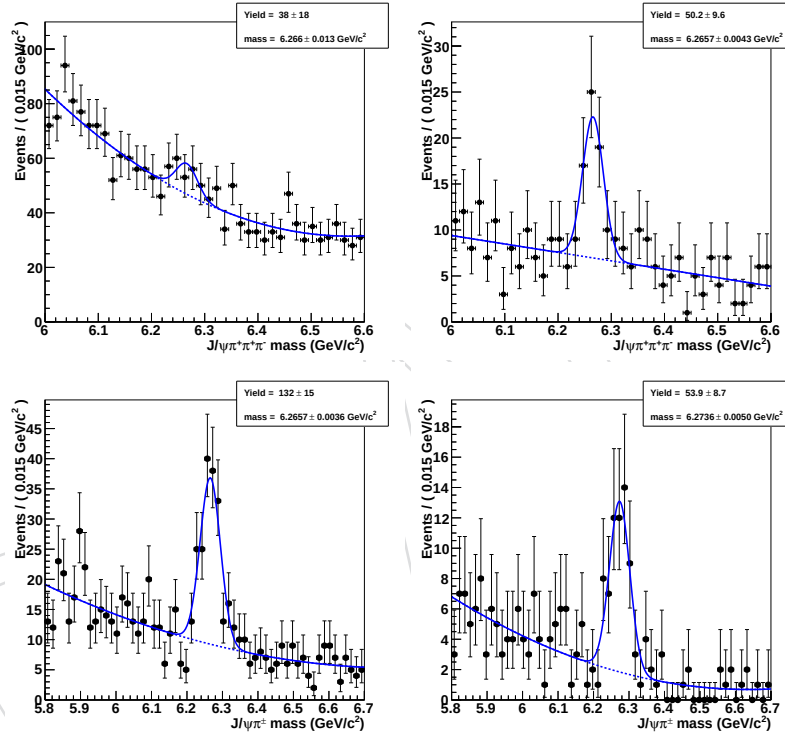


Figure 44: $B_c^\pm \rightarrow J/\psi \pi^\pm \pi^\pm \pi^\mp$ (top) and $B_c^\pm \rightarrow J/\psi \pi^\pm$ (bottom) signals in the low (left) and high (right) L_{xy}/σ_{xy} regions.

Results are consistent with a $\chi^2 = 0.91$.

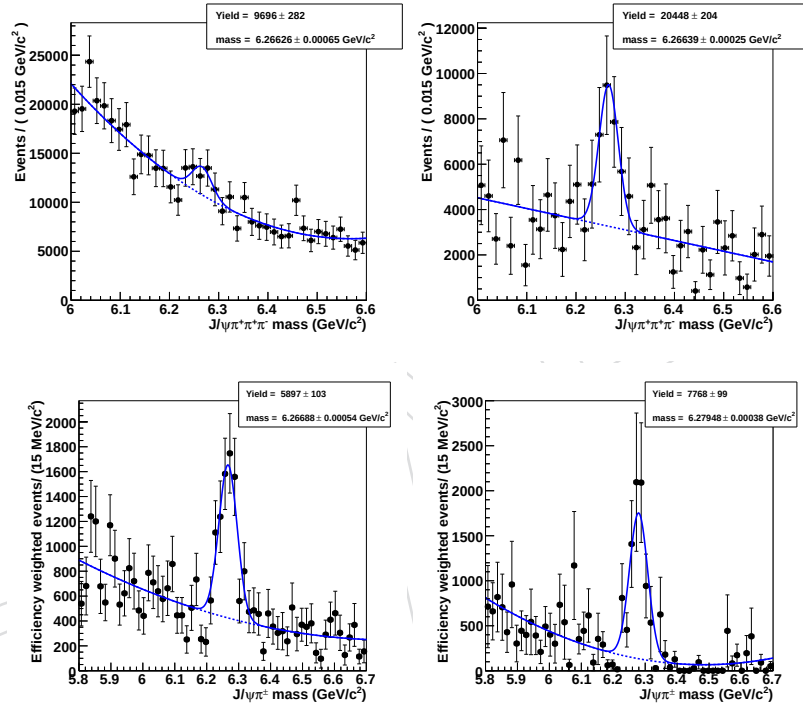


Figure 45: $B_c^\pm \rightarrow J/\psi \pi^\pm \pi^\pm \pi^\mp$ (top) and $B_c^\pm \rightarrow J/\psi \pi^\pm$ (bottom) efficiency corrected yields in the low (left) and high (right) two L_{xy}/σ_{xy} regions.

10.2.1.4 Split in low and high B_c Pt

The samples have been split in two sub-samples according to the B_c transverse momentum ($15 < p_T \leq 28$ and $p_T > 28$). The signals for the $B_c^\pm \rightarrow J/\psi \pi^\pm \pi^\pm \pi^\mp$ and $B_c^\pm \rightarrow J/\psi \pi^\pm$ channels are shown in Fig. 46; the corresponding efficiency corrected yields are collected in Fig. 47 and the ratio evaluation in the two samples is summarized in Table 19.

	$Y_{3\pi}$	Y_{B_c}	R_{B_c}
$15 < p_T \leq 28$	10491 ± 3854	6229 ± 673	1.68 ± 0.64
$p_T > 28$	1413 ± 338	440 ± 91	3.21 ± 1.02

Table 19: Ratio measurement in the two p_T regions.

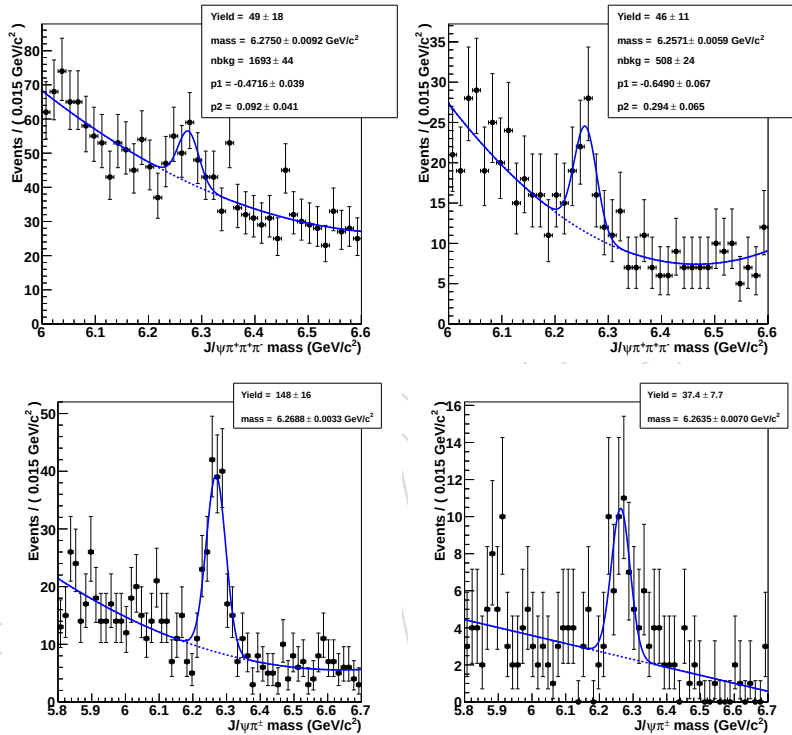


Figure 46: $B_c^\pm \rightarrow J/\psi \pi^\pm \pi^\pm \pi^\mp$ (top) and $B_c^\pm \rightarrow J/\psi \pi^\pm$ (bottom) signals in the low (left) and high (right) $B_c^\pm$ transverse momentum region.

In this case, we have $\chi^2 > 1$; however, when following the procedure described in Sec. 12.4, no residual systematic uncertainty is found.

The above split-sample studies have been carried out fixing the signal width to the one inferred from MC to have more reliable fit results. We verified that, by leaving all the parameters free when possible, no systematics was present. We have also performed the fits by fixing and relaxing different parameters: the outcome is always consistent with no systematics.

10.2.2 Fit Variants

As described in Sec. 12.5, the distribution of efficiency corrected data has been fitted in different mass ranges and with different parametrizations. Results for $Y_{3\pi}$ are summarized in Table 20; all fit parameters are free to float. The corresponding plots are collected in App. 12.5.1.3. The total systematic uncertainty on the ratio due to the fit variants is 8.1%.

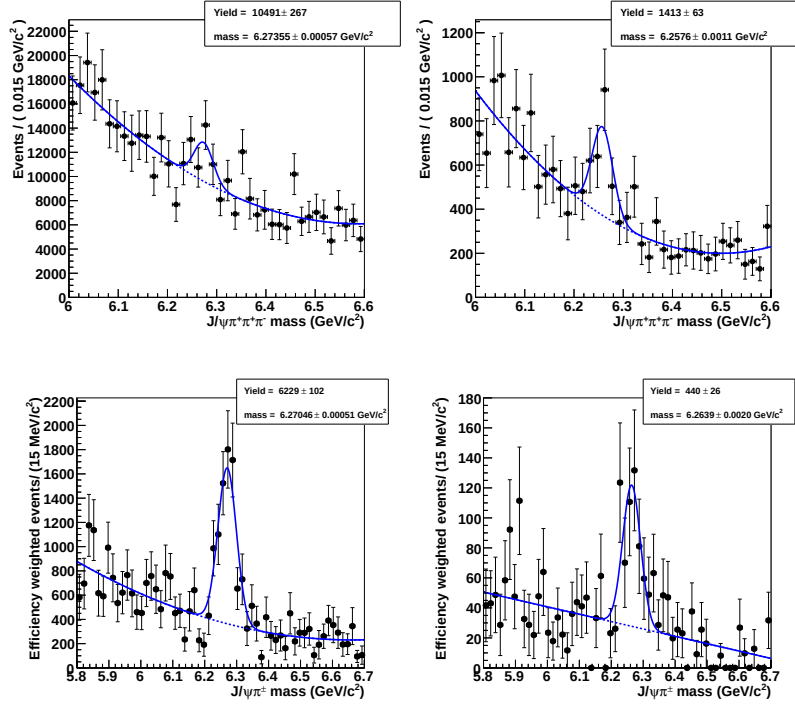


Figure 47: $B_c^\pm \rightarrow J/\psi\pi^\pm\pi^\pm\pi^\mp$ (top) and $B_c^\pm \rightarrow J/\psi\pi^\pm$ (bottom) efficiency corrected distributions in the low (left) and high (right) $B_c^\pm$ transverse momentum region.

	Signal PDF	Bkg PDF	Range (GeV/c ²)	Y_{B_c}
a	Gaussian	1 st order Chebyshev polynomial	6.13-6.4	15471 ± 4690
b	Gaussian	2 st order Chebyshev polynomial	6.10-6.4	16890 ± 6673
c	Gaussian	2 st order Chebyshev polynomial	6.1-6.505	17853 ± 5951
d	Gaussian	2 st order Chebyshev polynomial	5.8-6.7	14745 ± 4115
e	Gaussian	3 st order Chebyshev polynomial	6-6.6	17947 ± 5422
f	Gaussian	3 st order Chebyshev polynomial	5.8-6.7	15973 ± 4357

Table 20: Fit Variant for $Y_{3\pi}$

10.2.3 Finite MC sample size

As well as for $B_c^\pm \rightarrow J/\psi\pi^\pm$ and $B^\pm \rightarrow J/\psi K^\pm$, the efficiency correction is derived from the simulation (see Sec. 10.1). The limited size of the MC sample directly affects the accuracy of the efficiency determination. To propagate this uncertainty to the efficiency corrected yield $Y_{3\pi}$, the following procedure is adopted:

1. N efficiency curves are generated with variations of the parameter values as sampled from a multivariate Gaussian p.d.f. constructed from the fit covariance matrix;
2. the analysis is run N-times, data are weighted with a new efficiency function each time and $Y_{3\pi}$ is computed;
3. the resulting $Y_{3\pi}$ distribution is fit with a Gaussian, whose width is the systematic uncertainty associated to the finite size of the MC sample.

2000 pseudo-experiments have been generated and the resulting systematic uncertainty on $Y_{3\pi}$ is 3.9%. Propagating this uncertainty and the corresponding error on Y_{B_c} (evaluated in Sec.9.1.3) on the ratio measurement, a systematic contribution of 4.1% is accounted for.

10.2.4 Efficiency fit function

The 7-dimensional efficiency function has been parametrized with a polynomial function as described in Eq. 5. In order to check if the choice of this form introduces a systematic effect in the $Y_{3\pi}$ evaluation, data have been weighted according to the binned efficiency distribution obtained from MC. The efficiency has been measured in six 4-GeV/c² bins for each of the seven submasses and for each of the three trigger paths separately. From the fit to efficiency corrected data, $Y_{3\pi}$ is measured to be 15945 ± 4680 and $R_{B_c} = 2.53 \pm 0.79$. The difference between this result and the central value quoted in the analysis is assumed as systematic uncertainty.

10.2.5 Arbitration criterion

When multiple candidates are reconstructed in an event, the one with the highest p_T is selected. Since the number of candidates depends on the final state multiplicity, the arbitration allows for the rejection of a consistent fraction of background in the case of the multibody decay $B_c^\pm \rightarrow J/\psi \pi^\pm \pi^\pm \pi^\mp$.

In order to evaluate a possible systematic error due to the arbitration in this channel, the analysis has been performed removing the arbitration and leaving the rest of the analysis unchanged. This gives an yield of 117 ± 47 , which, once rescaled with the proper relative efficiency $((117 \pm 47) \cdot \epsilon = 106 \pm 43$, with $\epsilon = 0.91$ as measured from Monte Carlo), can be compared with our baseline yield of 92 ± 27 .

The larger error of the non-arbitrated signal is obviously related to the much higher background contamination.

These results are strongly correlated since the arbitrated sample, Y, is fully contained in the non-arbitrated sample, X.

To properly compare these two results, we calculated the weighted mean of the signal yields of the two disjoint samples Y and Z, where $Z=X-Y$, and assessed the systematics from the difference of our baseline result and the computed average. To this extent, we performed a simultaneous fit to the Y and Z mass spectra with the following model:

$$\text{Yield}(Y) \cdot \text{Gauss}(\text{mass}_{B_c}, \sigma_{\text{mass}_{B_c}}) + \text{Bkg}_Y \quad (7)$$

for the Y fit, and

$$\frac{\epsilon(Y)}{\epsilon(Z)} \cdot \text{Yield}(Y) \cdot \text{Gauss}(\text{mass}_{B_c}, \sigma_{\text{mass}_{B_c}}) + \text{Bkg}_Z \quad (8)$$

for the Z fit. $\epsilon(Y)$ and $\epsilon(Z)$ are the Y and Z sample efficiencies. The backgrounds, Bkg_Y and Bkg_Z , are described by two independent second-order Chebyshev polynomials; all the parameters are free and the efficiency ratio has a gaussian constraint with mean and error provided by MonteCarlo.

The fit converges to an average $\text{Yield}(Y) = 91.8 \pm 26$, which differs from our baseline $\text{Yield}(Y) = 92 \pm 27$ by 0.2 %, negligible w.r.t other systematics.

548 In Fig. 48 the results of the simultaneous fit are shown.

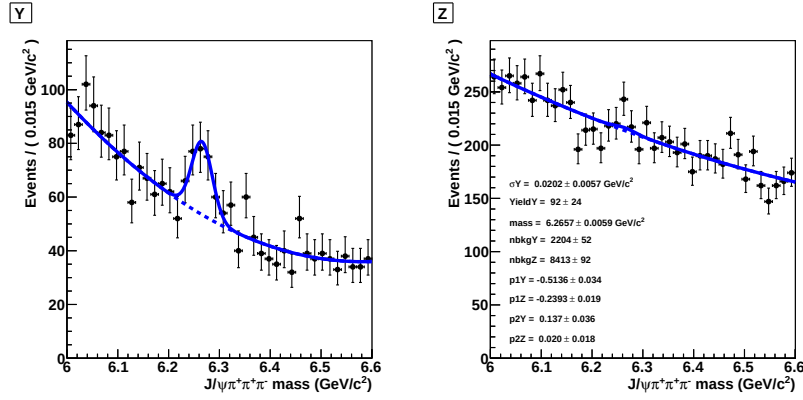


Figure 48: Results of the simultaneous fits.

549 10.2.6 B_c Lifetime uncertainty

550 As already pointed out, the MC sample has been produced with the $B_c^\pm$ lifetime taken from the
 551 PDG [19]. However, the LHCb Collaboration has recently published a new result: $\tau_{B_c} = 0.509$
 552 $\pm 0.008 \pm 0.012$ ps [1]. This is the most precise $B_c^\pm$ lifetime measurement published so far (see
 553 also Sec. 12.1). The efficiency has been evaluated changing the $B_c^\pm$ lifetime in the simulation to
 554 cover the range determined from the PDG average minus one standard deviation to the new
 555 LHCb measurement. Two values for the ratio are measured and their difference wrt the central
 556 value is taken as the associated systematic uncertainty. The uncertainty results to be $^{+1.6\%}_{-0.4\%}$.

557 10.2.7 Tracking efficiency

558 In this measurement two different multiplicity final states are compared. Assuming a tracking
 559 efficiency uncertainty for each pion track of 3.9% [20], a global 7.8% error has to be included in
 560 the final systematic evaluation.

561 10.3 Result

562 In summary, the ratio R_{B_c} has been measured to be

$$R_{B_c} = \frac{BR(B_c^\pm \rightarrow J/\psi \pi^\pm \pi^\pm \pi^\mp)}{BR(B_c^\pm \rightarrow J/\psi \pi^\pm)} = 2.55 \pm 0.80(stat) \pm 0.33(syst)^{+0.04}_{-0.01}(\tau(B_c)) \quad (9)$$

11 Conclusions

The CMS analysis of the $B_c^\pm \rightarrow J/\psi \pi^\pm$ and $B^\pm \rightarrow J/\psi K^\pm$ decays based on the 7 TeV data has allowed the measurement of the ratio:

$$\frac{\sigma(B_c^\pm) \times \text{Br}(B_c^\pm \rightarrow J/\psi \pi^\pm)}{\sigma(B^\pm) \times \text{Br}(B^\pm \rightarrow J/\psi K^\pm)} = (0.48 \pm 0.05(\text{stat}) \pm 0.03(\text{syst}) \pm 0.05(\tau_{B_c})) \times 10^{-2} \quad (10)$$

in a rapidity region complementary to that investigated by the LHCb collaboration [10].

The analysis of the $B_c^\pm \rightarrow J/\psi \pi^\pm \pi^\pm \pi^\mp$ decay mode has also allowed determination of the ratio:

$$\frac{\text{Br}(B_c^\pm \rightarrow J/\psi \pi^\pm \pi^\pm \pi^\mp)}{\text{Br}(B_c^\pm \rightarrow J/\psi \pi^\pm)} = 2.55 \pm 0.80(\text{stat}) \pm 0.33(\text{syst})_{-0.01}^{+0.04}(\tau_{B_c}) \quad (11)$$

which is in good agreement with the only other available estimate from the same LHCb experiment [4].

12 Appendix

12.1 A note on the L_{xy}/σ_{xy} distribution in the five-body decay channel.

In this section we discuss the signal L_{xy}/σ_{xy} evolution for the five-body decay and how it can be modified by the non-trivial decay dynamics and different lifetimes.

We recall that the $B_c^\pm \rightarrow J/\psi \pi^\pm \pi^\pm \pi^\mp$ decay can involve intermediate resonant states; indeed, a visual inspection of the $\pi^\pm \pi^\pm \pi^\mp$ and $\pi^\pm \pi^\mp$ data mass projections reveals presence of $a_1^\pm(1260)$ and $\rho^\circ(770)$ in the decay (Fig. 33). The quantitative determination of resonant contributions and their interferences in the decay would require a sophisticated amplitude analysis which is not feasible with the available statistics. However, since the reconstruction efficiency for this five-body decay could be affected by the decay dynamics, a model-independent efficiency treatment was indeed developed and explained in 10.1.

To this extent, it is interesting to compare the $B_c^\pm \rightarrow J/\psi \pi^\pm \pi^\pm \pi^\mp$ data with Monte Carlo under two different hypotheses: $J/\psi \pi^\pm \pi^\pm \pi^\mp$ non-resonant decay MC and resonant $J/\psi a_1$ decay MC.

In the left plot of Fig. 49 we show the ratio of data and MC yields as a function of the L_{xy}/σ_{xy} cut for the $J/\psi 3\pi$ non-resonant decay MC.

By a way of contrast, in the right plot of Fig. 49, we show the analogous evolution for the resonant $J/\psi a_1$ MC and selecting data in the a_1 region only.

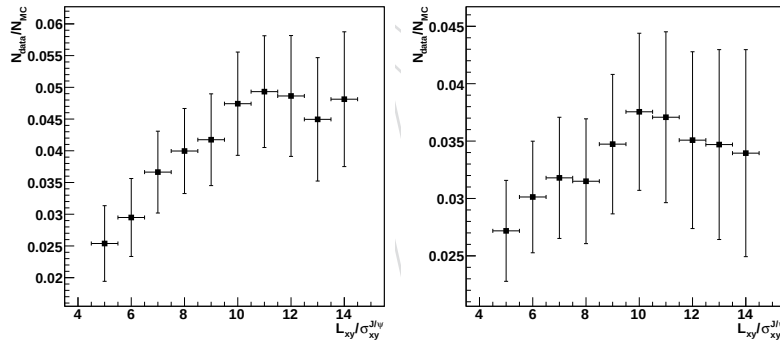


Figure 49: Ratio between data and MC signal yields as a function of L_{xy}/σ_{xy} cut for the $B_c^\pm \rightarrow J/\psi \pi^\pm \pi^\pm \pi^\mp$ channel. Left: the $B_c^\pm \rightarrow J/\psi \pi^\pm \pi^\pm \pi^\mp$ non-resonant MC sample is used to evaluate the MC signal yield; right: the resonant $B_c \rightarrow J/\psi a_1$ sample is used to evaluate the MC signal yield.

The consistency between data and MC improves by a factor of ~ 1.5 : from 0.025-0.05 for the first case to 0.027-0.037 for the second one.

At this point, if we insert in the $J/\psi a_1$ MC the new B_c lifetime measurement by LHCb [1], somewhat higher than the PDG value, the data/MC agreement improves since the ratio evolution plot in Fig. 50 practically flattens.

All in all, we recall that these dynamics effects are properly taken into account in the analysis by weighting the events with the efficiency calculated on the decay Dalitz plot (10.1).

A possible residual systematics due to the L_{xy}/σ_{xy} cut embedded in the trigger is investigated (as we do for all the other analysis cuts) through the split sample technique, checking the statistical consistency of the low and high L_{xy}/σ_{xy} independent samples (see 10.2.1.3).

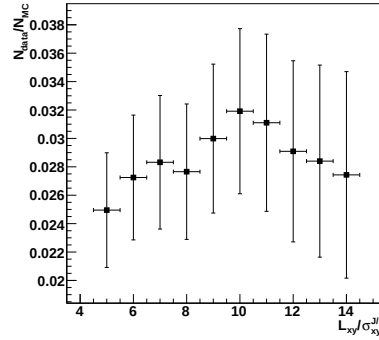


Figure 50: Ratio between data and resonant $J/\psi a_1$ MC signal yields as a function of L_{xy}/σ_{xy} cut for the $B_c^\pm \rightarrow J/\psi \pi^\pm \pi^\pm \pi^\mp$ channel, when the lifetime in the MC is reweighed to be $\tau(B_c) = 0.51$ ps.

No evidence of systematics is found.

A systematic uncertainty is instead added to our R_{B_c} ratio (see 10.2.6) due to the different lifetimes as quoted in the PDG and by LHCb.

12.2 Trigger turn-on studies

This analysis has been performed on data collected by the HLT paths HLT_Dimuon7_Jpsi_displaced, HLT_DoubleMu3p5_Jpsi_displaced and HLT_DoubleMu4_Jpsi_displaced, which have already been described in details in Sec.4.1.

In order to check if the simulation well describes the HLT online selection, trigger turn-on curves have been investigated both in data and in MC. This study has been performed for the HLT_Dimuon7_Jpsi_displaced and the HLT_DoubleMu4_Jpsi_displaced triggers, assuming the HLT_DoubleMu3p5_Jpsi_displaced path quite similar to the last one.

The main requirements implemented at the HLT level are:

- $p_T(J/\psi) > 6.9$ GeV/c
- $L_{xy}/\sigma_{xy} > 3$
- $\cos\alpha > 0.9$
- C.L. of the J/ψ vertex $> 0.005, 0.1$ and 0.15 respectively

The inclusive J/ψ trigger HLT_Dimuon0_Jpsi has been taken as a reference to study the first three items. This trigger path requires two muons lying in the J/ψ mass window and a minimum vertex probability.

The turn-on wrt the J/ψ transverse momentum has been studied first. Let's define:

- $N(J/\psi)_{Dimuon7}$ = number of reconstructed J/ψ firing HLT_Dimuon7_Jpsi_displaced and requiring the HLT_Dimuon0_Jpsi trigger bit
- $N(J/\psi)_{DoubleMu4}$ = number of reconstructed J/ψ firing HLT_DoubleMu4_Jpsi_displaced and requiring the HLT_Dimuon0_Jpsi trigger bit
- $N(J/\psi)_{Dimuon0}$ = number of reconstructed J/ψ requiring the HLT_Dimuon0_Jpsi trigger bit and satisfying $L_{xy}/\sigma_{xy} > 3$ and $\cos\alpha > 0.9$ (the single muon p_T is also required to be greater than 4 GeV/c when the HLT_DoubleMu4_Jpsi_displaced is under investigation)

The behavior of the ratio $\frac{N(J/\psi)_{Dimuon7}}{N(J/\psi)_{Dimuon0}}$ and $\frac{N(J/\psi)_{DoubleMu4}}{N(J/\psi)_{Dimuon0}}$ as a function of the J/ψ transverse momentum is shown for data and MC in Fig.51.

The ratio $\frac{N(J/\psi)_{Dimuon7}}{N(J/\psi)_{Dimuon0}}$ reaches a plateau at 7.1 GeV/c both in data and MC. The ratio $\frac{N(J/\psi)_{DoubleMu4}}{N(J/\psi)_{Dimuon0}}$ is already flat at 7 GeV/c due to the requirement on the single muon transverse momentum ($p_T > 4$ GeV/c). The requirement $p_T(J/\psi) \geq 7.1$ GeV/c is therefore introduced in the offline analysis to ensure a uniform behavior of the trigger for data and MC.

To investigate the turn-on wrt L_{xy}/σ_{xy} , let's define:

- $N(J/\psi)_{Dimuon7}$ = number of reconstructed J/ψ firing the HLT_Dimuon7_Jpsi_displaced and requiring the HLT_Dimuon0_Jpsi trigger bit
- $N(J/\psi)_{DoubleMu4}$ = number of reconstructed J/ψ firing the HLT_DoubleMu4_Jpsi_displaced and requiring the HLT_Dimuon0_Jpsi trigger bit
- $N(J/\psi)_{Dimuon0}$ = number of reconstructed J/ψ requiring the HLT_Dimuon0_Jpsi trigger bit and satisfying $p_T(J/\psi) > 6.9$ and $\cos\theta > 0.9$ (the single muon p_T is also required to be greater than 4 GeV/c when the HLT_DoubleMu4_Jpsi_displaced is under investigation)

The ratios $\frac{N(J/\psi)_{Dimuon7}}{N(J/\psi)_{Dimuon0}}$ and $\frac{N(J/\psi)_{DoubleMu4}}{N(J/\psi)_{Dimuon0}}$ are evolved as a function of L_{xy}/σ_{xy} in Fig.52. The behavior is flat starting from $L_{xy}/\sigma_{xy} = 5$ for both the HLT paths. Therefore the requirement $L_{xy}/\sigma_{xy} \geq 5$ is applied in the offline analysis.

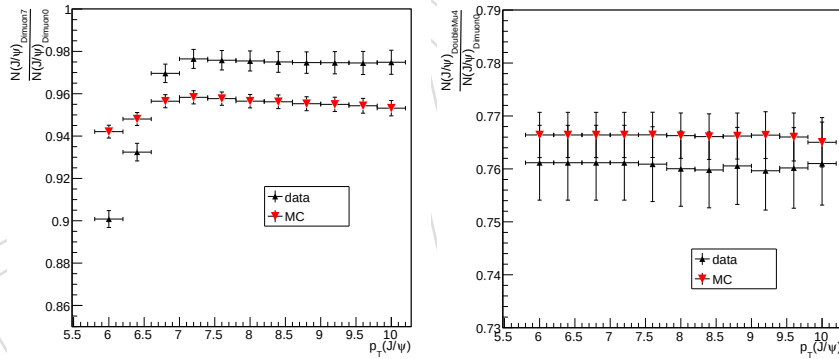


Figure 51: Number of reconstructed J/ψ firing the HLT_Dimuon7_Jpsi_displaced (left) or HLT_DoubleMu4_Jpsi_displaced (right) trigger normalized to the J/ψ collected by the inclusive trigger, evolved as a function of $p_T(J/\psi)$.

In order to check the turn-on curve wrt $\cos\alpha$, we define:

- $N(J/\psi)_{Dimuon7}$ = number of reconstructed J/ψ firing the HLT_Dimuon7_Jpsi_displaced and requiring the HLT_Dimuon0_Jpsi trigger bit
- $N(J/\psi)_{DoubleMu4}$ = number of reconstructed J/ψ firing the HLT_DoubleMu4_Jpsi_displaced and requiring the HLT_Dimuon0_Jpsi trigger bit
- $N(J/\psi)_{Dimuon0}$ = number of reconstructed J/ψ requiring the HLT_Dimuon0_Jpsi trigger bit and satisfying $p_T(J/\psi) > 6.9$ and $L_{xy}/\sigma_{xy} > 3$ (the single muon p_T is also required to be greater than 4 GeV/c when the HLT_DoubleMu4_Jpsi_displaced is under investigation)

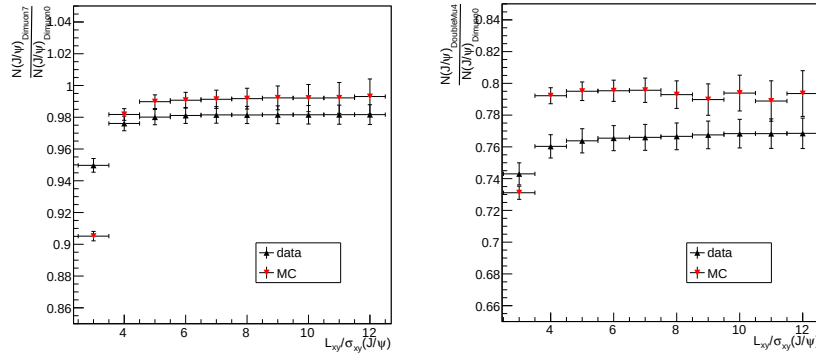


Figure 52: Number of reconstructed J/ψ firing the HLT_Dimuon7_Jpsi_displaced (left) or HLT_DoubleMu4_Jpsi_displaced (right) trigger normalized to the J/ψ collected by the inclusive trigger, evolved as a function of L_{xy}/σ_{xy} .

655 The ratios $\frac{N(J/\psi)_{Dimuon7}}{N(J/\psi)_{Dimuon0}}$ and $\frac{N(J/\psi)_{DoubleMu4}}{N(J/\psi)_{Dimuon0}}$ are evolved as a function of $\cos\alpha$ in Fig.53. No significant discrepancy is found between the two curves. The requirement $\cos\alpha > 0.9$ is therefore
 656 repeated in the offline analysis.
 657

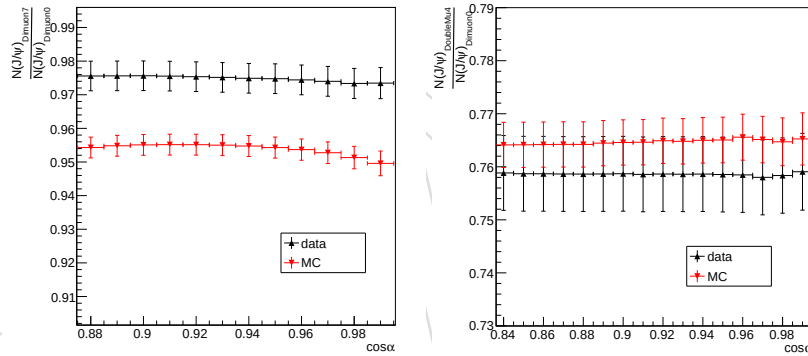


Figure 53: Number of reconstructed J/ψ firing the HLT_Dimuon7_Jpsi_displaced (left) or HLT_DoubleMu4_Jpsi_displaced (right) trigger normalized to the J/ψ collected by the inclusive trigger, evolved as a function of $\cos\alpha$.

658 Finally, the turn-on wrt the dimuon vertex confidence level is studied. The reference trigger
 659 HLT_Dimuon0_Jpsi_NoVertexing, which does not contain any cuts on the dimuon vertex prob-
 660 ability, is available in data only from the 2E33 trigger menu, therefore this study is performed
 661 only for the HLT_DoubleMu4_Jpsi_displaced path. We define:

- 662 • $N(J/\psi)_{DoubleMu4}$ = number of reconstructed J/ψ firing the HLT_DoubleMu4_Jpsi_displaced
 663 and requiring the HLT_Dimuon0_Jpsi_NoVertexing trigger bit
- 664 • $N(J/\psi)_{Dimuon0_NoVtx}$ = number of reconstructed J/ψ requiring the HLT_Dimuon0_Jpsi_NoVertexing
 665 trigger bit and satisfying $p_T(J/\psi) > 6.9$, $L_{xy}/\sigma_{xy} > 3$, $\cos\alpha > 0.9$ and $p_T(\mu) > 4$
 666 GeV/c.

667 The ratio $\frac{N(J/\psi)_{DoubleMu4}}{N(J/\psi)_{Dimuon0_NoVtx}}$ evolved as a function of the vertex CL is shown in Fig.54. Both data
 668 and MC turn-on curves have a similar increasing behavior: the offline requirement is set to
 669 $CL > 0.15$, as it is in the online.

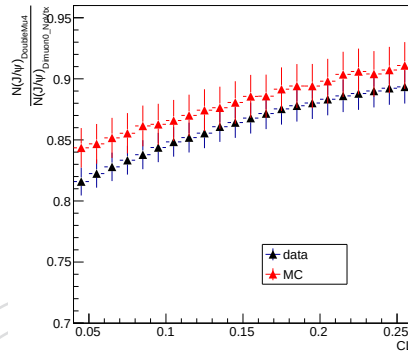


Figure 54: Number of reconstructed J/ψ firing the HLT_DoubleMu4_Jpsi_displaced trigger normalized to the J/ψ collected by the inclusive trigger, evolved as a function of the dimuon vertex probability.

12.3 Data and MC comparison for each trigger path for B^+ signal

In Sec. 8 data and MC distribution were compared for the whole available data sample. The high $B^\pm$ statistics allows for comparing data and simulation in each of the three considered HLT paths. Plots are collected in Fig. 55, 56 and 57.

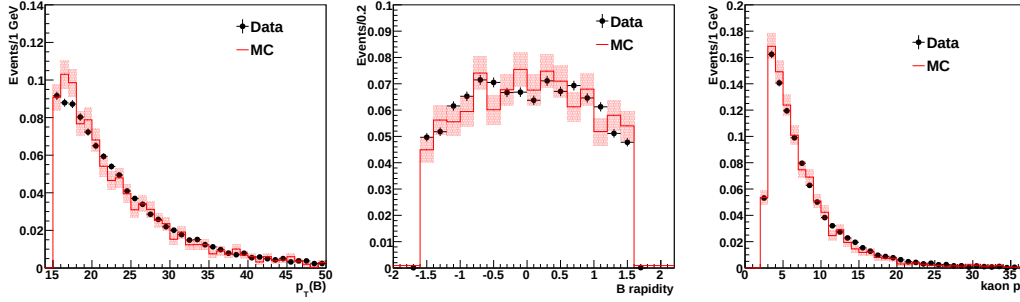


Figure 55: Data distribution (black) of B^+ p_T , y and kaon p_T are compared to MC (red) for the HLT_Dimuon7-Jpsi_Displaced trigger period.

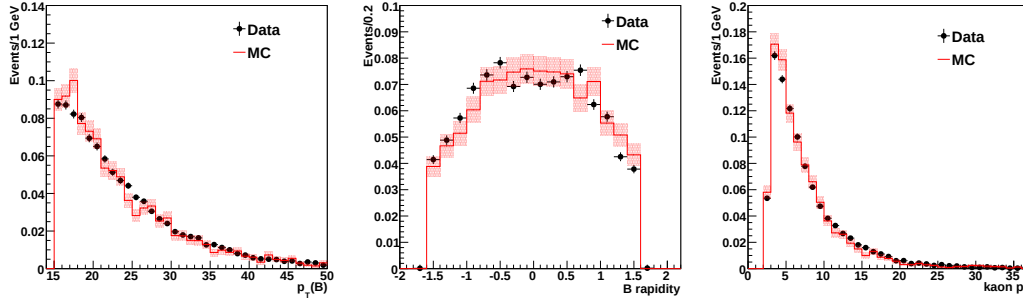


Figure 56: Data distribution (black) of B^+ p_T , y and kaon p_T are compared to MC (red) for the HLT_DoubleMu3p5-Jpsi_Displaced trigger period.

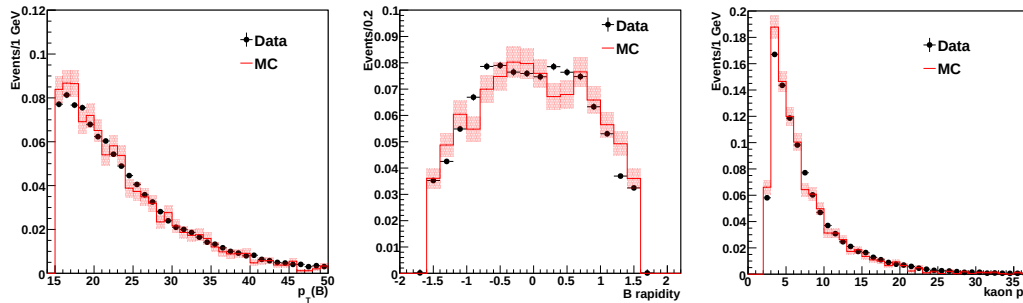


Figure 57: Data distribution (black) of B^+ p_T , y and kaon p_T are compared to MC (red) for the HLT_DoubleMu4-Jpsi_Displaced trigger period.

12.4 Split sample systematics

The data set is split in N independent samples: x_i individual measurements are performed with the associated errors σ_i , where i goes from 1 to N . If the N independent measurements are statistically consistent there is no evidence for a split sample systematic error. To test the statistical consistency a χ^2 with $N-1$ degrees of freedom is calculated through the form

$$\chi^2 = \sum_{i=1}^N \frac{(x_i - \bar{x})^2}{\sigma_i^2} \quad (12)$$

where $\bar{x}$ is the weighted average

$$\bar{x} = \frac{\sum_{i=1}^N x_i / \sigma_i^2}{\sum_{i=1}^N 1 / \sigma_i^2} \quad (13)$$

with an associated statistical error $\bar{\sigma}$

$$\bar{\sigma} = \frac{1}{\sqrt{\sum_{i=1}^N 1 / \sigma_i^2}} \quad (14)$$

If $\chi^2 / (N - 1) \leq 1$, then the measurements are consistent with each other within their errors. However, if $\chi^2 / (N - 1) > 1$, it is assumed that the split sample true errors are all underestimated because of an unknown systematic problem. If each subsample statistical error σ_i is scaled to

$$\sigma_i \sqrt{\frac{\chi^2}{(N - 1)}} \quad (15)$$

then it is guaranteed that $\chi^2 / (N - 1) = 1$ for the new value of σ_i . In addition $\bar{\sigma}$ will be increased by the same factor and become a scaled error $\tilde{\sigma}$

$$\tilde{\sigma} = \bar{\sigma} \sqrt{\frac{\chi^2}{(N - 1)}} = \sqrt{\frac{\bar{x}^2 - \bar{x}^2}{N - 1}} \quad (16)$$

where $\bar{x}^2$ is defined as

$$\bar{x}^2 = \frac{\sum_{i=1}^N x_i^2 / \sigma_i^2}{\sum_{i=1}^N 1 / \sigma_i^2} \quad (17)$$

When this scaled error is smaller than the statistical error from the fit of the unsplit data set (σ_{stat}), any difference between the split samples can be interpreted as being compatible with a statistical fluctuation. The final systematic error σ_{sys} is evaluated as

$$\begin{aligned} \sigma_{sys} &= \sqrt{\tilde{\sigma}^2 - \sigma_{stat}^2} & \text{if } \tilde{\sigma} > \sigma_{stat} \\ \sigma_{sys} &= 0 & \text{if } \tilde{\sigma} \leq \sigma_{stat} \end{aligned} \quad (18)$$

12.5 Fit variant systematics

The method to evaluate the systematic uncertainty relies on the fit variant approach [21] where the fitting conditions are varied, within reasonable limits. Only fits with sufficiently good fit quality and χ^2 are retained. The results from each different fit are considered to be equally valid, and the statistical errors on the measurements correlated. Since each fit is *a priori* as likely, it is given equal weight and the estimate of the systematic error is simply the standard deviation of the set of variants. The systematic error is then evaluated as in Eq. 19

$$\sigma_{sys} = \sqrt{\frac{\sum_{i=1}^N x_i^2 - N\bar{x}^2}{N-1}} \quad (19)$$

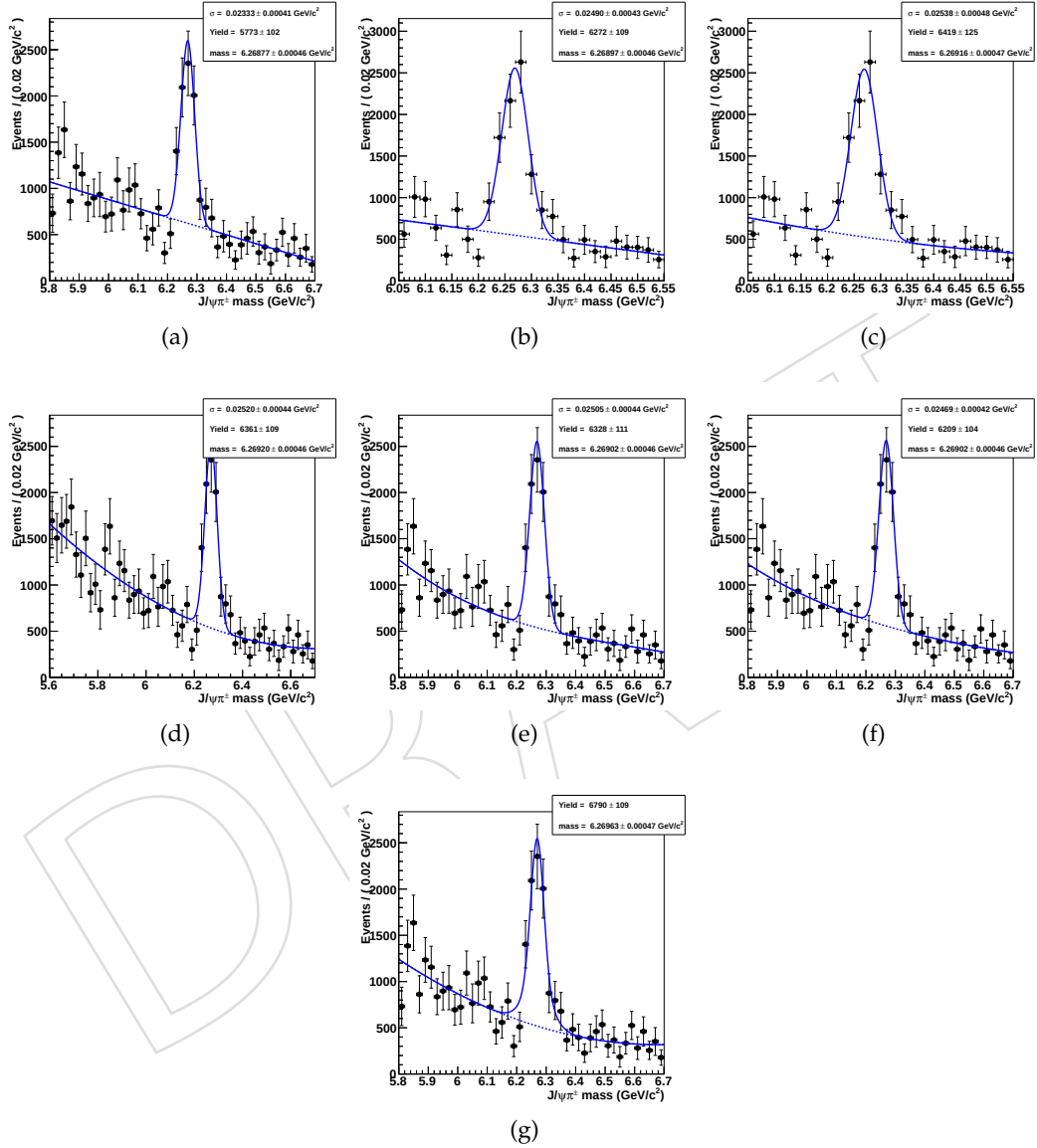
where x_i are the individual measurements and $\bar{x}$ is their mean.

12.5.1 Fit results

We summarize here the results for the different fits on the efficiency corrected data distributions for $B_c^\pm \rightarrow J/\psi\pi^\pm$, $B^\pm \rightarrow J/\psi K^\pm$ and $B_c^\pm \rightarrow J/\psi\pi^\pm\pi^\pm\pi^\mp$ respectively.

12.5.1.1 Y_{B_c} Fit Variant Plots of the different fits performed on the Y_{B_c} distribution are shown in Fig. 58 and correspond to:

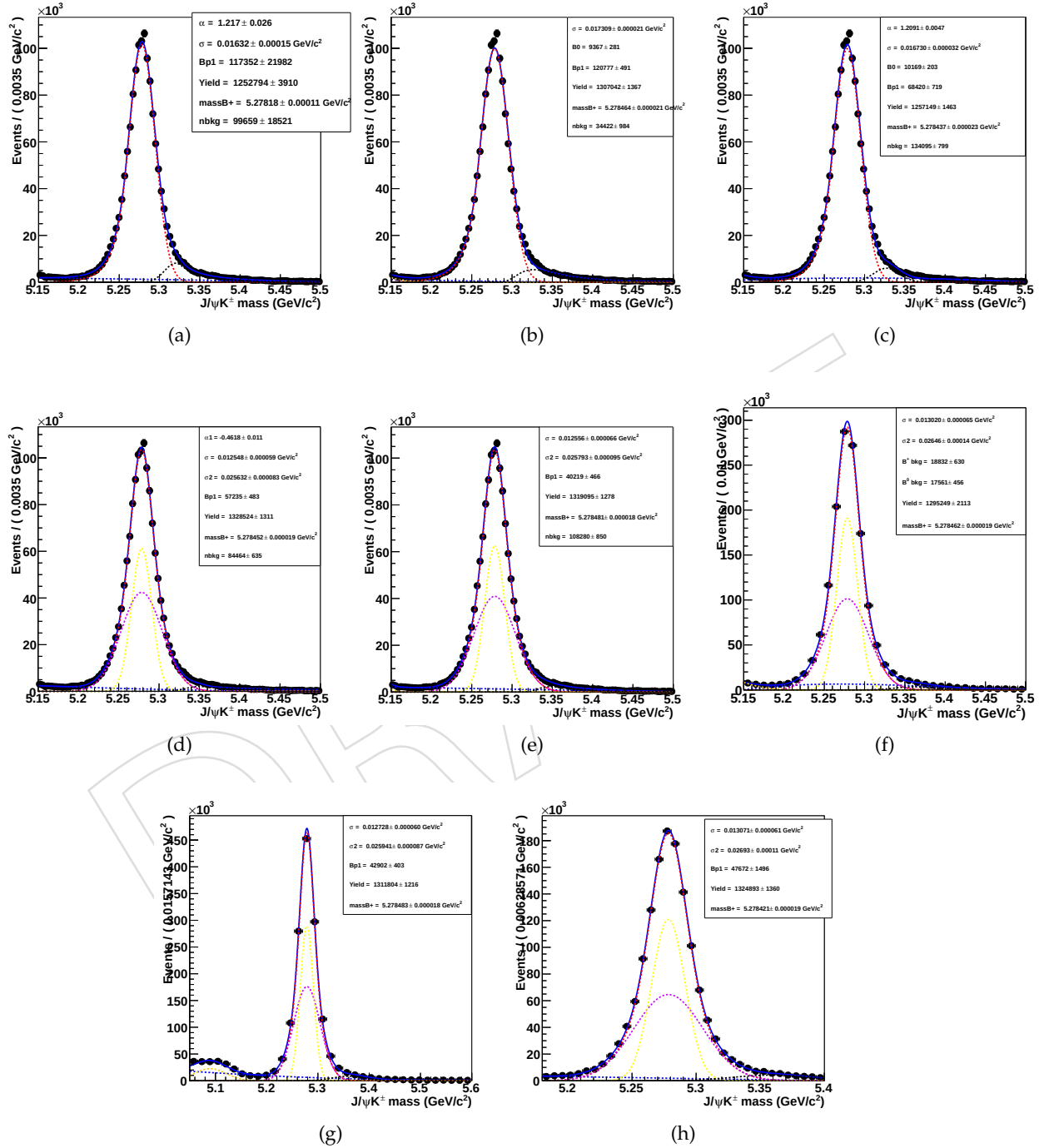
- Gaussian + 1st order Chebyshev polynomial (5.8-6.7 GeV/c²) (a)
- Gaussian + 1st order Chebyshev polynomial (6.05-6.55 GeV/c²) (b)
- Gaussian + 2nd order Chebyshev polynomial (6.05-6.55 GeV/c²) (c)
- Gaussian + 2nd order Chebyshev polynomial (5.6-6.7 GeV/c²) (d)
- Gaussian + 3rd order Chebyshev polynomial (5.8-6.7 GeV/c²) (e)
- Gaussian + Exponential (5.8-6.7 GeV/c²) (f)
- Double Gaussian + 2st order Chebyshev polynomial (5.8-6.7 GeV/c²) (g)

Figure 58: Fit variants for Y_{B_c} .

12.5.1.2 Υ_B Fit Variant

As stated in Sec. 9.1.2, the fit on the $J/\psi K$ invariant mass has been performed with different parametrizations and ranges. The B^0 component is always parametrized with a gaussian. Plots, in the following order, are collected in Fig. 59.

- Crystal Ball (signal) + 2^{nd} order Chebyshev pol. (comb. bkg) + Crystal Ball ($B \rightarrow J/\psi\pi$), Range(5.15-5.5 GeV/c²) (a)
- Crystal Ball (signal) + 2^{nd} order Chebyshev pol. (comb. bkg) + Landau ($B \rightarrow J/\psi\pi$), Range(5.15-5.5 GeV/c²) (b)
- Crystal Ball (signal) + 2^{nd} order Chebyshev pol. (comb. bkg) + gaussian ($B \rightarrow J/\psi\pi$), Range(5.15-5.5 GeV/c²) (c)
- Double Gaussian (signal) + 2^{nd} order Chebyshev pol. (comb. bkg) + Crystal Ball($B \rightarrow J/\psi\pi$), Range(5.15-5.5 GeV/c²) (d)
- Double Gaussian (signal) + 2^{nd} order Chebyshev pol. (comb. bkg) + Landau ($B \rightarrow J/\psi\pi$), Range(5.15-5.5 GeV/c²) (e)
- Double Gaussian (signal) + 3^{rd} order Chebyshev pol. (comb. bkg) + gaussian ($B \rightarrow J/\psi\pi$), Range(5.15-5.5 GeV/c²) (f)
- Double Gaussian (signal) + 2^{nd} order Chebyshev pol. (comb. bkg) + gaussian ($B \rightarrow J/\psi\pi$), Range(5.05-5.6 GeV/c²) (g)
- Double Gaussian (signal) + 2^{nd} order Chebyshev pol. (comb. bkg) + gaussian ($B \rightarrow J/\psi\pi$), Range(5.18-5.4 GeV/c²) (h)

Figure 59: Fit variants for Y_B .

12.5.1.3 $\Upsilon_{3\pi}$ Fit Variant Plots of the different fits performed on the distribution of $\Upsilon_{3\pi}$ are shown in Fig. 60 and correspond to:

- Gaussian + 1st order Chebyshev polynomial (6.13-6.4 GeV/c²) (a)
- Gaussian + 2st order Chebyshev polynomial (6.10-6.4 GeV/c²) (b)
- Gaussian + 2nd order Chebyshev polynomial (6.1-6.505 GeV/c²) (c)
- Gaussian + 2nd order Chebyshev polynomial (5.8-6.7 GeV/c²) (d)
- Gaussian + 3rd order Chebyshev polynomial (6 - 6.6 GeV/c²) (e)
- Gaussian + 3rd order Chebyshev polynomial (5.8-6.7 GeV/c²) (f)

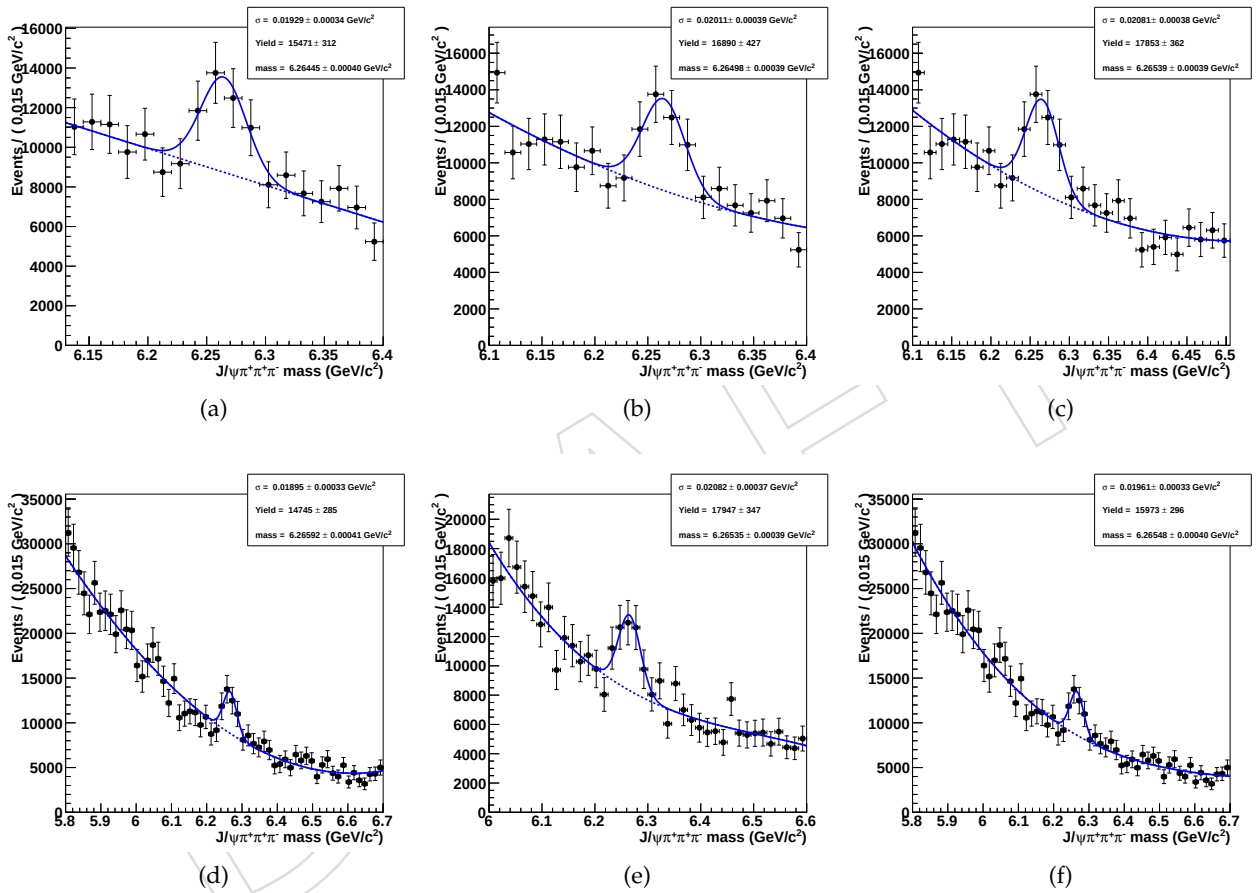


Figure 60: Fit variants for $\Upsilon_{3\pi}$.

12.6 Signal stability studies

12.6.1 $B_c^\pm \rightarrow J/\psi \pi^\pm \pi^\pm \pi^\mp$ signal

The evolution of the yield and of $S/\sqrt{S+B}$ (where both S and B are evaluated on data) as a function of selection cuts has been studied in order to check the stability of the B_c signal. Only one variable has been varied at a time. Plots are shown in Fig. 61.

In order to check if the MC simulation well describes the data, a further test have been performed: one cut is evolved at a time both in data and MC and the evolution of the ratio N_{data}/N_{MC} (where N_{data} is the signal yield in data and N_{MC} is the signal yield in MC) is investigated. The evolutions, collected in Fig.62, are sufficiently flat, as expected.

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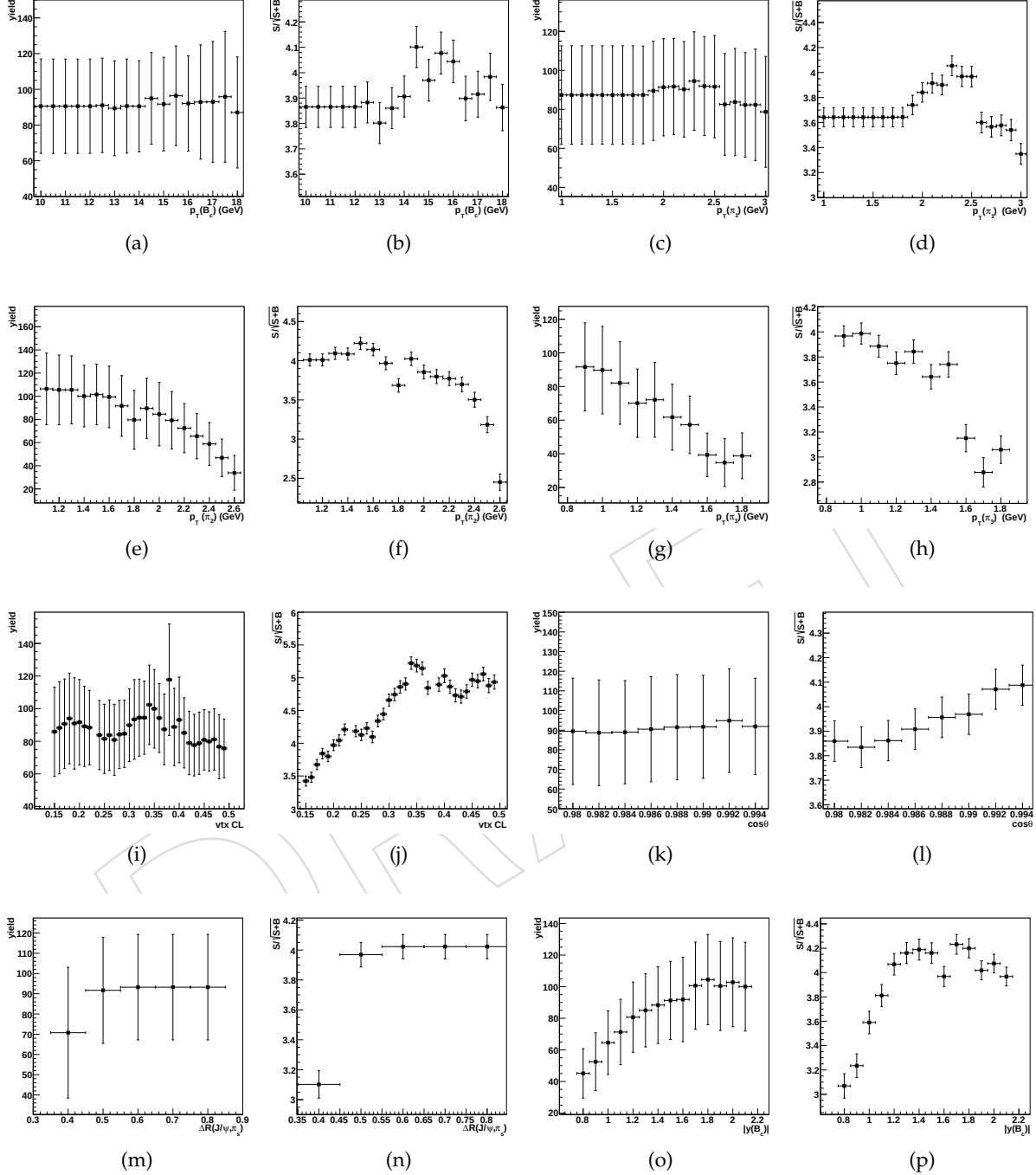


Figure 61: Yield and $S/\sqrt{S+B}$ behavior when one cut is evolved at a time for the $B_c^\pm \rightarrow J/\psi \pi^\pm \pi^\pm \pi^\mp$ channel. The plots are shown in pairs, the former shows the trend of the yield while the latter shows the trend of $S/\sqrt{S+B}$. Both S and B are evaluated on data. The considered variables are: $p_T(B_c)$; $p_T(\pi_1)$, $p_T(\pi_2)$ and $p_T(\pi_3)$, where the three pions are sorted by their p_T ; CL of the secondary vertex; $\cos\theta$; $\Delta R(J/\psi, \pi_5)$; $y(B_c)$.

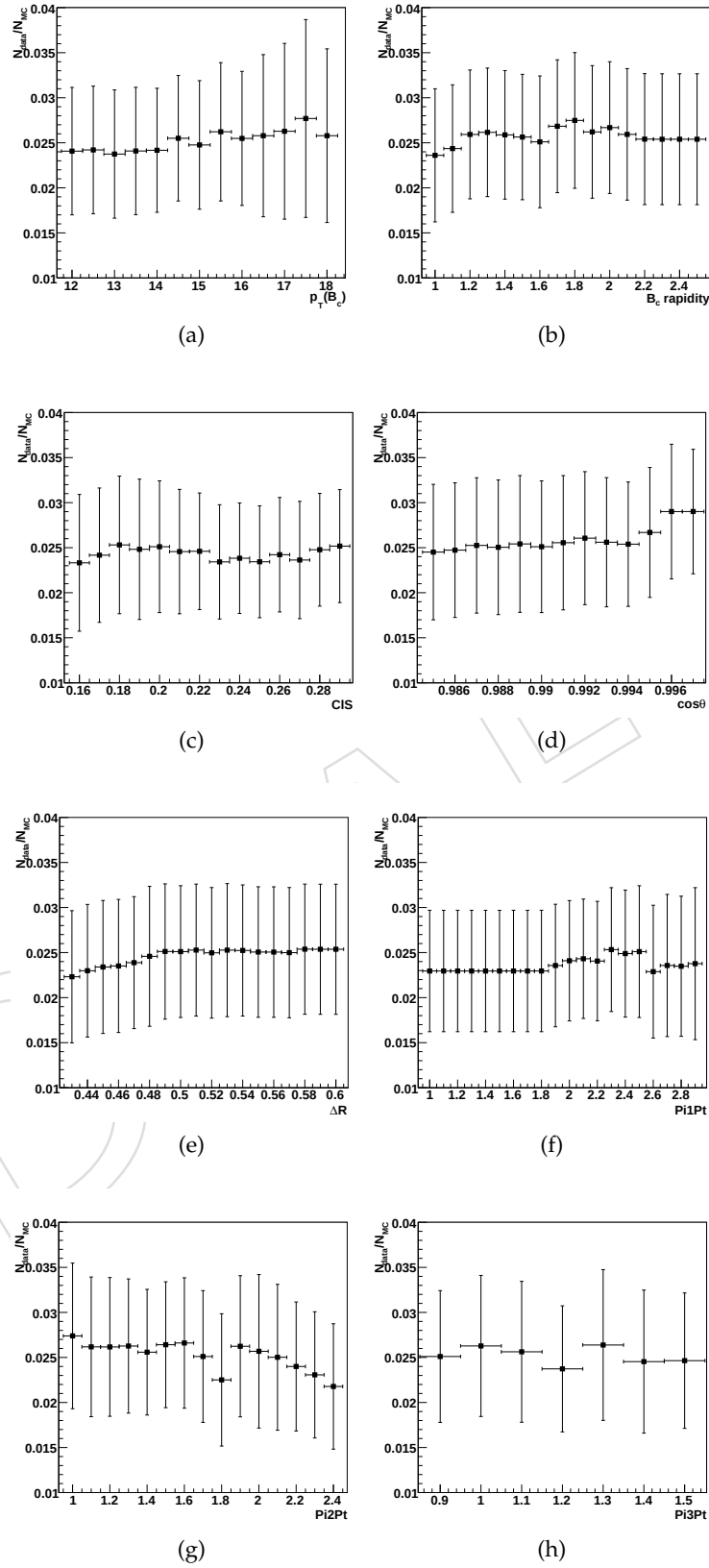


Figure 62: Evolution of the ratio N_{data}/N_{MC} for the $B_c^\pm \rightarrow J/\psi \pi^\pm \pi^\pm \pi^\mp$ channel as a function of the analysis cuts. The considered variables are: $p_T(B_c)$, (B_c) rapidity, CL of the secondary vertex, $\cos\theta$, $\Delta R(J/\psi, \pi_S)$, $p_T(\pi_1)$, $p_T(\pi_2)$ and $p_T(\pi_3)$.

12.6.2 $B_c^\pm \rightarrow J/\psi \pi^\pm$ signal

The evolution of the yield and of $S/\sqrt{S+B}$ (where S and B are evaluated on data) as a function of selection cuts is shown in Fig. 63.

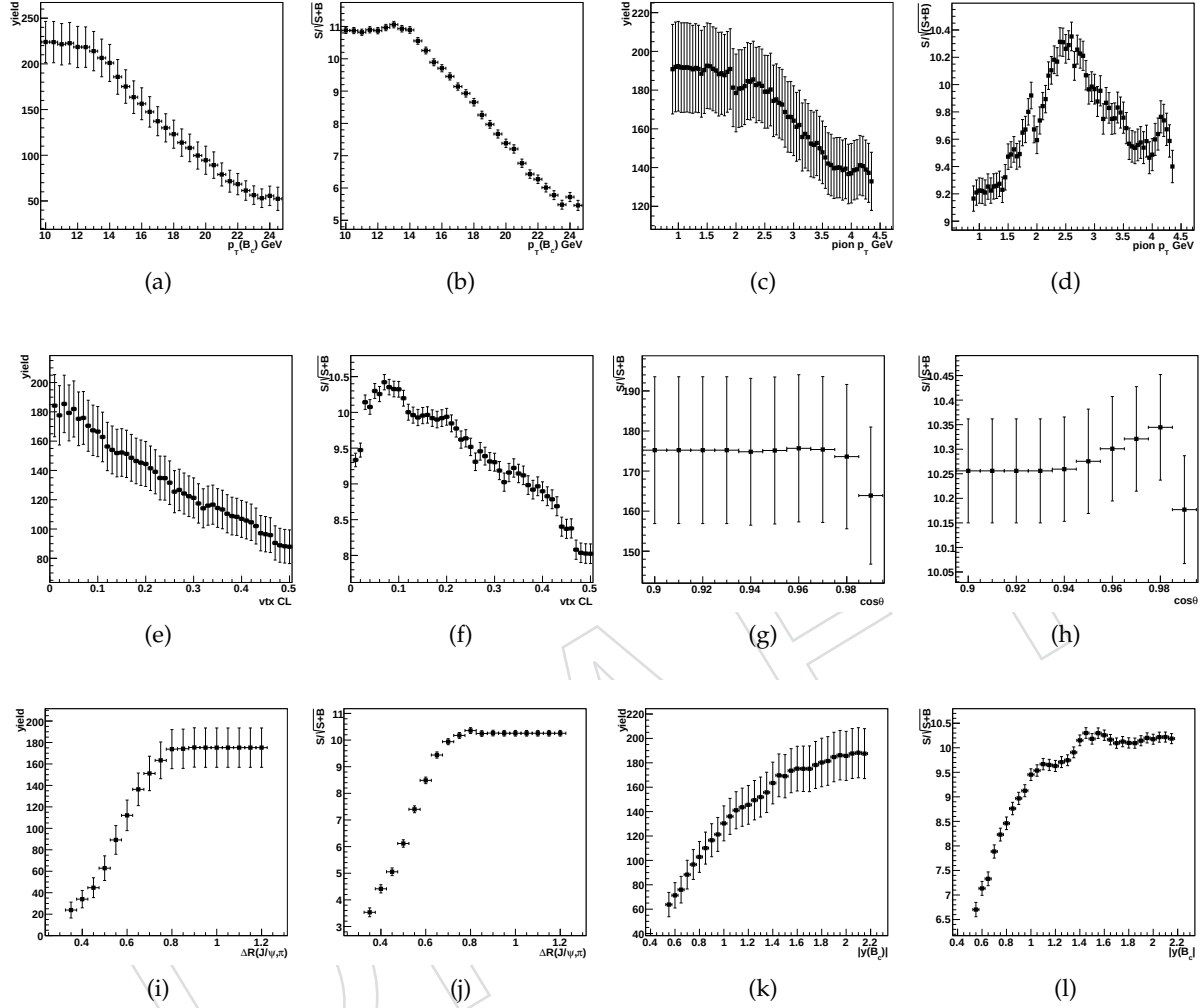


Figure 63: Yield and $S/\sqrt{S+B}$ behavior when one cut is evolved at a time for the $B_c^\pm \rightarrow J/\psi \pi^\pm$ channel. The plots are shown in pairs, the former shows the trend of the yield while the latter shows the trend of $S/\sqrt{S+B}$. Both S and B are evaluated on data. The considered variables are: $p_T(B_c)$; $p_T(\pi)$; CL of the secondary vertex; $\cos\theta$; $\Delta R(J/\psi, \pi)$; $y(B_c)$.

In Fig.64 one cut is evolved at a time both in data and MC and the evolution of the ratio N_{data}/N_{MC} (where N_{data} is the signal yield in data and N_{MC} is the signal yield in MC) is investigated. The evolutions are flat as expected.

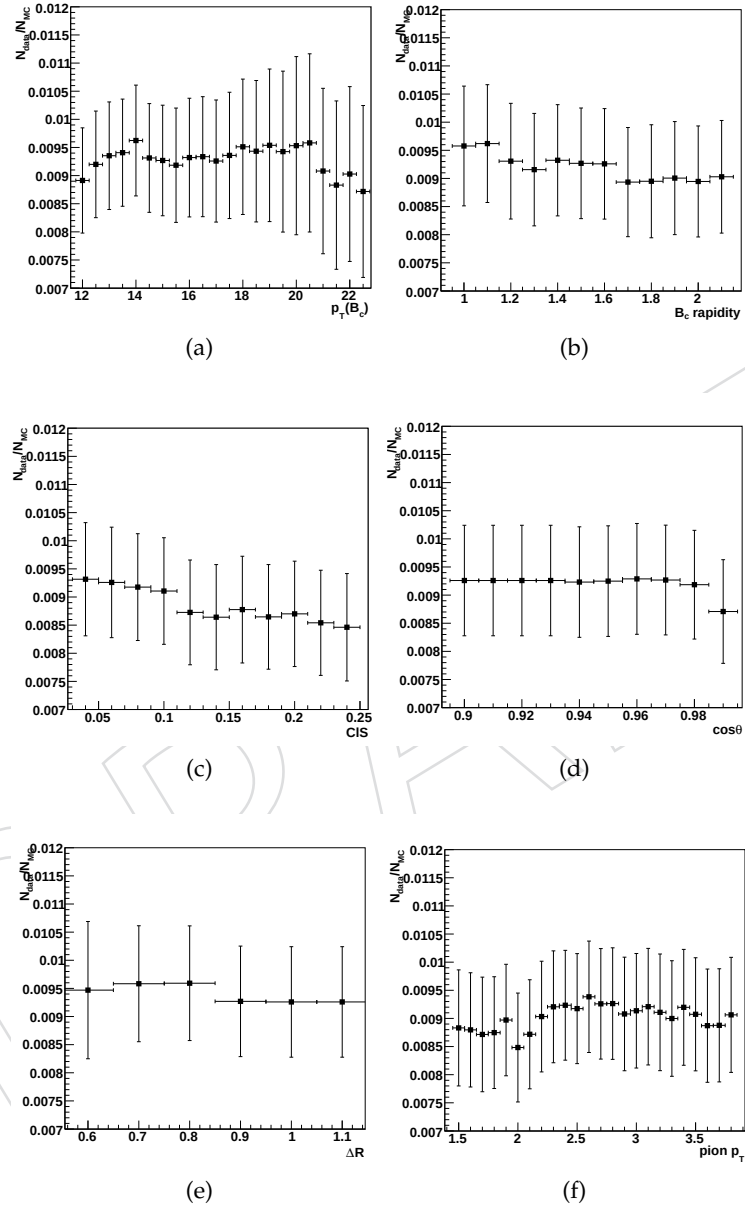


Figure 64: Evolution of the ratio N_{data}/N_{MC} for the $B_c^\pm \rightarrow J/\psi \pi^\pm$ channel as a function of the analysis cuts. The considered variables are: $p_T(B_c)$, (B_c) rapidity, CL of the secondary vertex, $\cos\theta$, $\Delta R(J/\psi, \pi)$ and $p_T(\pi)$.

12.7 Additional stability checks on $R_{c/u}$

The $R_{c/u}$ ratio has been also studied as a function of the value of the principal analysis cuts, such as B_c (B^+) vertex confidence level, pion/kaon p_T , $\cos\theta$ and ΔR . Results are shown in Fig. 65.

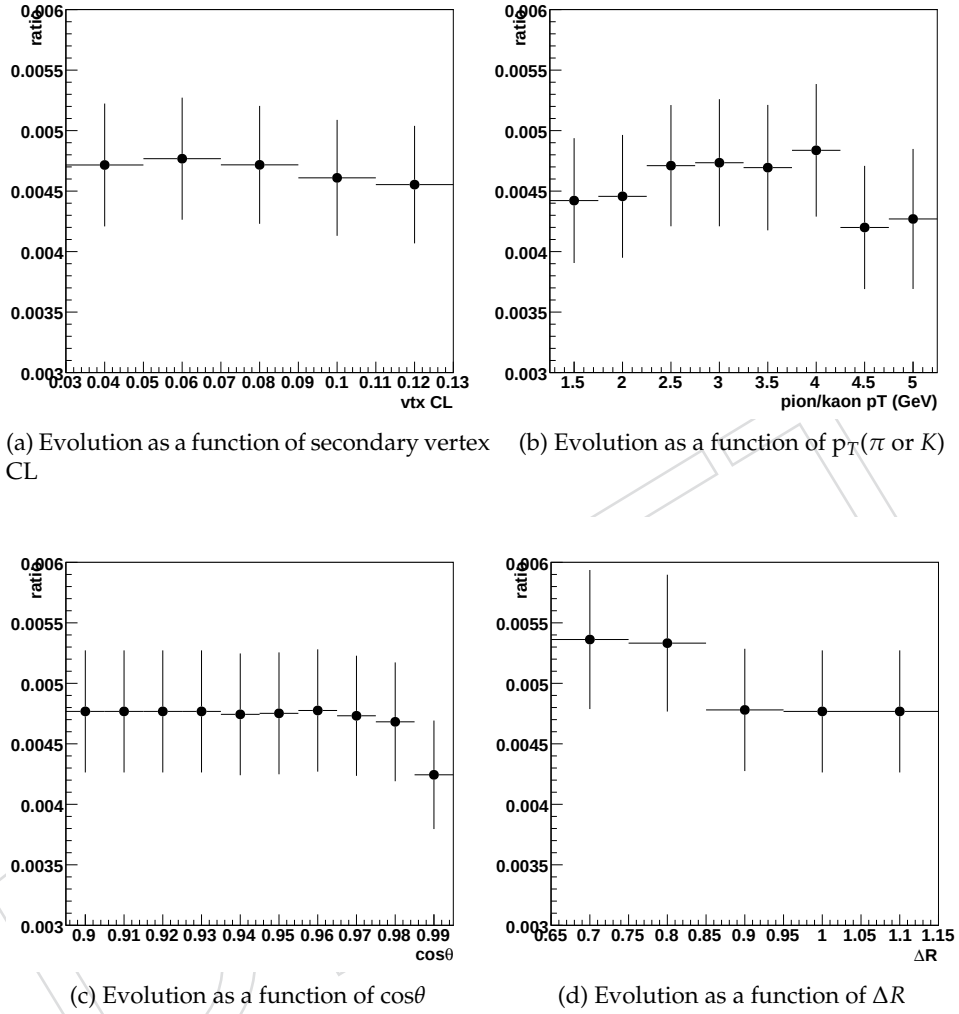


Figure 65: Behavior of $R_{c/u}$ when one cut is evolved for both of the decay channels.

12.8 Efficiency study in the $B_c^\pm \rightarrow J/\psi a_1(1260)$ simulation

The efficiency function parameters have been computed on the a_1 resonant MC as well. The resulting efficiency functions, one for each HLT path, are superimposed and projected on the 7 submasses in Fig.66. The corresponding $Y_{3\pi}$ has been evaluated following the procedure explained in Sec.10.1, $Y_{3\pi} = 17187 \pm 5044$, and the ratio results to be

$$R_{Bc} = \frac{BR(B_c^\pm \rightarrow J/\psi \pi^\pm \pi^\pm \pi^\mp)}{BR(B_c^\pm \rightarrow J/\psi \pi^\pm)} = 2.73 \pm 0.85 \quad (20)$$

No relevant discrepancy in the ratio determination is experienced when using the a_1 MC sample. It is worth recalling that the approach used to evaluate the efficiency (cfr Sec.10.1) makes the determination of the ratio essentially independent on the decay process model. We quote the ratio value as inferred from the NR MC sample since it covers all the phase space configurations. The systematic error already estimated would cover possible residual differences due to the limited MC sample statistics.

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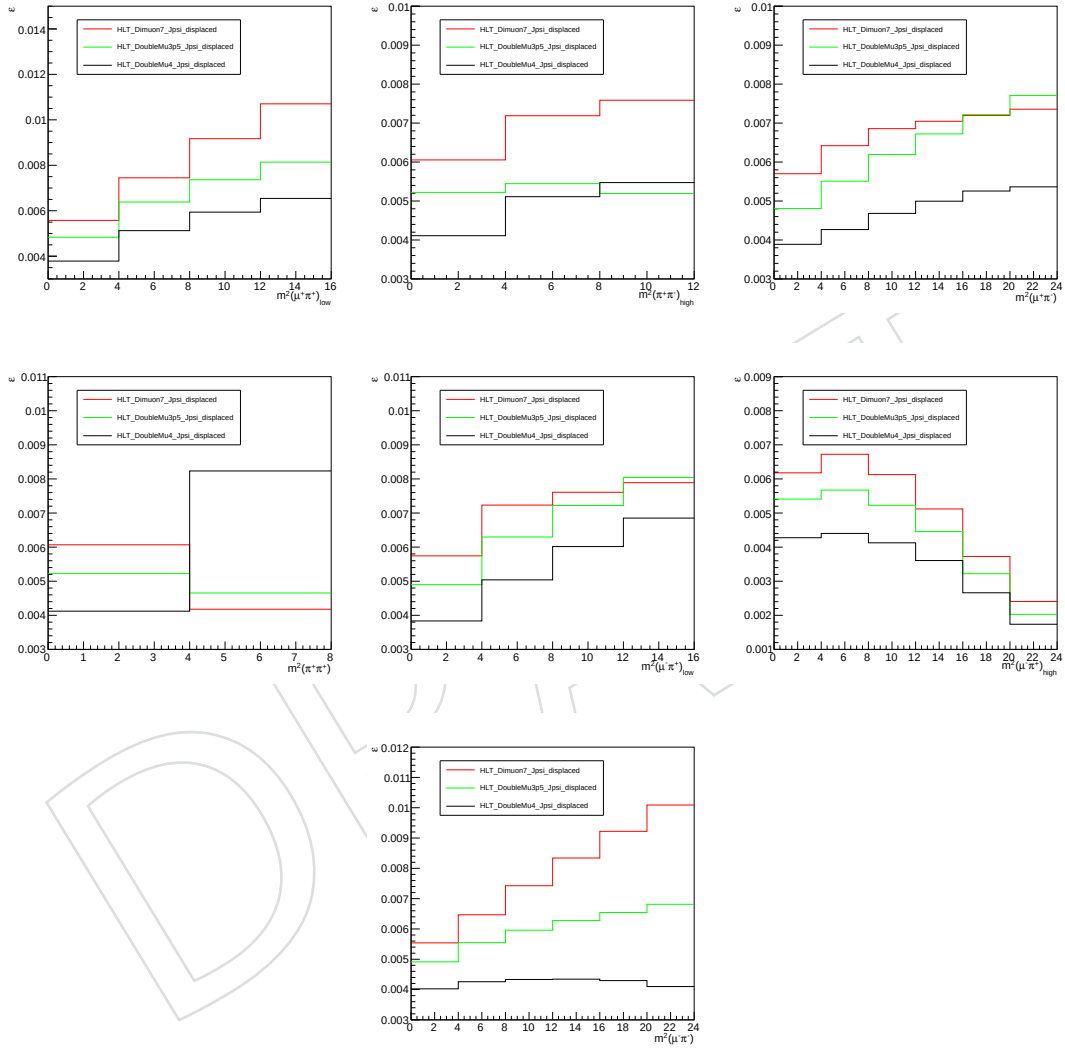


Figure 66: Projection of the efficiency function on the 7 two-body invariant masses used in the fit for each of the three HLT paths used in the analysis.

References

- [1] LHCb Collaboration, “Measurement of the B_c^+ meson lifetime using $B_c^+ \rightarrow J/\psi \mu^+ \nu X$ decays”, *JHEP* **74** (2014) 2839, doi:10.1140/epjc/s10052-014-2839-x.
- [2] A. Berezhnoy et al., “ B_c meson at LHC”, *Phys.Atom.Nucl* **60** (1997) 1729.
- [3] CDF Collaboration, “Observation of the B_c meson in $p\bar{p}$ collisions at $\sqrt{s} = 1.8$ TeV”, *Phys. Rev. Lett.* **81** (1998) 2432.
- [4] LHCb Collaboration, “First observation of the decay $B_c^\pm \rightarrow J/\psi \pi^\pm \pi^\pm \pi^\mp$ ”, *Phys.Rev.Lett.* **108** (2012) 251802, doi:http://dx.doi.org/10.1103/PhysRevLett.108.251802.
- [5] LHCb Collaboration, “Observation of the decay $B_c^+ \rightarrow J/\psi K^+ K^- \pi^+$ ”, *JHEP* **1311** (2013) 094, doi:10.1007/JHEP11(2013)094.
- [6] LHCb Collaboration, “Observation of the decay $B_c^+ \rightarrow B_s^0 \pi^+$ ”, *Phys. Rev. Lett.* **111** (2013) 181801, doi:10.1103/PhysRevLett.111.181801.
- [7] LHCb Collaboration, “Observation of $B_c^+ \rightarrow J/\psi D_s^+$ and $B_c^+ \rightarrow J/\psi D_s^{*+}$ ”, *Phys. Rev.D* **87** (2013) 112012, doi:10.1103/PhysRevD.87.112012.
- [8] LHCb Collaboration, “Observation of the decay $B_c^+ \psi(2S) \pi^+$ ”, *Phys. Rev. D.* **87** (2013) 071103, doi:10.1103/PhysRevLett.111.181801.
- [9] CDF Collaboration, “Measurement of the B_c^- meson lifetime in the decay $B_c^- \rightarrow J/\psi \pi^-$ ”, *Phys. Rev. D.* **87** (2013) 011101.
- [10] LHCb Collaboration, “Measurement of B_c^+ production and mass with the $B_c^+ \rightarrow J/\psi \pi^+$ decay”, *Phys.Rev.Lett.* **109** (2012) 232001.
- [11] C. Chang, C. Driouchi, P. Eerola, and X. Wu, “BCVEGPY: an event generator for hadronic production of the B_c meson”, *Comput.Phys.Communication* **159** (2004) 33, doi:10.1016/j.cpc.2004.02.005.
- [12] J. Chang, C Wang and X. Wu, “BCVEGPY2.0: An upgraded version of the generator BCVEGPY with the addition of hadroproduction of the P-wave B_c states”, *Comput.Phys.Communication* **174** (2006) 11, doi:10.1016/j.cpc.2005.09.008.
- [13] T. Sjöstrand, S. Mrenna, and P. Skands, “Pythia 6.4 physics and manual”, *JHEP* **0605** (2006) 026, doi:10.1088/1126-6708/2006/05/026.
- [14] R. Field, “Early LHC underlying event data-findings and surprises”, technical report, (2010).
- [15] D. Lange, “The EvtGen particle decay simulation package”, *Nucl.Instrum.Meth A* **462** (2001) 152, doi:10.1016/S0168-9002(01)00089-4.
- [16] Geant 4 Collaboration, “Geant4 toolkit for simulation of HEP experiments”, *Nucl.Instrum.Meth A* **502** (2003) 666, doi:10.1016/S0168-9002(03)00538-2.
- [17] G. Forden and D. Saxon, “Improving vertex position determination using a kinematic fit”, *Nucl. Instrum. Meth. A* **248** (1986) 439, doi:10.1016/0168-9002(86)91031-4.

- 799 [18] K. Prokofiev and T. Speer, "Vertex Fitting in the CMS tracker", CMS Internal Note
800 2006/032, Physik Institut Universitat Zurich, (2006).
- 801 [19] J. Beringer e al. (Particle Data Group), "Review of Particle Physics", *Phys. Rev. D* **86**
802 (2012) and 2013 partial update for the 2014 edition (URL:<http://pdg.lbl.gov>) 010001.
- 803 [20] CMS Collaboration, "Measurement of Tracking Efficiency", CMS Physics Analysis
804 Summary CMS-PAS-TRK-10-002, (2010).
- 805 [21] FOCUS Collaboration, "Study of the Cabibbo-suppresses decay mode $D^0 \rightarrow \pi^- \pi^+$ and
806 $D^0 \rightarrow K^- K^+$ ", *Phys. Lett. B* **555** (2003) 167, doi:[10.1016/S0370-2693\(03\)00053-4](https://doi.org/10.1016/S0370-2693(03)00053-4).

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